Technical University of Munich ABB AG Institute of Flight System Dynamics







Physics-Informed Machine Learning for Interpretable Fault Diagnosis in PMSM-Driven Belt Transmission Systems

–

Physikalisch informiertes maschinelles Lernen für die interpretierbare Fehlerdiagnose in PMSM-getriebenen Riemenantriebssystemen

Master's Thesis

Author: Chengkai Shen

Matriculation Number: 03779882

Supervisors: Dr.-Ing. Agnes Steinert
M.Sc. Zoe Mbikayi

Examiner: Prof. Dr.-Ing. Florian Holzapfel

2026-04-29

Statutory Declaration

I, Chengkai Shen, declare on oath towards the Institute of Flight System Dynamics of Technical University of Munich, that I have prepared the present Master's Thesis independently and with the aid of nothing but the resources listed in the bibliography.

This thesis has neither as-is nor similarly been submitted to any other university.

Garching, 2026-04-29

Chengkai Shen

Abstract

Permanent magnet synchronous motor-driven belt transmission systems are widely used in industrial automation, where mechanical faults cause progressive degradation and unplanned downtime. Conventional fault diagnosis methods either rely on dedicated sensors unavailable in standard servo drive installations or lack the physical interpretability required for safety-critical applications. This thesis presents two physics-informed fault diagnosis methods that operate exclusively on the motor current and encoder angle already integrated in the servo drive, requiring no additional instrumentation. Both methods follow a three-stage structure: physical modeling of the rotor dynamics, healthy system identification via physics-informed neural networks solving an inverse problem for the unknown physical parameters, and fault diagnosis through sparse identification of fault-induced deviations from the healthy baseline. The first method formulates the dynamics in terms of the angular velocity and applies sequentially thresholded least squares regression to a numerically differentiated fault residual as a separate post-processing step. The second method reformulates the dynamics in terms of the angular position, predicts the trigonometric projections of the encoder angle using a sinusoidal representation network, recovers all kinematic quantities via automatic differentiation, and embeds the sparse fault identification directly into the neural network training loop. Both methods are validated experimentally on a servo test bench with a toothed belt transmission under multiple fault scenarios and motion profiles. The healthy parameters identified by both methods agree within three percent. In the fault diagnosis stage, the speed-based method correctly identifies the dominant fault signatures but retains spurious terms caused by numerical differentiation noise, while the angle-based method yields substantially sparser and physically consistent fault identifications. The results demonstrate that coupling physics-informed learning with sparse identification within a single differentiable framework improves both the sensitivity and the interpretability of fault diagnosis using only standard servo drive signals.

Kurzfassung

Permanentmagnet-Synchronmotor-getriebene Zahnriemenübertragungssysteme sind in der industriellen Automatisierung weit verbreitet, wo mechanische Fehler zu fortschreitender Degradation und ungeplanten Stillstandszeiten führen. Herkömmliche Fehlerdiagnosemethoden erfordern entweder zusätzliche Sensorik, die in Standard-Servoantriebsinstallationen nicht verfügbar ist, oder ihnen fehlt die physikalische Interpretierbarkeit, die für sicherheitskritische Anwendungen erforderlich ist. Diese Arbeit stellt zwei physikalisch informierte Fehlerdiagnosemethoden vor, die ausschließlich mit dem Motorstrom und dem Gebersignal arbeiten, die bereits im Servoantrieb integriert sind, ohne zusätzliche Instrumentierung zu benötigen. Beide Methoden folgen einer dreistufigen Struktur: physikalische Modellierung der Rotordynamik, Identifikation des gesunden Systems mittels physikalisch informierter neuronaler Netze durch Lösung eines inversen Problems für die unbekannten physikalischen Parameter, und Fehlerdiagnose durch dünnbesetzte Identifikation fehlerbedingter Abweichungen von der gesunden

Basislinie. Die erste Methode formuliert die Dynamik in Bezug auf die Winkelgeschwindigkeit und wendet eine sequentiell schwellenwertbasierte Kleinste-Quadrate-Regression auf ein numerisch differenziertes Fehlerresiduum als separaten Nachverarbeitungsschritt an. Die zweite Methode reformuliert die Dynamik in Bezug auf die Winkelposition, sagt die trigonometrischen Projektionen des Geberwinkels mittels eines sinusoidalen Repräsentationsnetzwerks vorher, rekonstruiert alle kinematischen Größen über automatische Differentiation und bettet die dünnbesetzte Fehleridentifikation direkt in die Trainingsschleife des neuronalen Netzes ein. Beide Methoden werden experimentell an einem Servoprüfstand mit Zahnriemenübertragung unter mehreren Fehlerszenarien und Bewegungsprofilen validiert. Die von beiden Methoden identifizierten gesunden Parameter stimmen innerhalb von drei Prozent überein. In der Fehlerdiagnosephase identifiziert die geschwindigkeitsbasierte Methode die dominanten Fehlersignaturen korrekt, behält jedoch durch numerisches Differenzierungsrauschen verursachte Störterme bei, während die winkelbasierte Methode wesentlich dünnbesetztere und physikalisch konsistentere Fehleridentifikationen liefert. Die Ergebnisse zeigen, dass die Kopplung von physikalisch informiertem Lernen mit dünnbesetzter Identifikation innerhalb eines einzigen differenzierbaren Frameworks sowohl die Empfindlichkeit als auch die Interpretierbarkeit der Fehlerdiagnose unter ausschließlicher Verwendung standardmäßiger Servoantriebssignale verbessert.

Table of Contents

List of Figures vi

List of Tables viii

List of Algorithms ix

Table of Acronyms x

Table of Symbols xi

1 Introduction 1
- 1.1 Problem Description and Motivation . . . 1
- 1.2 Proposed Methods and Contributions . . . 2
- 1.3 Structure . . . 4

2 State of the Art 6
- 2.1 Signal-Based Approaches . . . 6
- 2.2 Model-Based Approaches . . . 7
- 2.3 Data-Driven Approaches . . . 7
- 2.4 Physics-Informed Machine Learning . . . 8
- 2.5 Research Gap . . . 10

3 Preliminaries 11
- 3.1 Rotation-Speed-Based Rotor System Modeling . . . 13
- 3.2 Rotation-Angle-Based Rotor System Modeling . . . 13
  - 3.2.1 Second-Order Angular Position Formulation . . . 13
  - 3.2.2 Trigonometric Angle Representation and Analytic Differentiation . . . 14
- 3.3 Physics-informed Neural Networks . . . 16
  - 3.3.1 Physics-Informed Neural Network for Forward Problem . . . 16
  - 3.3.2 Physics-Informed Neural Network for Inverse Problem . . . 19
  - 3.3.3 Normalization . . . 20
- 3.4 Sparse Identification of Nonlinear Dynamics . . . 22
- 3.5 Sinusoidal Representation Networks . . . 24

4 Rotation-Speed-Based Interpretable Fault Diagnosis Method 26
- 4.1 Physical Model for PINNs-based Healthy System Identification . . . 27
- 4.2 Healthy System Identification . . . 28
- 4.3 Fault Diagnosis . . . 32

5 Rotation-Angle-Based Interpretable Fault Diagnosis Method 37
- 5.1 Physical Model for PINNs-based Healthy System Identification . . . 39
- 5.2 Healthy System Identification . . . 41

5.3 Fault Diagnosis . . . 44
5.3.1 Fault Diagnosis Procedure . . . 45
5.3.2 Fault Diagnosis Formulation . . . 47
5.3.3 Practical Considerations and Discussion . . . 49

6 Experiment Setup 53
6.1 Experimental Test Bench . . . 53
6.2 Experimental Design . . . 55
6.3 Experiment Result . . . 56

7 Result and Discussion 61
7.1 Result of Rotation-Speed-Based Interpretable Fault Diagnosis . . . 61
7.1.1 Data Preprocessing . . . 61
7.1.2 Result of Rotation-Speed-Based Healthy System Identification . . . 62
7.1.3 Result of Rotation-Speed-Based Fault Diagnosis . . . 67
7.2 Result of Rotation-Angle-Based Interpretable Fault Diagnosis . . . 73
7.2.1 Data Preprocessing . . . 73
7.2.2 Result of Rotation-Angle-Based Healthy System Identification . . . 74
7.2.3 Result of Rotation-Angle-Based Fault Diagnosis . . . 76
7.3 Discussion of Diagnostic Results . . . 81
7.3.1 Comparative Assessment of the Two Methods . . . 81
7.3.2 Practical Insights from Implementation . . . 82
7.3.3 Current Limitations . . . 86

8 Conclusion 88
8.1 Summary . . . 88
8.2 Future Work . . . 89

Acknowledgements I

References III

List of Figures

3–1 Prototype photograph of the test bench 12
3–2 Simplified dynamic model of the PMSM-driven synchronous belt transmission system 12
3–3 (a) Raw encoder angle with 2π-wrapping discontinuities. (b) Continuous trigonometric projections $\cos\theta(t)$ and $\sin\theta(t)$. 15
3–4 General workflow of PINNs for forward problems 18
3–5 General workflow of PINNs for inverse problems 20

4–1 Workflow of the speed-based fault diagnosis method: (i) rotor system modeling, (ii) PINN-based healthy system identification, (iii) STLSQ sparse regression on the fault residual. 27
4–2 Hyperbolic tangent activation function $\tanh(x)$. 29
4–3 Softplus function $\zeta(x) = \ln(1 + e^x)$. 31

5–1 Workflow of the angle-based fault diagnosis method: (i) rotor system modeling, (ii) SIREN-based PINN healthy system identification, and (iii) PINN-embedded sparse fault identification with alternating ℓ_0/ℓ_1 sparsity promotion. 38

6–1 Experimental test bench. (A) Transmission mechanism. (B) Servo drive. (C) Disassembled motor with small pulley. (D) Inertial load. (E) Control cabinet. 53
6–2 Fault Terms Configuration. (A) Parallel misalignment between the drive and load-side shafts. (B) Tooth interference simulated by wire across 3 consecutive grooves on the load-side pulley. 55
6–3 Relative Movement in healthy scenario - load-side rotation distances of $+720°$, accelerations of 10 000 °/s^2, maximum velocity of 2400 °/s. 57
6–4 Relative Movement in misalignment scenario - load-side rotation distances of $+720°$, accelerations of 10 000 °/s^2, maximum velocity of 2400 °/s. 57
6–5 Relative Movement in tooth interference scenario - load-side rotation distances of $+720°$, accelerations of 10 000 °/s^2, maximum velocity of 2400 °/s. 58
6–6 Constant Speed Movement in healthy scenario - target velocity of 1440 °/s, acceleration of 2400 °/s^2, and without jerk. 58
6–7 Constant Speed Movement in misalignment scenario - target velocity of 1440 °/s, acceleration of 2400 °/s^2, and without jerk. 59
6–8 Constant Speed Movement in tooth interference scenario - target velocity of 1440 °/s, acceleration of 2400 °/s^2, and without jerk. 59

7–1 Forward simulation validation on a relative movement profile: load-side rotation distance of $+720°$, acceleration of 10 000 °/s^2, maximum velocity of 2400 °/s. Relative RMSE: 0.429 %. 64
7–2 Forward simulation validation on a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s^2, without jerk limitation. Relative RMSE: 3.552 %. Top: speed comparison; bottom: relative simulation error. 66

7–3 Relative RMSE of forward ODE simulation against measured speed for all 70 healthy trajectories, ranked in descending order. The red and blue dashed lines indicate thresholds of 3 % and 1 %, respectively. 67
7–4 Identified fault coefficients for parallel misalignment under a relative movement profile: load-side rotation distance of $+720^\circ$, acceleration of 10 000 °/s^2, maximum velocity of 2400 °/s. 69
7–5 Identified fault coefficients for parallel misalignment under a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s^2, without jerk limitation. 69
7–6 Identified fault coefficients for tooth interference under a relative movement profile: load-side rotation distance of $+720^\circ$, acceleration of 10 000 °/s^2, maximum velocity of 2400 °/s. 70
7–7 Identified fault coefficients for tooth interference under a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s^2, without jerk limitation. . 70
7–8 Validation of the identified fault model for tooth interference under the relative movement profile. Top: measured rotation speed versus the trajectory reconstructed by integrating the complete identified dynamics. Middle: measured angular acceleration versus the predicted complete derivative (healthy model plus sparse fault terms). Bottom: relative rotation speed error over time. Relative RMSE: 1.83 %. . . 72
7–9 Relative RMSE of forward ODE simulation against measured rotation speed for all 70 healthy trajectories using the angle-based identified parameters, ranked in descending order. The red and blue dashed lines indicate thresholds of 3 % and 1 %, respectively. 76
7–10 Identified fault coefficients for parallel misalignment under a relative movement profile: load-side rotation distance of $+720^\circ$, acceleration of 10 000 °/s^2, maximum velocity of 2400 °/s. 78
7–11 Identified fault coefficients for parallel misalignment under a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s^2, without jerk limitation. 78
7–12 Identified fault coefficients for tooth interference under a relative movement profile: load-side rotation distance of $+720^\circ$, acceleration of 10 000 °/s^2, maximum velocity of 2400 °/s. 79
7–13 Identified fault coefficients for tooth interference under a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s^2, without jerk limitation. . 80

List of Tables

6–1 Test Bench Hardware Components 54

List of Algorithms

4–1 Rotation-Speed-Based Healthy System Identification 32
4–2 Speed-Based Fault Diagnosis via STLSQ 35
5–1 Angle-Based Healthy System Identification 44
5–2 Angle-Based Fault Diagnosis via PINNs-Based Sparse Identification 52

Table of Acronyms

Acronym	Description
AD	Automatic Differentiation
FNN	Feedforward Neural Network
ML	Machine Learning
MSE	Mean Squared Error
ODE	Ordinary Differential Equation
PIML	Physics-Informed Machine Learning
PINNs	Physics-Informed Neural Networks
PLC	Programmable Logic Controller
PMSM	Permanent Magnet Synchronous Motor
ReLU	Rectified Linear Unit
RK4	Fourth-Order Runge–Kutta
RMS	Root Mean Square
RMSE	Root Mean Square Error
SINDy	Sparse Identification of Nonlinear Dynamics
SIREN	Sinusoidal Representation Network
STLSQ	Sequentially Thresholded Least Squares
STRidge	Sequentially Thresholded Ridge Regression

Table of Symbols

Latin Letters

Symbol	Description
a	perturbation amplitude
$\mathbf{b}_i$	bias vector of the i-th layer
B	viscous friction coefficient
B^*	scaled viscous friction coefficient
$\hat{B}$	identified viscous friction coefficient
$c(t)$	cosine projection of the shaft angle, $\cos\theta(t)$
$\hat{c}(t^*)$	neural network prediction of $c(t)$
d	input dimension
E_1	number of Phase I training epochs (angle-based method)
E_2	number of Phase II training epochs (angle-based method)
E_3	number of Stage II retraining epochs
f	harmonic frequency order
$F(x)$	belt force as a function of displacement
$\mathcal{F}$	differential operator of the governing equation
$\mathbf{h}_i$	hidden activation vector of the i-th layer
h_k	Savitzky–Golay convolution coefficient
J	total moment of inertia reflected to the motor shaft
J^*	scaled moment of inertia
$\hat{J}$	identified moment of inertia
J_L	load-side inertia
k	equivalent belt stiffness
k_1, k_2, k_3	linear, quadratic, and cubic belt stiffness coefficients
K	number of alternating ℓ_0/ℓ_1 loops
L	number of network layers
$\mathcal{L}_{\text{data}}$	data fidelity loss
$\mathcal{L}_{\text{geometry}}$	unit-circle geometry constraint loss
$\mathcal{L}_{\text{physics}}$	physics residual loss
$\mathcal{L}_{\text{h}}$	total loss for healthy system identification (angle-based)
$\mathcal{L}_{\text{f}}$	total loss for fault diagnosis Stage I
$\mathcal{L}_{\gamma}$	total loss for fault diagnosis Stage II
$\mathcal{L}_{\text{total}}$	total loss for healthy system identification (speed-based)
m	Savitzky–Golay half-window size, window length is $2m+1$
M	number of collocation points
n	number of input features to a network layer; harmonic order
N	number of observation (measurement) points
N_1	number of Phase I training iterations (speed-based method)
N_2	number of Phase II training iterations (speed-based method)
p	Savitzky–Golay polynomial degree

$\mathcal{P}(\boldsymbol{\xi})$	sparsity-promoting penalty function
r_m	motor-side pulley radius
r_L	load-side pulley radius
$R_f(t)$	fault residual
$\mathcal{R}^*$	normalized physics residual (speed-based)
$\mathcal{R}^*_\theta$	normalized physics residual (angle-based)
$\mathcal{R}_{\mathsf{f}}$	augmented fault physics residual
$\mathcal{R}_{\mathsf{geo}}$	unit-circle geometry residual
$s(t)$	sine projection of the shaft angle, $\sin\theta(t)$
$\hat{s}(t^*)$	neural network prediction of $s(t)$
t	physical time
t^*	normalized time, $t^* \in [0, 1]$
t_0	start time of the data segment
t_{end}	end time of the data segment
t_s	time span of the data segment, $t_s = t_{\text{end}} - t_0$
T	end of the temporal domain
T_c	Coulomb friction torque
T_c^*	scaled Coulomb friction torque
$\hat{T}_c$	identified Coulomb friction torque
$T_m(t)$	motor torque
$T_m^*(t)$	normalized motor torque
$u(t)$	external (forcing) input to the governing ODE
$\mathbf{W}_i$	weight matrix of the i-th layer
w_d	loss weight for the data loss
w_g	loss weight for the geometry loss
w_p	loss weight for the physics loss
w_{lasso}	loss weight for the ℓ_1 penalty
x	belt elastic displacement
x_0	nominal belt elastic displacement
$y(t)$	general system response

Greek Letters

Symbol	Description
α	weighting coefficient for physics loss; ridge regularization parameter
$\boldsymbol{\beta}$	vector of physical parameters, $\{J, B, T_c\}$
$\boldsymbol{\beta}^*$	vector of scaled physical parameters, $\{J^*, B^*, T_c^*\}$
$\hat{\boldsymbol{\beta}}$	vector of identified physical parameters, $\{\hat{J}, \hat{B}, \hat{T}_c\}$
$\boldsymbol{\gamma}$	fault amplitude coefficient vector
$\hat{\boldsymbol{\gamma}}$	identified (sparse) fault coefficient vector
Δt	sampling interval
Δ_{T_m}	min-max range of the motor torque
Δx	fault-induced perturbation of belt displacement

$\zeta(x)$	softplus function, $\zeta(x) = \ln(1 + e^x)$
η	learning rate
$\theta(t)$	angular position of the motor shaft
$\theta_m(t)$	angular position of the motor shaft (with explicit subscript)
$\theta_L(t)$	angular position of the load-side shaft
$\boldsymbol{\Theta}$	candidate library matrix (SINDy / fault diagnosis)
λ	sparsity threshold (STLSQ)
$\boldsymbol{\xi}$	fault coefficient vector (speed-based method)
σ_ω	standard deviation of the angular velocity signal
σ_{T_m}	standard deviation of the motor torque signal
$\sigma_{\dot{\theta}}$	standard deviation of the reconstructed angular velocity
τ_0	hard thresholding parameter (angle-based method)
$\boldsymbol{\phi}$	neural network weights and biases
$\boldsymbol{\phi}^*$	frozen neural network weights (Stage II)
$\boldsymbol{\varphi}$	phase parameter vector for harmonic fault terms
$\boldsymbol{\varphi}^*$	frozen phase parameters (Stage II)
$\omega(t)$	angular velocity of the motor shaft
$\omega^*(t^*)$	normalized angular velocity
ω_0	SIREN frequency hyperparameter
$\omega_m(t)$	angular velocity of the motor shaft (with explicit subscript)
$\omega_L(t)$	angular velocity of the load-side shaft

Operators and Functions

Symbol	Description
$\dot{(\cdot)}$	first time derivative, $\mathrm{d}/\mathrm{d}t$
$\ddot{(\cdot)}$	second time derivative, $\mathrm{d}^2/\mathrm{d}t^2$
$(\cdot)_s$	derivative with respect to normalized time t^*
$\mathrm{sgn}(\cdot)$	signum function
$\mathrm{atan2}(\cdot,\cdot)$	four-quadrant arctangent function
$\mathrm{std}(\cdot)$	standard deviation
$\|\cdot\|_0$	ℓ_0 norm (number of non-zero entries)
$\|\cdot\|_1$	ℓ_1 norm
$\|\cdot\|_2$	ℓ_2 norm

Subscripts and Superscripts

Symbol	Description
$(\cdot)_m$	motor side
$(\cdot)_L$	load side
$(\cdot)^*$	normalized or scaled quantity
$(\cdot)_s$	derivative with respect to normalized time t^*
$\hat{(\cdot)}$	estimated or network-predicted quantity
$(\cdot)_{\text{healthy}}$	healthy baseline prediction

$(\cdot)_{\text{PINN}}$	PINN network output
$(\cdot)_{\text{NN}}$	neural network output

1 Introduction

1.1 Problem Description and Motivation

Permanent magnet synchronous motor (PMSM)-driven belt transmission systems are widely employed in industrial automation, packaging machinery, and precision positioning applications, where they offer high torque density, accurate speed regulation, and low-maintenance operation. In these systems, a toothed synchronous belt transmits torque from a motor-side pulley to a load-side pulley through the meshing of belt teeth with pulley grooves. Over time, this mechanical interface is subject to progressive degradation under sustained dynamic loads. Common fault modes include tooth interference, where foreign debris becomes trapped in the belt–pulley mesh and disrupts the normal tooth engagement, and shaft misalignment, where a parallel offset between the drive and load-side shafts introduces asymmetric belt loading and uneven tooth wear. These faults increase friction, accelerate belt degradation, and introduce periodic torque disturbances into the drivetrain. If left undetected, they lead to reduced positioning accuracy, increased energy consumption, and ultimately unplanned downtime or sudden belt failure. Early and accurate fault detection is therefore critical to ensure operational reliability, reduce maintenance costs, and prevent safety hazards.

In recent years, data-driven Machine Learning (ML) techniques have demonstrated significant improvements over conventional methods in fault detection accuracy and adaptability. However, despite their effectiveness, purely data-driven approaches often operate as black boxes, lacking the physical interpretability essential for safety-critical industrial applications where understanding fault mechanisms is as important as detecting them. In parallel, classical frequency-domain fault diagnosis methods, such as spectral analysis and envelope detection, are primarily designed for steady-speed operating conditions. When the motor operates under variable-speed profiles involving acceleration and deceleration phases, the fault-characteristic frequencies shift over time, causing these methods to degrade considerably. Furthermore, many existing diagnostic approaches rely on multiple dedicated sensors, such as accelerometers, torque transducers, or load-side encoders, to capture fault-related signatures. In practice, standard PMSM transmission systems are often equipped only with the sensors integrated into the servo drive, namely a rotary encoder and a current sensor, making multi-sensor strategies difficult to deploy without additional hardware cost and installation effort. These challenges motivate the need for a diagnostic framework that is physically interpretable, robust under variable-speed operation, and capable of working with the limited sensor information already available in the system.

Physics-Informed Machine Learning (PIML) offers a promising direction to address these limitations by combining physical laws with data-driven models. Among PIML approaches, Physics-Informed Neural Networks (PINNs) are particularly relevant, as they embed known governing equations directly into the training loss function as soft constraints. This enables PINNs to identify system parameters and detect deviations from expected behavior even when data are limited or noisy. Crucially, PINNs can extract diagnostic information solely from the sensors al-

ready integrated into the servo drive, namely the rotary encoder and the current sensor, without requiring any additional instrumentation. This is a significant practical advantage, because the external sensors typically used in conventional fault diagnosis are both expensive and difficult to retrofit. For example, industrial-grade piezoelectric accelerometers from manufacturers such as Kistler or PCB Piezotronics typically cost several hundred euros per unit, and an inline rotary torque transducer such as the HBM T210 can cost well over one thousand euros, not including the signal conditioning electronics, cabling, and mechanical integration effort required for each additional sensor. In a production environment with dozens or hundreds of belt-driven axes, equipping every drive with dedicated vibration and torque sensors quickly becomes prohibitively expensive. By contrast, the encoder and current sensor used in the proposed approach are already present in standard servo motors and drives at no additional cost. Moreover, because the learned models remain consistent with the underlying physics, their outputs are inherently interpretable and can be traced back to physically meaningful quantities. These properties, minimal sensor requirements, physical interpretability, and robustness under variable-speed operation, make PINNs well suited for fault diagnosis in PMSM-driven belt transmission systems deployed in real industrial environments.

1.2 Proposed Methods and Contributions

This thesis presents two physics-informed fault diagnosis methods for PMSM-driven synchronous belt transmission systems, both operating exclusively on the electrical and rotational signals already available in the servo drive, namely the motor current and the encoder angle, without requiring any additional instrumentation.

Both methods follow a common three-stage structure: (i) physical modeling of the rotor dynamics, (ii) healthy system identification via PINNs, and (iii) fault diagnosis through sparse identification of fault-induced deviations from the healthy baseline. In the healthy system identification stage, PINNs are applied to solve an inverse problem: the neural network simultaneously approximates the measured system trajectory and identifies the unknown physical parameters of the rotor dynamics by minimizing a composite loss that combines data fidelity with the residual of the governing Ordinary Differential Equation (ODE). The identified healthy parameters then define a baseline model against which fault-induced deviations are detected in the subsequent stage.

The two methods differ fundamentally in the choice of state variable, network architecture, differentiation strategy, and fault identification procedure, leading to distinct trade-offs in noise robustness and diagnostic precision.

The first method, presented in Chapter 4, is formulated in terms of the angular velocity. The neural network directly predicts the speed signal as supplied by the servo drive, which is internally computed from the encoder readings via numerical differentiation. Once the healthy parameters are identified, a fault residual is computed offline by subtracting the healthy model

prediction from the measured angular acceleration. A physics-motivated candidate fault library is assembled, comprising friction-related terms, polynomial speed-dependent terms, and Fourier harmonics at integer and fractional multiples of the shaft angle to account for the transmission ratio. The Sequentially Thresholded Least Squares (STLSQ) algorithm is then applied to this library as a standalone post-processing step, yielding a sparse and interpretable set of fault terms.

The second method, presented in Chapter 5, reformulates the dynamics directly in terms of the angular position. Rather than predicting the noisy speed signal, the network outputs continuous trigonometric projections of the shaft angle, from which the velocity and acceleration are recovered analytically via automatic differentiation, entirely bypassing the noise-prone numerical differentiation chain of the first method. A Sinusoidal Representation Network (SIREN) is adopted to faithfully represent the inherently oscillatory nature of these signals. The fault identification stage departs from the two-step design of the first method: rather than applying sparse regression as a separate post-processing step, candidate fault terms with trainable amplitudes and phases are embedded directly into the physics residual of the PINN. Sparsity is enforced through an alternating ℓ_0/ℓ_1 scheme within the gradient-based optimization loop, resulting in an end-to-end fault identification pipeline without any separate post-processing step.

The methods are evaluated on an existing B&R PMSM servo test bench transmitting motion to an inertial load through an Optibelt toothed synchronous belt. The experimental protocol, including the fault configurations and motion profiles, was designed and carried out as part of this work. Two fault scenarios are studied: parallel misalignment and tooth interference (debris entrapment). Experiments are conducted under both variable-speed and constant-speed motion profiles across multiple speed and acceleration configurations, covering the range of operating conditions encountered in practice.

The main contributions of this thesis are as follows:

- **Minimal-sensor fault diagnosis using only servo drive signals.** Both methods rely exclusively on two signals that are universally available in any standard PMSM servo installation, namely the motor control current and the encoder angle, with no accelerometers, torque transducers, acoustic sensors, or any other dedicated instrumentation. This makes the proposed framework directly deployable on existing industrial drives without hardware modification or additional cost.
- **Physics-informed healthy system identification from minimal measurements.** A PINN-based inverse problem formulation is developed that identifies the physical parameters of a PMSM-driven belt transmission system directly from the two available servo drive signals, without requiring dedicated identification experiments or additional sensors.
- **Rotation-speed-based interpretable fault diagnosis.** A two-step diagnostic framework is proposed in which PINN-identified healthy parameters are used to isolate a fault residual, and STLSQ sparse regression over a physics-motivated candidate library identifies the fault type and produces interpretable, coefficient-level estimates of its magnitude that

can be related to fault severity.

- **Rotation-angle-based interpretable fault diagnosis with end-to-end sparse identification.** A SIREN-based PINN formulation is developed that operates directly on the raw encoder angle, eliminates all numerical differentiation through analytic derivative reconstruction, and embeds sparse identification of fault terms directly into the PINN training loop, without any separate post-processing step.
- **Experimental design and validation under realistic industrial conditions.** The fault configurations and motion profiles are designed and executed on the available test bench, covering parallel misalignment and tooth interference faults across variable-speed and constant-speed operating profiles. The experimental evaluation demonstrates that the angle-based method yields more accurate and robust fault identification by virtue of its tighter coupling between physics-informed learning and sparse identification.

1.3 Structure

The remainder of this thesis is organized as follows.

Chapter 2 reviews existing fault diagnosis approaches for rotating machinery and transmission systems, examining signal-based, model-based, and data-driven methods with respect to their sensor requirements, interpretability, and robustness under variable-speed operation. Physics-informed machine learning approaches are then discussed, leading to the identification of the specific research gap addressed by this work.

Chapter 3 establishes the theoretical foundations required for the subsequent methodology chapters. The rotational dynamics of the PMSM-driven belt transmission are formulated as both a first-order ODE in angular velocity and a second-order ODE in angular position, providing the alternative formulations underlying the two proposed methods. A trigonometric representation of the encoder angle is introduced to handle the 2π-wrapping discontinuity, along with analytic expressions for reconstructing the angular velocity and acceleration. The chapter further introduces the PINN framework for inverse problems, the physics-aware normalization strategy, the SINDy and STLSQ sparse regression methodology, and the SIREN architecture.

Chapter 4 presents the first proposed fault diagnosis method. The rotation speed is selected as the network output, and a three-stage workflow is developed comprising physical modeling, PINN-based healthy system identification with a two-phase training strategy, and STLSQ sparse regression applied to a fault residual computed from the identified healthy baseline. A physics-motivated candidate fault library is designed to capture typical transmission fault mechanisms.

Chapter 5 develops the second proposed method, which builds upon and extends the first. Motivated by the limitations of the speed-based formulation identified during its implementation, in particular the noise amplification inherent in numerical differentiation and the separation

between physics-informed learning and sparse identification, the second method reformulates the dynamics in terms of the angular position. A SIREN-based PINN predicts the continuous trigonometric projections of the encoder angle, and all kinematic quantities are recovered through automatic differentiation. The fault diagnosis stage embeds the candidate fault terms directly into the augmented physics residual, with an alternating ℓ_0/ℓ_1 scheme enforcing sparsity within the PINN training loop.

Chapter 6 describes the hardware platform and experimental protocol. The B&R PMSM servo test bench, equipped with a synchronous belt transmission at a 1:3 pulley ratio, provides the motor current and encoder angle as the only measurement signals. The experimental protocol, designed and executed as part of this thesis, covers three system conditions across multiple motion profile configurations spanning both variable-speed and constant-speed operation.

Chapter 7 reports the implementation details, experimental results, and comparative analysis of both methods, followed by a synthesis that discusses practical insights gained during implementation and addresses the current limitations of the work.

Chapter 8 concludes the thesis by summarizing the main contributions and outlining directions for future work.

2 State of the Art

This chapter reviews the existing literature on fault diagnosis methods for rotating machinery and transmission systems. Section 2.1 covers traditional signal-based approaches. Section 2.2 discusses model-based approaches. Section 2.3 reviews data-driven methods and their limitations. Section 2.4 presents PIML approaches, including PINNs and sparse identification methods. Section 2.5 identifies the research gap addressed by this work.

Fault diagnosis in transmission systems can be broadly categorized into signal-based, model-based, and data-driven approaches [10, 16]. It is important to note that these categories are not independent: model-based and data-driven methods both rely on measured signals as their input, and therefore inherit the capabilities and limitations of the underlying sensing infrastructure. What distinguishes the three categories is not the source of information, but rather how that information is processed and interpreted. A closely related and practically significant difference lies in the sensor requirements. Signal-based methods can often operate with a single sensor type, such as a vibration accelerometer or a current probe, while model-based approaches typically require multiple synchronized measurements, for example simultaneous torque, position, and current signals, to constrain the physical model and isolate fault-related parameter changes. Data-driven methods are the most flexible in terms of input modality, but their diagnostic performance generally scales with the number and diversity of sensor channels available for training.

2.1 Signal-Based Approaches

Signal-based methods utilize signal processing techniques such as empirical mode decomposition [11] and spectral analysis to extract features indicative of potential faults [8]. While these methods are widely adopted in industry due to their relatively straightforward implementation and modest sensor requirements, their diagnostic precision depends strongly on expert experience in selecting appropriate signal features and interpreting the results. Empirical mode decomposition, for instance, is effective at separating nonstationary signals into intrinsic mode functions, making it suitable for extracting transient fault-related components from vibration data. However, the method suffers from mode mixing, where signals of similar frequency content are incorrectly grouped into the same mode, which can obscure subtle fault signatures in the presence of broadband noise. Spectral analysis, on the other hand, provides a clear frequency-domain representation of periodic fault patterns and is well suited for identifying characteristic fault frequencies such as gear mesh frequencies or bearing defect frequencies under constant-speed conditions. Its main limitation is the assumption of stationarity: when the motor operates under variable-speed profiles involving acceleration and deceleration phases, the fault-characteristic frequencies shift over time. Classical frequency-domain techniques such as envelope analysis and order tracking attempt to address this through resampling in the angular domain, but their effectiveness degrades when speed variations are rapid or when the encoder

resolution is insufficient to support accurate angular resampling.

2.2 Model-Based Approaches

Model-based approaches leverage physical understanding to identify mechanical faults [5, 24]. By constructing analytical or numerical models of the system, deviations between the model prediction and the measured behavior can be used as fault indicators. A key strength of model-based methods is that the diagnostic output is directly linked to physical parameters, such as stiffness, damping, or mass imbalance, which provides inherent interpretability and allows engineers to reason about fault severity and root cause. However, this interpretability comes at the cost of increased sensor demands: to reliably estimate physical parameters through inverse identification, model-based methods typically require multiple synchronized measurement channels, such as simultaneous acquisition of motor current, shaft position, and torque or vibration signals. Each additional sensor adds hardware cost, installation complexity, and potential points of failure. Beyond sensor requirements, these methods also demand detailed prior knowledge of the system's geometry, material properties, and boundary conditions, which may not always be available or may change over time due to wear and aging. Furthermore, they struggle with persistent nonlinearities and unmodeled dynamics, such as backlash, friction, or contact phenomena in belt-pulley meshing, and often require advanced, system-specific noise reduction techniques to achieve reliable performance. As a result, model-based approaches tend to work well for systems with well-characterized dynamics and controlled operating conditions, but their accuracy and generalizability degrade in complex industrial transmission systems where the modeling assumptions are only approximately satisfied.

2.3 Data-Driven Approaches

In recent years, data-driven approaches have attracted significant attention in the field of machinery fault diagnosis [14, 15]. Like model-based methods, data-driven approaches fundamentally depend on the quality and richness of the measured signals fed into them. However, their sensor requirements differ in character: rather than needing specific physical quantities to constrain a model, data-driven methods benefit from a high volume and diversity of sensor channels, as additional input features generally improve the discriminative power of the learned representations. In practice, state-of-the-art deep learning frameworks for fault diagnosis often fuse data from vibration accelerometers, acoustic emission sensors, temperature probes, and current measurements simultaneously, which yields strong diagnostic performance but creates a significant gap between laboratory setups and standard industrial installations where only a limited set of sensors is available.

Developments in deep learning, such as transfer learning [39] and representation learning [32], have demonstrated strength in complex rotating systems, particularly when dealing with strong nonlinearities and large volumes of operational data. Transfer learning is particularly valuable

when labeled fault data are scarce for a specific machine, as it allows a model trained on one system or operating condition to be adapted to another with limited retraining. This makes it a practical option for industrial scenarios where each machine may have slightly different characteristics. Representation learning, meanwhile, excels at automatically discovering compact and discriminative feature representations from raw sensor signals, reducing the reliance on hand-crafted feature engineering and enabling end-to-end diagnostic pipelines.

Despite these advantages, purely data-driven methods face two fundamental limitations. First, they lack physical interpretability, operating as black-box models that provide diagnostic outputs without revealing the underlying fault mechanisms. This is a critical drawback for mission-critical infrastructure where understanding why a fault is diagnosed is as important as the diagnosis itself. For example, a deep neural network may correctly classify a vibration pattern as indicating misalignment, but it cannot explain whether the misalignment is parallel or angular, nor can it quantify the physical severity in terms of offset distance or angular deviation. This absence of mechanistic insight limits the ability of maintenance engineers to make informed decisions about corrective actions. Second, these methods typically require large amounts of labeled faulty data for training, which is often infeasible in practice since faulty operating conditions are rare, costly to reproduce, and potentially dangerous to induce deliberately. While techniques such as data augmentation and generative adversarial networks have been proposed to mitigate data scarcity, they introduce synthetic samples whose fidelity to real fault behavior is difficult to guarantee, potentially leading to models that perform well on augmented datasets but fail under actual fault conditions.

2.4 Physics-Informed Machine Learning

To overcome the interpretability gap of purely data-driven methods, PIML provides a promising paradigm for integrating physical knowledge into learning architectures. [26] introduced PINNs, which embed governing differential equations into the training loss function, enabling both forward simulation and inverse parameter identification. PINNs have since been applied across a wide range of scientific and engineering domains. A central advantage of PINNs is their ability to enforce physical consistency even in data-sparse regimes: because the governing equations act as a regularizer during training, the network is constrained to solutions that are physically plausible, which reduces overfitting and improves generalization compared to purely data-driven alternatives. However, PINNs are not without limitations. Training can be sensitive to the balance between data-fitting and physics-residual loss terms, and convergence difficulties have been reported for stiff or multiscale dynamical systems where the governing equations span widely different time scales.

In the context of rotating systems, PINNs have been applied to condition monitoring of rotating shafts and shown to achieve improved robustness and better physical consistency compared to purely data-driven alternatives [22]. This work demonstrated that embedding the shaft dynamics into the loss function enables accurate parameter estimation even from noisy vibration

measurements, which is a significant advantage over classical system identification methods that require careful preprocessing and noise filtering. Furthermore, PINN-based frameworks have demonstrated the capability to accurately quantify the severity of demagnetization faults in PMSM drive systems [21]. Their approach showed that PINNs can distinguish between different levels of magnet degradation by identifying changes in the electromagnetic model parameters, providing not only fault detection but also severity estimation grounded in the motor's physical model. However, both of these approaches relied on direct measurements of the quantity of interest, namely vibration data from accelerometers or electromagnetic signals from dedicated sensor arrays, which limits their applicability to systems where only standard servo drive instrumentation is available.

As a complementary approach, Sparse Identification of Nonlinear Dynamics (SINDy) discovers governing equations directly from data by assuming that system dynamics can be expressed as sparse combinations of physically meaningful basis functions [6]. The strength of SINDy lies in its transparency: the identified model is an explicit mathematical expression whose terms can be directly interpreted as physical forces, damping, or coupling effects. This makes SINDy particularly attractive for fault diagnosis, where the appearance of a new term or a significant change in a coefficient can be directly associated with a specific fault mechanism. However, SINDy's performance depends on the quality of the candidate library and on accurate numerical differentiation of measured signals, which can amplify noise and lead to spurious terms in the identified model. Other symbolic regression methods, such as AI Feynman [31, 33], also extract concise mathematical expressions to explain system behavior. While AI Feynman can discover more complex functional relationships than the linear-in-parameters formulation of SINDy, it is computationally more expensive and less straightforward to constrain toward physically motivated candidate expressions.

Of particular relevance to this work, [17, 18] proposed a hierarchical physics-informed neural network for rotor system health assessment, demonstrating the feasibility of combining ODE identification from noisy measurements with interpretable fault diagnosis through sparse regression. Their approach showed that the combination of PINNs for baseline parameter identification and SINDy-inspired sparse regression for fault term discovery can yield transparent and physically meaningful diagnostic results. This two-step framework is conceptually compelling because it separates the task of learning the healthy system dynamics from the task of identifying fault-induced deviations, which simplifies both the training process and the interpretation of results. However, their implementation relied on vibration measurements from accelerometers mounted directly on the rotor housing, and the fault scenarios considered were limited to rotor-specific faults such as mass imbalance and rub-impact. The transferability of this framework to belt transmission systems with different fault modes and sensor constraints has not been investigated.

2.5 Research Gap

Despite the advances reviewed above, several challenges remain for fault diagnosis in PMSM-driven transmission systems. Many existing diagnostic approaches rely on multiple dedicated sensors, such as vibration transducers, temperature probes, and acoustic sensors, to capture fault-related signatures. In standard PMSM configurations, however, typically only electrical measurements (e.g., motor current) and encoder signals for position and speed feedback are available, especially in cost-sensitive industrial applications. Consequently, transmission faults such as misalignment, tooth interference, or bearing defects must be diagnosed indirectly from electrical and kinematic signals, which contain fault information only through electromechanical coupling. This indirect observation makes fault detection more challenging, as mechanical fault signatures are attenuated and mixed with electrical noise and motor dynamics.

Furthermore, while the hierarchical PINN framework proposed by [17, 18] demonstrated promising results, their approach relied on direct vibration measurements from accelerometers, which are not available in standard PMSM transmission configurations. In addition, their method applies SINDy as a separate post-processing step after the PINNs have been trained, meaning the sparse identification operates on the PINNs' output rather than being jointly optimized with them. This sequential separation may limit the method's ability to discover subtle fault terms when the signal-to-noise ratio is low or when the PINNs' approximation introduces residual errors that propagate into the sparse regression.

The present work addresses these gaps by extending the physics-informed fault diagnosis framework to PMSM-driven synchronous belt transmission systems using solely the electrical and rotational signals already available in the servo drive. Two methods are developed and compared. The first, presented in Chapter 4, follows a two-step procedure in which PINNs identify the healthy system and a standalone SINDy-based sparse regression is subsequently applied to extract fault-related terms. The second, presented in Chapter 5, embeds the sparse identification directly into the PINNs training framework, allowing the neural network and the sparse fault coefficients to be jointly optimized. As will be shown in the experimental evaluation, the second method yields more accurate and robust fault identification, demonstrating that tightly coupling physics-informed learning with sparse identification outperforms the separated two-step approach.

3 Preliminaries

This chapter provides the theoretical background necessary for the fault diagnosis framework developed in later chapters. The central object of study is a Permanent Magnet Synchronous Motor (PMSM) driving a mechanical load through a synchronous belt transmission. To diagnose faults in this system in a physically interpretable manner, one must first understand the governing dynamics of the rotating shaft, know how to embed these dynamics into a data-driven learning process, and have a principled way to extract sparse, human-readable fault signatures from the learned model. Accordingly, this chapter begins by deriving the equations of motion for the motor-side rotational dynamics, formulated both as a first-order equation in angular velocity and as a second-order equation in angular position. These equations capture the interplay between the driving torque, inertia, viscous friction, and Coulomb friction that governs the shaft motion under healthy conditions. The chapter then introduces PINNs, which allow these mechanical equations to be enforced as soft constraints during training so that unknown physical parameters such as inertia and friction coefficients can be identified directly from measured data. Finally, the Sparse Identification of Nonlinear Dynamics (SINDy) framework is reviewed, which provides a systematic approach to selecting a minimal set of physically meaningful terms from a predefined candidate library, enabling deviations from the healthy baseline to be attributed to specific fault mechanisms.

The test bench constitutes a PMSM driving a mechanical load through a synchronous belt transmission. A prototype photograph of the test bench is presented in Figure 3–1, and detailed descriptions are given in Section 6.1.

For the purpose of dynamic modeling, the transmission system is abstracted as shown in Figure 3–2. The motor applies a driving torque T_m to the small pulley with radius r_m, which transmits motion to the large pulley with radius r_L through a toothed synchronous belt. The belt elasticity is represented by an equivalent stiffness k. The load-side inertia J_L is concentrated at the large pulley. The motor-side angular position and velocity are denoted by θ_m and ω_m, and the corresponding load-side quantities by θ_L and ω_L.

In such a system, the rotational dynamics are governed by the interplay between the driving torque produced by the motor and the resistive forces acting on the rotating shaft, including viscous friction and Coulomb friction. Due to sensor limitations, only the motor-side shaft dynamics are accessible for measurement. The load-side speed is not directly measured. In principle, the motor-side and load-side dynamics are each described by their own angular position and velocity, as indicated in Figure 3–2. However, since no encoder or speed sensor is installed on the load side, only the motor-side quantities are accessible for measurement. In the remainder of this work, the motor-side angular position and velocity are therefore denoted simply as θ and ω, dropping the subscript m for notational convenience.



Figure 3–1: Prototype photograph of the test bench

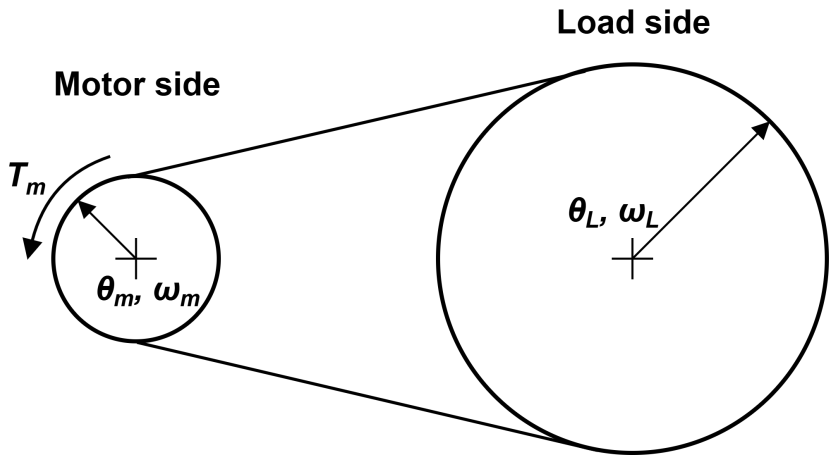


Figure 3–2: Simplified dynamic model of the PMSM-driven synchronous belt transmission system

3.1 Rotation-Speed-Based Rotor System Modeling

In the experimental setup, only the motor-side motion is directly measured, since no encoder or speed sensor is installed on the load side. Therefore, a control-oriented motor-side model is adopted. Assuming that the synchronous belt transmission is sufficiently rigid and that belt compliance, backlash, and slip are negligible, the transmission and load can be represented by equivalent inertia and damping referred to the motor shaft, as commonly done in rigid-model approximations of timing-belt-driven systems [9]. As a result, the overall mechanical subsystem is described by the following ODE:

$$J\dot{\omega}(t) + B\omega(t) + T_c \operatorname{sgn}(\omega(t)) = T_m(t) \ , \tag{3–1}$$

where $\omega(t)$ (rad/s) is the angular velocity of the motor shaft, $\dot{\omega}(t)$ (rad/s^2) is the angular acceleration, and $T_m(t)$ (N $\cdot$ m) is the motor torque, which is proportional to the motor current and serves as the sole control input to the system. The physical parameters are: J (kg $\cdot$ m^2), the total moment of inertia reflected to the motor shaft, so that $J\dot{\omega}(t)$ represents the inertial torque resisting changes in rotational speed; B (N $\cdot$ m $\cdot$ s/rad), the viscous friction coefficient, so that $B\omega(t)$ is the viscous friction torque proportional to speed; and T_c (N $\cdot$ m), the Coulomb friction torque, where $\operatorname{sgn}(\cdot)$ is the signum function capturing the direction-dependent nature of dry friction and $T_c \operatorname{sgn}(\omega(t))$ represents the constant-magnitude friction torque opposing motion.

3.2 Rotation-Angle-Based Rotor System Modeling

3.2.1 Second-Order Angular Position Formulation

The system dynamics can be expressed in terms of the angular position $\theta(t)$ (rad) of the motor shaft. Recognizing that the angular velocity and acceleration are related to the angular position by $\omega(t) = \dot{\theta}(t)$ and $\dot{\omega}(t) = \ddot{\theta}(t)$, the equation of motion becomes a second-order ordinary differential equation in θ:

$$J\ddot{\theta}(t) + B\dot{\theta}(t) + T_c \operatorname{sgn}(\dot{\theta}(t)) = T_m(t) \ , \tag{3–2}$$

where $\dot{\theta}(t)$ (rad/s) is the angular velocity and $\ddot{\theta}(t)$ (rad/s^2) is the angular acceleration. The motor torque $T_m(t)$ (N $\cdot$ m) is proportional to the motor current and serves as the sole control input to the system. The physical parameters are: J (kg $\cdot$ m^2), the total moment of inertia reflected to the motor shaft, so that $J\ddot{\theta}(t)$ represents the inertial torque resisting changes in rotational speed; B (N $\cdot$ m $\cdot$ s/rad), the viscous friction coefficient, so that $B\dot{\theta}(t)$ is the viscous friction torque proportional to speed; and T_c (N $\cdot$ m), the Coulomb friction torque, where $\operatorname{sgn}(\cdot)$ is the signum function and $T_c \operatorname{sgn}(\dot{\theta}(t))$ represents the constant-magnitude friction torque opposing motion.

This formulation is physically equivalent to **(3–1)** but expressed in terms of the angular position and its derivatives. Beyond being naturally suited for position-tracking control objectives, the angle-based formulation offers two practical advantages.

First, the angular position $\theta(t)$ is the raw output of the rotary encoder and is therefore available at full sensor resolution, whereas the angular velocity $\omega(t)$ provided by the servo drive is internally computed from the encoder signal using numerical differentiation, which amplifies high-frequency noise and quantization artifacts. By formulating the dynamics directly in terms of $\theta(t)$ and delegating the differentiation to automatic differentiation within the neural network, the proposed method bypasses this noise amplification and operates on the highest-quality signal available in the system.

Second, the angular position provides a natural basis for capturing periodic fault signatures. Many transmission faults, such as tooth interference and misalignment, produce torque disturbances that repeat at specific multiples of the shaft rotation angle. Formulating the dynamics in terms of θ allows these angle-periodic patterns to be directly represented by trigonometric candidate functions in the fault library, which would be less straightforward in a speed-based formulation.

3.2.2 Trigonometric Angle Representation and Analytic Differentiation

The angle-based formulation established in **(3–2)** requires the angular velocity $\dot{\theta}(t)$ and the angular acceleration $\ddot{\theta}(t)$ as inputs to the physics residual. Although the angular position $\theta(t)$ itself does not appear explicitly in **(3–2)**, both derivatives must be obtained from $\theta(t)$, since the encoder provides the angular position as the only kinematic measurement. However, using the raw encoder angle directly as the neural network output is problematic due to its discontinuous nature. The rotary encoder on the test bench outputs the absolute shaft angle $\theta(t)$, which is inherently periodic: it wraps from 2π back to 0 (or from π to $-\pi$) at every full revolution, introducing discontinuities that corrupt any finite-difference estimate of the angular velocity and acceleration. Moreover, this periodicity poses difficulties for the neural network models developed in later sections, whose inputs and outputs shall be continuous. This subsection introduces a trigonometric representation that resolves these discontinuities and enables analytic differentiation of the angular position.

To address both issues, we represent the angular position by its trigonometric projections:

$$c(t) = \cos\theta(t)\ , \qquad s(t) = \sin\theta(t)\ , \tag{3–3}$$

which are continuous and bounded regardless of how many revolutions the shaft completes, and satisfy the fundamental identity:

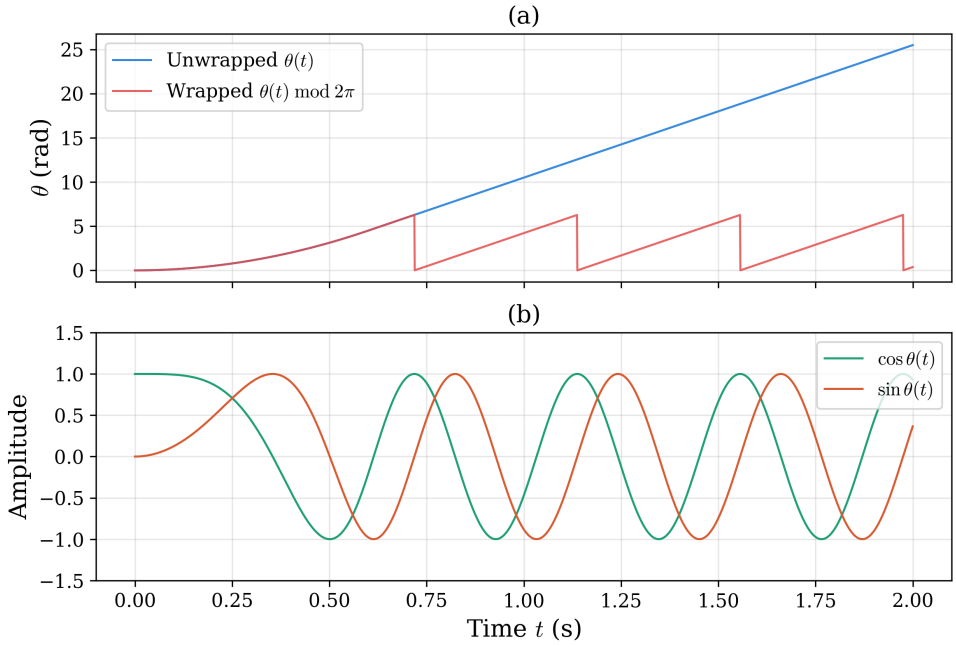


Figure 3–3: (a) Raw encoder angle with 2π-wrapping discontinuities. (b) Continuous trigonometric projections $\cos\theta(t)$ and $\sin\theta(t)$.

$$c^2 + s^2 = 1\ . \tag{3–4}$$

This $(\cos, \sin)$ representation eliminates the 2π-discontinuity at the signal level and provides a natural two-dimensional encoding of the cyclic angle that is well suited as input and output for neural network architectures.

The rotation angle can be recovered at any time via the four-quadrant arctangent function $\theta = \operatorname{atan2}(s,\, c)$. Unlike $\arctan(s/c)$, $\arcsin$, or $\arccos$, it uniquely resolves the angle over the full interval $(-\pi,\, \pi]$ without singularities.

Since $c(t)$ and $s(t)$ are smooth, the angular velocity and acceleration can be obtained analytically from the derivatives of these trigonometric signals. Starting from the relation $\theta = \operatorname{atan2}(s,\, c)$ and applying the chain rule, the angular velocity can be expressed in terms of c, s and their time derivatives $\dot{c}$, $\dot{s}$. A key advantage of this approach is that the discontinuous angle signal $\theta(t)$ never needs to be computed or handled explicitly; all necessary kinematic quantities are derived entirely from the continuous signals $c(t)$ and $s(t)$. The general form:

$$\dot{\theta} = \frac{c\,\dot{s} - s\,\dot{c}}{c^2 + s^2}\ , \tag{3–5}$$

which, by **(3–4)**, simplifies to:

$$\dot{\theta} = c\,\dot{s} - s\,\dot{c}\,. \tag{3–6}$$

Differentiating **(3–5)** with respect to t using the quotient rule yields the general form of the angular acceleration:

$$\ddot{\theta} = \frac{c\,\ddot{s} - s\,\ddot{c}}{c^2 + s^2} - 2\,\frac{(c\,\dot{s} - s\,\dot{c})(c\,\dot{c} + s\,\dot{s})}{(c^2 + s^2)^2}\,, \tag{3–7}$$

where $\ddot{c}$ and $\ddot{s}$ are the second time derivatives of the trigonometric signals. The first term captures the contribution from the acceleration of the raw signals, while the second term is a cross-coupling correction arising from the nonlinear mapping. Using the definitions $\dot{c} = -s\,\dot{\theta}$ and $\dot{s} = c\,\dot{\theta}$, which imply $c\,\dot{c} + s\,\dot{s} = c(-s\,\dot{\theta}) + s(c\,\dot{\theta}) = 0$, together with the unit-circle identity **(3–4)**, the second term vanishes and the expression reduces to:

$$\ddot{\theta} = c\,\ddot{s} - s\,\ddot{c}\,. \tag{3–8}$$

The simplified forms **(3–6)** and **(3–8)** rely on **(3–4)** holding exactly, which is guaranteed when c and s are computed from an exact angle via **(3–3)**. When c and s are instead produced by a neural network, this is not automatically satisfied. It is therefore enforced explicitly as a constraint during network training, ensuring both the validity of the analytic derivatives and the physical consistency of the predicted angle.

3.3 Physics-informed Neural Networks

PINNs integrate known physical laws, such as ordinary or partial differential equations, into the neural network loss function. By penalizing violations of the governing equations during training, PINNs enforce physical consistency even when measurement data are sparse or noisy. In this work, the mechanical model established in this section both captures the dominant dynamics of the test bench and provides the physics-based constraints for the PINN framework developed in later sections.

3.3.1 Physics-Informed Neural Network for Forward Problem

The standard PINNs framework addresses forward problems, where the governing equations and their parameters are fully known and the objective is to approximate the solution of the differential equation.

In a forward problem, the physical parameters β in the governing ODE are given, and the goal is to find a solution $y(t)$ that satisfies both the differential equation and the prescribed initial or

boundary conditions. The governing ODE is written in residual form as

$$\mathcal{F}[y(t),\, u(t);\, \boldsymbol{\beta}] = 0, \quad t \in [0, T]\,, \tag{3–9}$$

where $y(t)$ is the system response, $u(t)$ is a known external input (forcing term), $\boldsymbol{\beta}$ is the vector of physical parameters, and $\mathcal{F}$ is the differential operator defined by the governing equation. In this notation, the quantities before the semicolon ($y(t)$ and $u(t)$) are functions of time on which $\mathcal{F}$ acts, while the quantities after the semicolon ($\boldsymbol{\beta}$) are fixed parameters. As a concrete example, for a linear second-order system with mass m, damping c, and stiffness k driven by an external force $u(t)$, the governing ODE $m\ddot{y} + c\dot{y} + ky = u(t)$ is expressed in residual form as

$$\mathcal{F}[y(t),\, u(t);\, \boldsymbol{\beta}] = m\ddot{y}(t) + c\dot{y}(t) + ky(t) - u(t)\,, \tag{3–10}$$

with parameter vector $\boldsymbol{\beta} = (m,\, c,\, k)$. A neural network $y_{\text{PINN}}(t)$, parameterized by weights and biases $\boldsymbol{\phi}$, is trained to approximate the solution of **(3–9)**.

The time derivatives required by the governing equation, such as $\mathrm{d}y/\mathrm{d}t$ and $\mathrm{d}^2y/\mathrm{d}t^2$, are computed via Automatic Differentiation (AD) through the network's computational graph. AD is a technique for computing derivatives of functions defined by computer programs by systematically applying the chain rule to each elementary operation recorded in the computational graph. In contrast to symbolic differentiation, which manipulates mathematical expressions to produce a closed-form derivative and can suffer from expression swell, and to numerical differentiation, which approximates derivatives via finite differences and introduces truncation and round-off errors, AD evaluates derivatives to machine precision by tracing through the exact sequence of operations performed during the forward pass.

In this work, the PINNs are implemented in the DeepXDE library [19] using its PyTorch backend, so that whenever a time derivative appears in the physics residual, it is obtained by differentiating the network output with respect to the network input through the underlying AD engine, without any finite-difference approximation. An important clarification is that the exactness of AD refers to the exact application of the chain rule to the function implemented by the computational graph, not to exact recovery of the true physical derivative of the underlying system. If the network provides a faithful approximation of the measured trajectory, its AD derivative closely approximates the physical derivative of the measured signal; if the network fit is poor, over-smoothed, biased towards low frequencies due to spectral bias, or otherwise inconsistent with the true dynamics, its AD derivative will reflect these shortcomings. AD does not denoise measurement data or recover true derivatives from an imperfect fit; it returns the exact derivative of whatever function the network has learned to represent [7]. This observation motivates the two-phase training strategy adopted later in Section 4.2 and Section 5.2, in which the net-

work is first pre-trained on the data loss alone so that its output closely matches the measured trajectory before the physics residual, which depends on the AD derivatives, is activated.

As illustrated in Figure 3–4, the training loss consists of a measurement loss $\mathcal{L}_{\text{data}}$ that penalizes the mismatch between the network output and available observations, and an ODE residual loss $\mathcal{L}_{\text{physics}}$ that enforces the governing equation at collocation points distributed over the temporal domain. The data loss is defined as the mean squared error between the network output and the measured response at N observation points:

$$\mathcal{L}_{\text{data}}(\phi) = \frac{1}{N}\sum_{i=1}^{N} |y_{\text{PINN}}(t_i) - y_i|^2 \ , \tag{3–11}$$

where t_i denotes the time instant at which the system is measured and y_i is the corresponding measured response. The initial condition is implicitly enforced through the first data point (t_1, y_1). The physics loss enforces the governing ODE at M collocation points within the temporal domain $[0, T]$:

$$\mathcal{L}_{\text{physics}}(\phi; \beta) = \frac{1}{M}\sum_{j=1}^{M} |\mathcal{F}[y_{\text{PINN}}(t_j); \beta]|^2 \ . \tag{3–12}$$

These collocation points are time instants at which the ODE residual is evaluated, serving to enforce physical consistency across the entire temporal domain, including regions where no measurement data are available. They can be chosen as equally spaced grid points over $[0, T]$ or sampled randomly; they are independent of the measurement locations and do not require corresponding measured data.

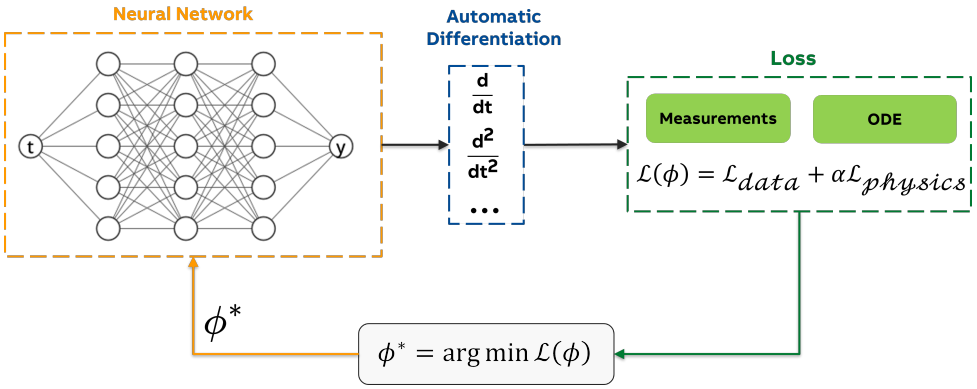


Figure 3–4: General workflow of PINNs for forward problems

Since the physical parameters β are known and fixed, only the network parameters ϕ are optimized by minimizing the weighted sum of the two loss terms:

$$\hat{\boldsymbol{\phi}} = \arg\min_{\boldsymbol{\phi}} \; \mathcal{L}_{\text{data}}(\boldsymbol{\phi}) + \alpha \, \mathcal{L}_{\text{physics}}(\boldsymbol{\phi}; \boldsymbol{\beta}) \; , \tag{3–13}$$

where α is a weighting coefficient that controls the relative importance of the two loss terms. The physics residual acts as a regularizer, constraining the learned solution to be consistent with the governing dynamics even in regions where measurement data are sparse or absent.

In the following subsection, the framework is extended to the inverse problem setting, where the physical parameters β are unknown and must be identified jointly with the network parameters from measured data.

3.3.2 Physics-Informed Neural Network for Inverse Problem

In the context of dynamical systems, an inverse problem refers to the task of inferring unknown model parameters from observed system responses, as opposed to the forward problem of predicting the response given known parameters [26]. PINNs are particularly suited for such inverse problems, as they incorporate the governing equations directly into the training loss, allowing the unknown physical parameters to be identified jointly with the network weights during optimization.

The structure of the problem is identical to the forward case introduced in Section 3.3.1: the governing ODE is still written in residual form as in **(3–9)**, a neural network $y_{\text{PINN}}(t)$ with parameters $\boldsymbol{\phi}$ still approximates the system response, and the data loss $\mathcal{L}_{\text{data}}(\boldsymbol{\phi})$ and physics loss $\mathcal{L}_{\text{physics}}(\boldsymbol{\phi}; \boldsymbol{\beta})$ retain the forms given in **(3–11)** and **(3–12)**, respectively. The derivatives appearing in $\mathcal{F}$ are again obtained via automatic differentiation through the network graph.

The key difference lies in the set of trainable variables. In the inverse setting, the physical parameters β are unknown and are treated as additional trainable variables alongside the network weights $\boldsymbol{\phi}$. Both are optimized jointly by minimizing the combined loss function:

$$\hat{\boldsymbol{\phi}}, \, \hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\phi}, \boldsymbol{\beta}} \; \mathcal{L}(\boldsymbol{\phi}, \boldsymbol{\beta}) = \mathcal{L}_{\text{data}}(\boldsymbol{\phi}) + \alpha \, \mathcal{L}_{\text{physics}}(\boldsymbol{\phi}, \boldsymbol{\beta}) \; , \tag{3–14}$$

where α is a weighting coefficient that controls the relative importance of the two loss terms [40]. Figure 3–5 illustrates the overall workflow.

Hence, by solving **(3–14)**, the PINN model simultaneously learns the system trajectory through $\boldsymbol{\phi}$ and identifies the physical parameters $\hat{\boldsymbol{\beta}}$. During training, the physics residual acts as a regularizer, guiding the network towards solutions that are consistent with the governing dynamics. In this work, the PINNs inverse problem framework is applied in two variants. In Chapter 4, the neural network directly outputs the angular velocity $\omega(t)$. In Chapter 5, the neural network outputs the trigonometric projections $\cos\theta(t)$ and $\sin\theta(t)$, from which the angular position and its derivatives are recovered analytically.

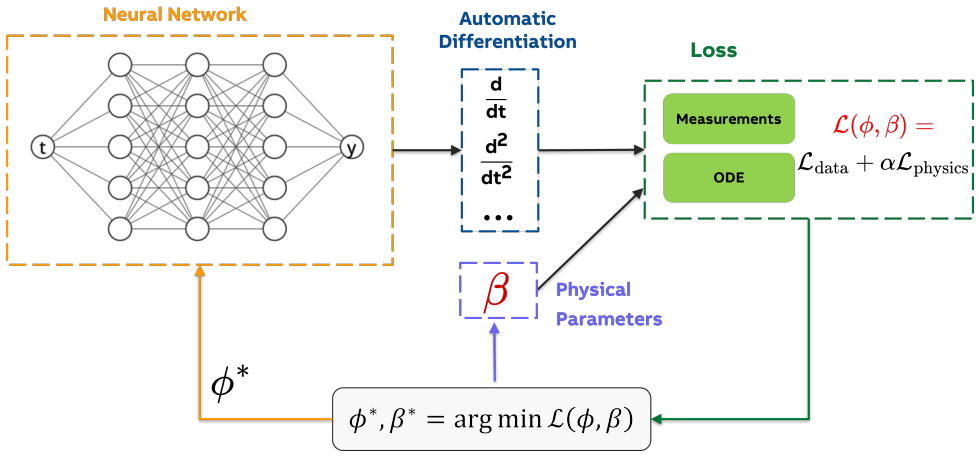


Figure 3–5: General workflow of PINNs for inverse problems

3.3.3 Normalization

Neural networks with bounded activation functions such as $\tanh$ or $\sin$ are most effective when their inputs and outputs lie within a moderate numerical range, typically of order unity. When raw physical signals with vastly different magnitudes and units are fed directly into the network, certain neurons may saturate in flat regions of the activation function, leading to vanishing gradients and slow or failed convergence. Similarly, if the terms in the physics residual span several orders of magnitude, the optimizer struggles to balance them, causing some physical parameters to dominate the loss landscape while others remain poorly identified. These numerical stability considerations motivate the normalization strategy adopted in this work, which draws on the data preprocessing framework proposed by Xu et al. [35] for physics-informed neural networks in fluid mechanics and adapts it to the PINN-based system identification of rotational dynamics. This normalization plays a central role in the healthy system identification stage, where accurate and balanced estimation of multiple physical parameters is required.

The system response $y(t)$ and the external input $u(t)$ are scaled by dividing each signal segment by its corresponding standard deviation:

$$y^*(t) = \frac{y(t)}{\sigma_y}\,, \qquad u^*(t) = \frac{u(t)}{\sigma_u}\,, \tag{3–15}$$

where σ_y and σ_u denote the standard deviations computed from each data segment. Division by the standard deviation is one of several common normalization strategies; alternatives include min-max scaling to $[0, 1]$ or full Z-score normalization that additionally subtracts the mean. In this work, only the standard deviation is used as the scaling factor and the mean is not subtracted, which avoids the need to track constant offset terms when converting the physical parameters back to their dimensional form. [35]

The time variable is treated differently from the physical signals: rather than scaling by its

standard deviation, it is linearly mapped to the unit interval:

$$t^* = \frac{t - t_0}{t_s} \in [0,\, 1]\ , \tag{3–16}$$

where t_0 is the start time of the segment, t_{end} is the end time, and $t_s = t_{\mathrm{end}} - t_0$ is the time span. The reason for this different treatment is that time is the sole input to the neural network and must be mapped to a fixed, well-defined domain to ensure consistent activation ranges across segments of different duration. Dividing by the standard deviation would preserve the segment's temporal distribution but not guarantee a bounded input range, whereas mapping to $[0, 1]$ ensures that the network input always starts at zero and ends at one regardless of the segment length. Since the neural network operates on the normalized time t^*, derivatives computed via automatic differentiation are taken with respect to t^* rather than the physical time t. By the chain rule, the relationship between derivatives in the two time coordinates is obtained as follows. Since $t^* = (t - t_0)/t_s$, one has $\mathrm{d}t^*/\mathrm{d}t = 1/t_s$. Applying the chain rule to any function f gives:

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\mathrm{d}f}{\mathrm{d}t^*} \cdot \frac{\mathrm{d}t^*}{\mathrm{d}t} = \frac{1}{t_s}\frac{\mathrm{d}f}{\mathrm{d}t^*}\ , \tag{3–17}$$

and applying it twice yields the second-order derivative:

$$\frac{\mathrm{d}}{\mathrm{d}t} = \frac{1}{t_s}\frac{\mathrm{d}}{\mathrm{d}t^*}\ , \qquad \frac{\mathrm{d}^2}{\mathrm{d}t^2} = \frac{1}{t_s^2}\frac{\mathrm{d}^2}{\mathrm{d}t^{*2}}\ . \tag{3–18}$$

The time span t_s therefore appears as an additional scaling factor in every time derivative within the physics residual and must be accounted for during both training and parameter conversion. The specific normalization applied to the neural network inputs and outputs differs between the two identification methods and is detailed in Chapter 4 and Chapter 5, respectively.

The fundamental principle of PINNs is to learn from both measured data and physical laws simultaneously. For this joint learning to succeed, two levels of balancing must be ensured:

- **Between losses:** The data loss and the physics residual loss must operate on comparable scales; otherwise, one loss component dominates the gradient updates and the other is effectively ignored.
- **Within the physics residual:** For the inverse problem of parameter identification, the individual terms within the physics residual must also be of comparable magnitude. When the physical parameters span very different orders of magnitude, the optimizer tends to prioritize terms with larger contributions, making it difficult to identify all parameters with equal reliability. [35]

The normalization strategy adopted in this work addresses both levels of balancing simultaneously, forming a physics-aware normalization that accounts not only for the signal ranges but also for the structure of the governing equation. By scaling the data so that all signals entering the network and the residual are of order unity, the data loss and physics loss naturally fall within a comparable range without manual weight adjustment. At the same time, because the scaling factors propagate through the ODE into the physical parameters, the scaled parameter counterparts are brought to similar magnitudes, eliminating the need for parameter-specific initialization. This leads to a more robust and generalizable training process, where all parameters can be identified reliably from a single, uniform setup.

3.4 Sparse Identification of Nonlinear Dynamics

Sparse identification of nonlinear dynamics (SINDy) is a data-driven method for discovering governing equations from measurement data by identifying sparse combinations of basis functions [6]. Unlike black-box models, SINDy produces explicit mathematical expressions that are physically interpretable, making it particularly suitable for fault diagnosis where understanding the fundamentals of faults is important.

SINDy assumes that most dynamical systems can be represented by few active terms from a larger library of candidate functions. For a general formulation of a dynamical model $\dot{y}(t) = f(y,t)$, the function f is considered to be sparse with respect to terms within a predefined library of basis functions $\boldsymbol{\Theta}(y,t)$. This library is typically constructed from polynomial terms or other physically meaningful functions, e.g.:

$$\boldsymbol{\Theta}(y,t) = [1,\ y,\ y^2,\ y^3,\ \ldots] \,. \tag{3–19}$$

The objective is to estimate a coefficient vector $\boldsymbol{\xi}$ that is sparse. In this context, sparsity means that most entries of $\boldsymbol{\xi}$ are exactly zero and only a small subset take non-zero values. Each entry of $\boldsymbol{\xi}$ corresponds to one candidate basis function in $\boldsymbol{\Theta}(y,t)$: a non-zero entry indicates that the associated term is active and contributes to the dynamics, while a zero entry removes the corresponding term from the model. A sparse $\boldsymbol{\xi}$ therefore selects a small number of active terms from the potentially much larger library and discards the rest, yielding a parsimonious representation in which each retained term can be directly associated with a specific physical effect. The degree of sparsity, that is, the number of non-zero entries relative to the library size, controls the trade-off between model complexity and fidelity to the data: too few active terms may fail to capture the dominant dynamics, while too many re-introduce redundancy and risk fitting measurement noise rather than genuine physical structure. With this notion of sparsity, the coefficient vector provides the approximation:

$$\dot{y} \approx \boldsymbol{\Theta}(y,t)\,\boldsymbol{\xi} \,. \tag{3–20}$$

The coefficient vector $\boldsymbol{\xi}$ is determined by solving a sparse regression problem that balances data fitting with a sparsity-promoting penalty:

$$\boldsymbol{\xi}^* = \arg\min_{\boldsymbol{\xi}} \|\dot{y} - \boldsymbol{\Theta}(y,t)\,\boldsymbol{\xi}\|_2^2 + \lambda\,\mathcal{P}(\boldsymbol{\xi})\ , \tag{3–21}$$

where λ is a regularization parameter controlling the trade-off between fitting accuracy and model complexity, and $\mathcal{P}(\boldsymbol{\xi})$ is a sparsity-promoting penalty. The entries of $\boldsymbol{\xi}$ that are non-zero are referred to as the *active terms*, since only they contribute to the identified model; the remaining entries are inactive and effectively removed from the library. Common choices for the penalty are:

$$\mathcal{P}(\boldsymbol{\xi}) = \begin{cases} \|\boldsymbol{\xi}\|_0 & (\ell_0\text{: number of non-zero coefficients}) \\ \|\boldsymbol{\xi}\|_1 & (\ell_1\text{: LASSO}) \\ \|\boldsymbol{\xi}\|_2^2 & (\ell_2\text{: ridge regression}) \end{cases} \tag{3–22}$$

To illustrate how these penalties act, consider a coefficient vector $\boldsymbol{\xi} = [0.8,\ 0,\ -0.3,\ 0,\ 0.1]^\top$. The ℓ_0 penalty counts the non-zero entries and yields $\|\boldsymbol{\xi}\|_0 = 3$, treating all three active terms equally regardless of their magnitude. The ℓ_1 penalty sums the absolute values and yields $\|\boldsymbol{\xi}\|_1 = 1.2$, penalizing larger coefficients more strongly and shrinking small coefficients towards zero. The ℓ_2 penalty sums the squared values and yields $\|\boldsymbol{\xi}\|_2^2 = 0.74$, which penalizes large coefficients even more aggressively than ℓ_1 but shrinks small coefficients only mildly, so that they typically remain non-zero rather than being eliminated.

The ℓ_0 penalty is the most natural choice as it directly enforces sparsity by counting the number of active terms. However, the ℓ_0 problem is combinatorial, meaning that finding the optimal solution requires enumerating subsets of the library and checking each one, and it is NP-hard, meaning that no algorithm is known that can solve it efficiently as the library size grows, making direct optimization intractable for libraries of practical size. The ℓ_1 penalty provides a convex surrogate that promotes sparsity through soft thresholding, and the ℓ_2 penalty regularizes the solution but does not enforce exact sparsity. These can also be combined as the elastic net penalty:

$$\mathcal{P}(\boldsymbol{\xi}) = \alpha\|\boldsymbol{\xi}\|_1 + (1-\alpha)\|\boldsymbol{\xi}\|_2^2\ . \tag{3–23}$$

An alternative approach, originally proposed alongside SINDy [6], is the Sequentially Thresholded Least Squares (STLSQ) algorithm, which approximates the ℓ_0 solution through an iterative procedure. The algorithm proceeds as follows: first, a standard least-squares solution is computed using all candidate functions in the library. Then, all coefficients whose absolute

values fall below a prescribed threshold λ are set to zero, effectively removing the corresponding terms from the active set. The least-squares problem is then solved again using only the remaining active terms, which adjusts the surviving coefficients to account for the removed terms. This alternation between least-squares fitting and hard thresholding is repeated until the set of active terms no longer changes between successive iterations. The key hyperparameter is the threshold λ: a small value retains more terms and risks overfitting, while a large value aggressively prunes the library and may discard physically relevant contributions. Compared to ℓ_1 relaxation, which shrinks all coefficients continuously towards zero and requires post-hoc truncation to achieve exact sparsity, STLSQ provides a more direct and interpretable control over the model complexity through a single threshold parameter and tends to produce cleaner selections of active terms, making it well suited for physics-based model discovery.

In the context of fault diagnosis, each non-zero coefficient in $\boldsymbol{\xi}$ signifies a specific physical effect. By comparing the terms identified under faulty conditions against the healthy baseline, the fault-related contributions can be explicitly extracted. Furthermore, the magnitude of the identified coefficients reflects the fault severity, and their presence or absence can be used for fault classification.

3.5 Sinusoidal Representation Networks

The rotation-angle-based method developed in Chapter 5 requires the neural network to output the trigonometric projections $\cos\theta(t)$ and $\sin\theta(t)$, which are inherently periodic and oscillatory signals. As the motor shaft rotates through multiple revolutions, these outputs exhibit rapid oscillations whose frequency increases with rotational speed. Representing such high-frequency periodic content is challenging for conventional neural networks: Rectified Linear Unit (ReLU)-based architectures produce piecewise linear mappings whose higher-order derivatives vanish almost everywhere, while smooth activations such as $\tanh$ suffer from spectral bias, the tendency to preferentially learn low-frequency components of a target signal [25]. This motivates the adoption of Sinusoidal Representation Network (SIREN), introduced by Sitzmann et al. [30], which employ periodic sine functions as activation functions throughout the network. By embedding periodicity directly into the network architecture, SIREN is naturally suited to represent the oscillatory encoder signals and their derivatives with high fidelity.

A SIREN consists of fully connected layers where each hidden layer applies a sine nonlinearity. Formally, given an input $\mathbf{x} \in \mathbb{R}^d$, the output of the i-th layer is defined as:

$$\mathbf{h}_i = \sin(\omega_0 \, \mathbf{W}_i \mathbf{h}_{i-1} + \mathbf{b}_i)\,, \tag{3–24}$$

where $\mathbf{h}_i$ denotes the hidden activation vector of the i-th layer, $\mathbf{W}_i$ and $\mathbf{b}_i$ are the weight matrix and bias vector of the i-th layer, respectively, and ω_0 is a hyperparameter that controls the frequency of the sinusoidal activations. For the first layer, $\mathbf{h}_0 = \mathbf{x}$. The final output layer may be

linear depending on the task.

A key property of SIREN is that the derivative of the network output with respect to its input is itself a SIREN. Since, for the i-th layer:

$$\frac{\partial}{\partial \mathbf{h}_{i-1}} \sin(\omega_0 \, \mathbf{W}_i \mathbf{h}_{i-1} + \mathbf{b}_i) = \omega_0 \, \mathbf{W}_i \cos(\omega_0 \, \mathbf{W}_i \mathbf{h}_{i-1} + \mathbf{b}_i) \,, \tag{3–25}$$

the derivatives remain smooth, well-defined, and representable within the same functional class. This property makes SIREN particularly well-suited for tasks that require supervision not only on the signal itself but also on its higher-order derivatives.

To ensure proper training convergence, Sitzmann et al. proposed a principled initialization scheme. The weights of the first layer are drawn uniformly as $W_1 \sim \mathcal{U}(-1/d,\, 1/d)$, where d is the input dimension, while subsequent layers are initialized as $W_i \sim \mathcal{U}\Big(-\sqrt{6/n}/\omega_0,\, \sqrt{6/n}/\omega_0\Big)$, with n being the number of input features to the respective layer. This initialization preserves the distribution of activations across layers during the forward pass.

Compared to conventional Feedforward Neural Network (FNN) employing non-periodic activations, SIREN offers distinct advantages for representing periodic or oscillatory signals. ReLU-based networks produce piecewise linear mappings whose derivatives are piecewise constant and whose higher-order derivatives vanish almost everywhere, making them unsuitable for tasks involving derivative supervision. Smooth activations such as sigmoid or tanh are differentiable to all orders but lack a periodic inductive bias, causing them to suffer from spectral bias—the tendency to preferentially learn low-frequency components of a target signal [25]. By embedding periodicity directly into its architecture, SIREN enables compact representations across a broad frequency spectrum while maintaining well-behaved derivatives throughout the network.

The adoption of SIREN in this work is motivated by the nature of the signals encountered in the modeling of the synchronous belt transmission system. As discussed in Section 3.2.2, the angular position of the motor is reconstructed from sine and cosine encoder signals via an arctangent operation, yielding an inherently periodic and continuous quantity. Consequently, both the angular position and the derived angular velocity exhibit strong periodic characteristics that are naturally aligned with the representational capacity of sinusoidal activation functions. A detailed treatment of how SIREN is integrated into the dynamics modeling framework is provided in Chapter 5.

4 Rotation-Speed-Based Interpretable Fault Diagnosis Method

This chapter presents the first of two fault diagnosis methods developed in this work. This method uses the angular velocity $\omega(t)$ as the neural network output. The speed signal is computed by the motor drive from the raw encoder readings and served directly as training data, with time as the neural network input.

The proposed fault diagnosis framework combines PINNs with sparse regression to achieve interpretable fault detection in a motor-driven synchronous belt transmission system. The framework follows a three-stage structure: (i) physical modeling of the rotor dynamics, (ii) identification of the healthy system using PINNs, and (iii) fault diagnosis through sparse regression on the residual dynamics.

The key motivation for this three-stage design is the separation of concerns between nominal system dynamics and fault-induced deviations. Under the assumption that faults manifest as relatively small perturbations on top of the dominant healthy dynamics, which is reasonable for incipient or moderate transmission faults such as those considered in this work, a purely data-driven approach applied directly to faulty measurements would conflate the dominant healthy behavior with the typically small fault signatures, making reliable fault identification difficult. By first establishing a physics-grounded model of the healthy system, the framework ensures that the subsequent fault analysis operates only on the residual dynamics that cannot be explained by the baseline model, which improves the sensitivity and interpretability of the fault detection (as demonstrated experimentally in Section 7.1.3). Furthermore, embedding the governing ODE into the identification process through PINNs constrains the solution to be physically consistent, reducing the risk of overfitting and enabling meaningful parameter estimates even from limited or noisy measurements.

The overall workflow is illustrated in Figure 4–1 and consists of three sequential stages: rotor dynamics system modeling, healthy system identification, and fault diagnosis. The main contributions of this chapter are: (i) a physics-informed inverse problem formulation for identifying the healthy system parameters J, B, and T_c directly from operational speed and torque measurements without dedicated identification experiments; (ii) a two-phase training strategy that stabilizes joint optimization of network weights and physical parameters; and (iii) an interpretable fault diagnosis stage that applies STLSQ sparse regression to the fault residual, enabling explicit identification of fault type and severity from a physics-motivated candidate library.

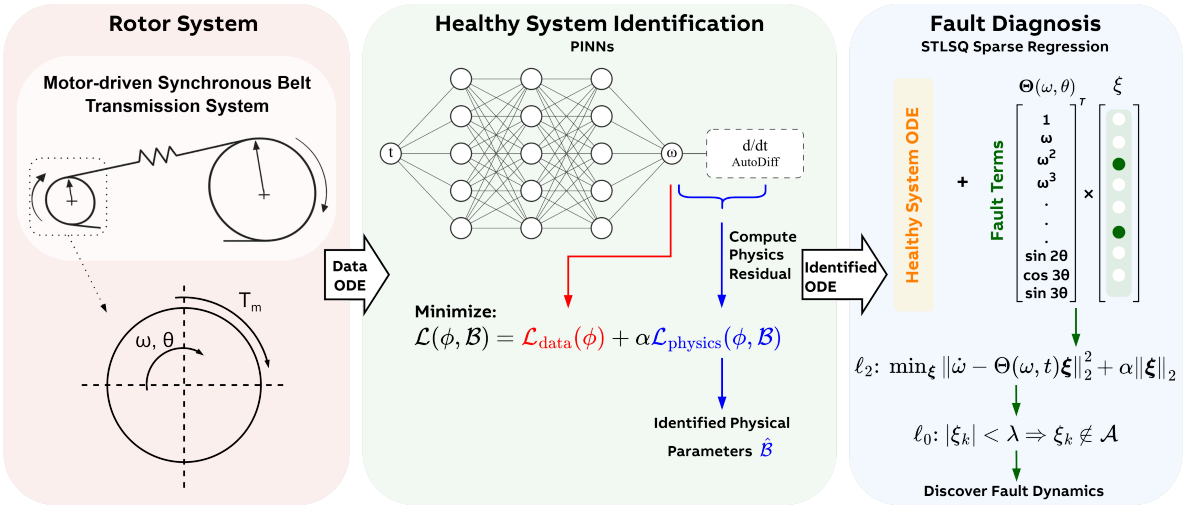


Figure 4–1: Workflow of the speed-based fault diagnosis method: (i) rotor system modeling, (ii) PINN-based healthy system identification, (iii) STLSQ sparse regression on the fault residual.

4.1 Physical Model for PINNs-based Healthy System Identification

The rotational dynamics of the motor shaft under healthy operating conditions are governed by the first-order ODE established in Equation (3–1), where the motor torque $T_m(t)$, proportional to the measured motor current, serves as the known external input to the system. In the PINN framework, the neural network predicts $\omega(t)$ from the input t, and the acceleration $\dot{\omega}(t)$ is obtained via automatic differentiation. The physics residual is evaluated at collocation points and penalized in the physics loss, while the physical parameters J, B, and T_c are treated as trainable variables optimized jointly with the network weights.

Following the normalization strategy described in Section 3.3.3, all quantities entering the residual are replaced by their scaled counterparts. Substituting $\omega(t) = \sigma_\omega\, \omega^*(t)$, $T_m(t) = \sigma_{T_m}\, T_m^*(t)$, and $\mathrm{d}/\mathrm{d}t = (1/t_s)\, \mathrm{d}/\mathrm{d}t^*$ into Equation (3–1) and dividing through by σ_{T_m} yields the normalized residual:

$$\mathcal{R}^*(t^*) = J^*\, \dot{\omega}^*(t^*) + B^*\, \omega^*(t^*) + T_c^*\, \operatorname{sgn}(\omega^*(t^*)) - T_m^*(t^*)\,, \tag{4–1}$$

where J^*, B^*, and T_c^* are the scaled, non-dimensional physical parameters related to their dimensional counterparts by:

$$J^* = \frac{J\, \sigma_\omega}{t_s\, \sigma_{T_m}}\,, \qquad B^* = \frac{B\, \sigma_\omega}{\sigma_{T_m}}\,, \qquad T_c^* = \frac{T_c}{\sigma_{T_m}}\,. \tag{4–2}$$

These scaled parameters are the external trainable variables in the healthy system identification stage, optimized jointly with the neural network weights during PINN training. Since the moment

of inertia, viscous friction coefficient, and Coulomb friction torque are all strictly positive physical quantities, a positivity constraint is imposed on J^*, B^*, and T_c^* during optimization to ensure physically meaningful estimates.

During training, the PINN operates entirely in the normalized space: the network input is $t^* \in [0, 1]$, the network output is $\omega^*(t^*)$, and the learnable physical parameters are J^*, B^*, and T_c^*. This ensures that all terms in the residual **(4–1)** and the data loss are of comparable magnitude, as discussed in Section 3.3.3.

After training converges, the dimensional physical parameters are recovered by inverting Equation (4–2):

$$J = \frac{J^* \, t_s \, \sigma_{T_m}}{\sigma_\omega} \, , \qquad B = \frac{B^* \, \sigma_{T_m}}{\sigma_\omega} \, , \qquad T_c = T_c^* \, \sigma_{T_m} \, . \tag{4–3}$$

These recovered parameters carry physical units: the moment of inertia J in kg · m^2, the viscous friction coefficient B in N · m · s/rad, and the Coulomb friction torque T_c in N · m. Together, they define the healthy baseline model against which fault-induced deviations are identified in the subsequent fault diagnosis stage.

4.2 Healthy System Identification

The healthy system identification stage formulates a physics-informed inverse problem as described in Section 3.3.2. PINNs are employed to simultaneously approximate the speed trajectory and estimate the scaled physical parameters $\boldsymbol{\beta}^* = \{J^*, B^*, T_c^*\}$ from measurement data recorded under healthy operating conditions. The network learns the mapping from normalized time t^* to scaled angular velocity $\omega^*(t^*)$, and the physical parameters are identified jointly with the network weights through the minimization of a composite loss function. This composite loss combines a Mean Squared Error (MSE) data fidelity term with an MSE physics residual term derived from the normalized governing ODE **(4–1)**:

$$\mathcal{L}_{\text{total}} = w_p \, \mathcal{L}_{\text{physics}} + w_d \, \mathcal{L}_{\text{data}} \, , \tag{4–4}$$

where $\mathcal{L}_{\text{physics}}$ is the mean squared physics residual evaluated at M collocation points within the time domain:

$$\mathcal{L}_{\text{physics}} = \frac{1}{M} \sum_{j=1}^{M} \left| \mathcal{R}^*(t_j^*) \right|^2 \, , \tag{4–5}$$

$\mathcal{L}_{\text{data}}$ is the mean squared error between the network prediction and the measured scaled

speed at N observation points:

$$\mathcal{L}_{\text{data}} = \frac{1}{N}\sum_{i=1}^{N} |\omega^*_{\text{NN}}(t^*_i) - \omega^*_i|^2 \ , \tag{4–6}$$

and w_p, w_d are the loss weights controlling the relative importance of each term. Through this joint optimization, accurate identification of the physical parameters is achieved without requiring separate system identification experiments.

Neural Network Architecture

The neural network takes the normalized time $t^* \in [0, 1]$ as the single input and outputs the scaled angular velocity $\omega^*(t^*)$. The architecture is a fully connected feedforward network. For an input t^*, the output is computed by propagating the signal through L hidden layers, each consisting of an affine transformation followed by an element-wise nonlinearity:

$$\mathbf{h}_i = \tanh(\mathbf{W}_i \mathbf{h}_{i-1} + \mathbf{b}_i) \, , \qquad i = 1, \ldots, L \ , \tag{4–7}$$

where $\mathbf{h}_0 = t^*$, $\mathbf{W}_i$ and $\mathbf{b}_i$ are the weight matrix and bias vector of the i-th layer, and the final scalar output is obtained by applying a linear map to $\mathbf{h}_L$. The hyperbolic tangent activation function (illustrated in Figure 4–2) is chosen for two reasons. First, $\tanh$ is smooth and infinitely differentiable, so that the higher-order derivatives required to evaluate the physics residual via automatic differentiation are well defined throughout the network. Second, $\tanh$ is bounded between -1 and 1, which keeps the activations of all hidden layers in a moderate numerical range and helps maintain stable gradient propagation during training.

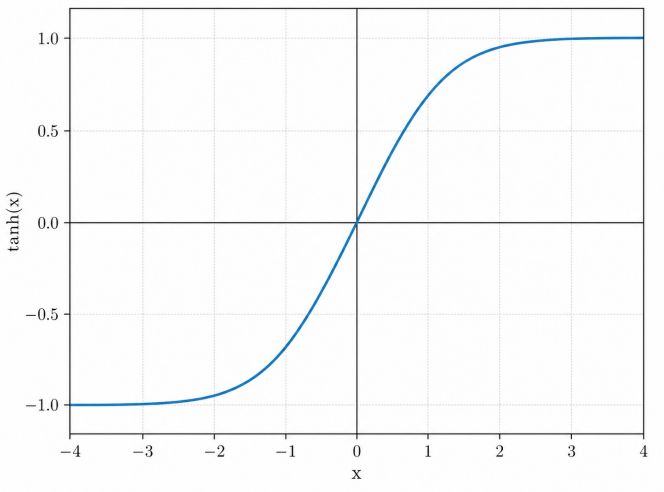


Figure 4–2: Hyperbolic tangent activation function $\tanh(x)$.

The motor torque $T_m^*(t^*)$, required for evaluating the physics residual **(4–1)**, is computed from the measured motor control current, to which it is proportional, and is provided as an exogenous input through radial basis function interpolation.

Two-Phase Training Strategy

Training is carried out in two sequential phases to avoid the adverse interaction between an untrained network and the physics-informed parameter estimation. In standard PINN training, the data loss and the physics loss are minimized simultaneously from the outset. In practice, this joint optimization can lead to conflicting gradient directions during the early training phase: the data loss drives the network towards fitting the measurements, while the physics loss penalizes residuals of the governing ODE [12]. When the network has not yet learned a reasonable approximation of the trajectory, its output and the associated automatic-differentiation derivatives bear little physical meaning. Consequently, the physics loss gradients computed from these unreliable derivatives can interfere with the data-fitting process, slowing convergence or driving the optimizer towards poor local minima. To mitigate this issue, a two-phase training strategy is adopted, following the rationale of data-guided PINNs proposed by Zhou and Wei [41].

In Phase I, the physics residual weight is set to $w_p = 0$ and the physical parameters J^*, B^*, T_c^* are excluded from the set of trainable variables. The network is trained purely on the data loss using the Adam optimizer [13] with a learning rate of 10^{-3}, which represents a commonly used balanced choice that is small enough to avoid divergence yet large enough to ensure timely convergence on the present problem. Training proceeds for N_1 epochs, until the data loss **(4–6)** has decreased to a sufficiently small value, indicating that the network output closely matches the measured trajectory. This pre-training step is necessary because the ODE residual involves products of the network derivatives with the physical parameters (see **(4–1)**). At random initialization, the network output does not yet match the measured trajectory, and the automatic differentiation derivatives therefore reflect the random weight configuration rather than the true system dynamics. Activating the physics residual at this stage is unfavorable for two reasons. First, the residual is evaluated on a trajectory that is not yet representative of the system, and its gradients with respect to the network weights and the physical parameters are correspondingly uninformative. Second, the data loss and the physics loss tend to produce gradients that differ in magnitude and direction, particularly during early training, a phenomenon documented in the PINN literature [37]; when both terms are activated simultaneously from random initialization, this conflict can stall convergence or bias the optimization toward a poor local minimum. Pre-training the network on the data loss alone ensures that, by the time the physics residual is activated, the network output and its derivatives carry information consistent with the measured dynamics and the gradient conflict is substantially reduced.

In Phase II, the physics residual is activated by setting $w_p > 0$, and the scaled physical parameters are registered as external trainable variables, optimized jointly with the network weights. Since the network already approximates the measured trajectory closely enough by the end of

Phase I, the physics residual and its gradients carry information consistent with the measured dynamics, enabling stable and efficient identification of the physical parameters. All three loss components contribute to the gradient, and training continues with the Adam optimizer at the same learning rate for N_2 epochs. A positivity constraint is enforced on J^*, B^*, and T_c^* through the softplus function (illustrated in Figure 4–3):

$$\zeta(x) = \ln(1 + e^x) \,, \tag{4–8}$$

which maps any real-valued trainable variable x to a strictly positive output, ensuring physically meaningful estimates throughout training. The convergence of the physical parameters is monitored by recording their values at fixed epoch intervals during training and checking that successive estimates change by less than a prescribed tolerance.

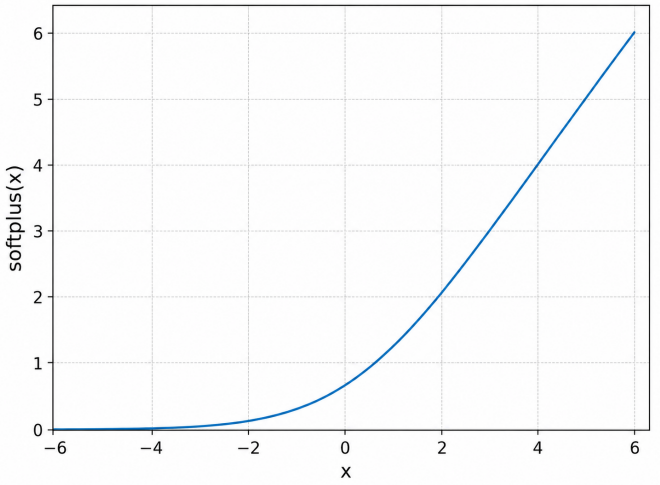


Figure 4–3: Softplus function $\zeta(x) = \ln(1 + e^x)$.

After Phase II training has finished, the scaled parameters $\boldsymbol{\beta}^* = \{J^*, B^*, T_c^*\}$ are converted back to their dimensional estimates $\hat{\boldsymbol{\beta}} = \{\hat{J}, \hat{B}, \hat{T}_c\}$ using the inverse scaling relations in Equation (4–3). Before being used for fault diagnosis, these identified parameters are validated by forward numerical simulation of the healthy ODE **(3–1)** and comparison against the measured trajectories across all healthy operating conditions, as detailed in Section 7.1.2. Once validated, they represent the identified healthy baseline of the system and are carried forward to the fault diagnosis stage, where deviations from this baseline are used to detect and characterize fault-induced dynamics.

The complete procedure is summarized in Algorithm 4–1.

Algorithm 4–1 Rotation-Speed-Based Healthy System Identification

Input (healthy operating conditions)**:** $\{\omega_i,\, T_{m,i},\, t_i\}_{i=1}^{N}$

Compute scaling factors σ_ω, σ_{T_m}, t_s and normalize data.
Initialize network $\omega^*_{\text{NN}}(t^*;\, \boldsymbol{\phi})$ and physical parameters $\boldsymbol{\beta}^* = \{J^*,\, B^*,\, T_c^*\}$.

Phase I: Data-driven pre-training
for epoch $= 1$ to N_1 **do**
$\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \eta\, \nabla_{\boldsymbol{\phi}}\, \mathcal{L}_{\text{data}}(\boldsymbol{\phi})$;
end

Phase II: Physics-informed training
for epoch $= N_1 + 1$ to $N_1 + N_2$ **do**
$\dot{\omega}^* \leftarrow \partial\, \omega^*_{\text{NN}} / \partial\, t^*$;
$\mathcal{R}^* \leftarrow J^*\, \dot{\omega}^* + B^*\, \omega^* + T_c^*\, \operatorname{sgn}(\omega^*) - T_m^*$;
$\boldsymbol{\phi},\, \boldsymbol{\beta}^* \leftarrow \boldsymbol{\phi} - \eta\, \nabla_{\boldsymbol{\phi}}\, \mathcal{L}_{\text{total}}, \quad \boldsymbol{\beta}^* - \eta\, \nabla_{\boldsymbol{\beta}^*}\, \mathcal{L}_{\text{total}}$;
end

$\hat{J},\, \hat{B},\, \hat{T}_c \leftarrow$ recover via Eq. **(4–3)**.

return $\hat{\boldsymbol{\beta}} = \{\hat{J},\, \hat{B},\, \hat{T}_c\}$

4.3 Fault Diagnosis

Once the healthy baseline parameters $\hat{\boldsymbol{\beta}} = \{\hat{J},\, \hat{B},\, \hat{T}_c\}$ are established, the fault diagnosis stage aims to discover additional dynamic terms that arise under faulty operating conditions. The key idea is to reformulate the problem so that the sparse regression operates on the fault-induced residual rather than on the full system dynamics, as introduced in Section 3.4.

Fault Residual Formulation

Under faulty conditions, the governing dynamics of the motor shaft deviate from the healthy baseline model: additional fault-induced torque contributions cause the measured speed trajectory to no longer be fully explained by the identified healthy ODE. To isolate these fault-induced contributions, the fault residual $R_f(t)$ is introduced as the difference between the measured angular acceleration and the acceleration predicted by the healthy model. The healthy model prediction is computed from the identified parameters $\hat{\boldsymbol{\beta}} = \{\hat{J},\, \hat{B},\, \hat{T}_c\}$ obtained in the healthy system identification stage, using the measured torque $T_m(t)$ and speed $\omega(t)$ as inputs:

$$\dot{\omega}_{\text{healthy}}(t) = \frac{1}{\hat{J}} \left[T_m(t) - \hat{B}\, \omega(t) - \hat{T}_c\, \operatorname{sgn}(\omega(t)) \right] , \tag{4–9}$$

which is the rearranged form of the healthy ODE **(3–1)** solved for the angular acceleration. The fault residual is then defined as:

$$R_f(t) = \dot{\omega}(t) - \dot{\omega}_{\text{healthy}}(t) \ , \tag{4–10}$$

where $\dot{\omega}(t)$ is obtained from the measured speed signal via numerical differentiation. Since automatic differentiation through the PINN is no longer available at this stage, a Savitzky–Golay filter is applied to the speed signal to simultaneously smooth the data and compute the derivative, reducing the amplification of high-frequency noise inherent in finite-difference schemes [27]. For each sample $\omega_i = \omega(t_i)$, the filter fits a polynomial of degree p to the $2m+1$ samples within a sliding window centered at t_i in the least-squares sense, and evaluates the derivative of the fitted polynomial at the window center. This is equivalent to a discrete convolution of the samples with a set of precomputed coefficients $\{h_k\}_{k=-m}^{m}$:

$$\dot{\omega}(t_i) \approx \frac{1}{\Delta t} \sum_{k=-m}^{m} h_k \, \omega_{i+k} \ , \tag{4–11}$$

where Δt is the sampling interval, $2m+1$ is the window length, and the coefficients h_k depend on the polynomial degree p and the window size; they are tabulated once and reused for every sample. The fault residual $R_f(t)$ captures all dynamic contributions not explained by the identified healthy ODE and serves as the target for the subsequent sparse identification.

Candidate Fault Library

A library of candidate fault terms $\boldsymbol{\Theta}(\omega,\, \theta)$ is constructed based on domain knowledge of typical transmission faults. The library comprises four groups of basis functions, each operating on a specific physical quantity:

$$\boldsymbol{\Theta}(\omega,\, \theta) = \Big[\underbrace{\operatorname{sgn}(\omega)}_{\text{Coulomb}},\ \underbrace{\omega,\ \omega^2,\ \omega^3}_{\text{polynomial}},\ \underbrace{\sin\theta,\ \cos\theta,\ \ldots,\ \sin 5\theta,\ \cos 5\theta}_{\text{Fourier harmonics}},\ \underbrace{\sin\tfrac{\theta}{3},\ \cos\tfrac{\theta}{3},\ \sin\tfrac{2\theta}{3},\ \cos\tfrac{2\theta}{3}}_{\text{fractional harmonics}} \Big] . \tag{4–12}$$

The Coulomb and polynomial terms capture speed-dependent fault effects such as changes in friction characteristics. The integer-frequency Fourier terms, constructed from the shaft angle $\theta(t)$, represent angle-periodic disturbances characteristic of faults such as parallel misalignment. The fractional-frequency harmonics account for the 1:3 transmission ratio between the motor and load pulleys, capturing load-side periodic faults such as tooth interference whose angular period is a multiple of the load-side revolution.

The STLSQ algorithm governing the sparse regression, introduced in Section 3.4, requires two hyperparameters: the sparsity threshold $\lambda > 0$, which controls the minimum coefficient magnitude retained in the active set, and the ridge regularization parameter $\alpha \geq 0$, which

penalizes large coefficients to improve numerical conditioning. A small λ retains more terms and risks overfitting, while a large λ aggressively prunes the library and may discard physically relevant fault contributions.

Sparse Regression

The fault residual is expressed as a sparse linear combination of the candidate library:

$$R_f(t) \approx \boldsymbol{\Theta}(\omega,\, \theta)\, \boldsymbol{\xi}\ , \tag{4–13}$$

where the coefficient vector $\boldsymbol{\xi}$ is determined by solving the sparse regression problem using the Sequentially Thresholded Least Squares algorithm. STLSQ iteratively performs least-squares regression and sets coefficients below a prescribed threshold λ to zero, yielding a parsimonious model where only the most significant fault terms are retained[6]. The columns of the library matrix are normalized prior to regression to prevent scale-dependent bias in the thresholding step.

A critical distinction from standard SINDy applications is that the regression target is not the full state derivative $\dot{\omega}(t)$ but the fault residual $R_f(t)$ defined in Eq. **(4–10)**. By subtracting the known healthy dynamics before fitting, the sparse regression only needs to identify the additional fault-related terms rather than re-discovering the entire system model. This separation improves the identifiability of small fault signatures that would otherwise be masked by the dominant healthy dynamics. It should be noted, however, that the fault residual $R_f(t)$ inherits the uncertainty of the identified healthy parameters $\hat{J}$, $\hat{B}$, and $\hat{T}_c$: any deviation between the identified parameters and their true values appears in $R_f(t)$ as a residual term that the sparse regression may interpret as a fault contribution.

The sparsity threshold λ controls the minimum coefficient magnitude retained after each iteration, and the ridge penalty α provides additional regularization to stabilize the least-squares solution. Both are hyperparameters selected prior to regression following a two-step procedure. First, the algorithm is applied to data from the healthy scenario, where no fault terms should be active: λ is then chosen as the smallest value at which all candidate coefficients are pruned to zero, ensuring that the threshold is large enough to suppress noise-driven spurious terms but no larger than necessary. Second, λ is verified on data from each fault scenario by checking that the identified active terms remain stable across repeated runs and across different motion profiles for the same fault type. The ridge penalty α is set to a small fixed value sufficient to stabilize the least-squares step when the candidate library contains nearly collinear terms, and the sensitivity of the identified coefficients to this value is checked by varying α over one order of magnitude. The specific values selected through this procedure are reported in Section 7.1.3.

Each non-zero coefficient ξ_k in the identified vector directly corresponds to a specific physical fault mechanism, providing an interpretable fault signature. The magnitude of ξ_k reflects the severity of the corresponding fault contribution, enabling both fault classification and severity assessment.

The complete fault diagnosis procedure is summarized in Algorithm 4–2.

Algorithm 4–2 Speed-Based Fault Diagnosis via STLSQ

Input (faulty operating conditions)**:** $\{\omega_i,\, T_{m,i},\, \theta_i,\, t_i\}_{i=1}^{N}$, healthy parameters $\hat{\boldsymbol{\beta}} = \{\hat{J},\, \hat{B},\, \hat{T}_c\}$, threshold λ, ridge penalty α.

$\dot{\omega} \leftarrow$ Savitzky–Golay differentiation of $\omega(t)$
$\dot{\omega}_{\text{healthy}} \leftarrow \frac{1}{\hat{J}}\left[T_m - \hat{B}\,\omega - \hat{T}_c\,\mathrm{sgn}(\omega)\right]$
$R_f \leftarrow \dot{\omega} - \dot{\omega}_{\text{healthy}}$
Construct candidate library $\boldsymbol{\Theta}(\omega,\, \theta)$ with Eq. **(4–12)**

for $k = 1, 2, \ldots$ until convergence **do**
  ℓ_2: $\boldsymbol{\xi} \leftarrow \arg\min_{\boldsymbol{\xi}} \|R_f - \boldsymbol{\Theta}\,\boldsymbol{\xi}\|_2^2 + \alpha\|\boldsymbol{\xi}\|_2^2$;
  ℓ_0: Remove ξ_k from active set $\quad \forall\, |\xi_k| < \lambda$;
  $\boldsymbol{\xi}_{\text{active}} \leftarrow \arg\min_{\boldsymbol{\xi}_{\text{active}}} \|R_f - \boldsymbol{\Theta}_{\text{active}}\,\boldsymbol{\xi}_{\text{active}}\|_2^2 + \alpha\|\boldsymbol{\xi}_{\text{active}}\|_2^2$;
end

return fault terms $\boldsymbol{\xi}$ and corresponding library entries

The speed-based method presented in this chapter establishes an interpretable fault diagnosis framework by combining PINNs-based healthy system identification with STLSQ sparse regression. By isolating fault-induced residual dynamics from the dominant healthy baseline, the approach enables sensitive detection and explicit characterization of fault mechanisms through the identified sparse coefficients.

Nevertheless, the method is fundamentally constrained by the quality of the speed signal: $\omega(t)$ is internally computed by the servo drive via numerical differentiation of the encoder readings, and the fault residual $R_f(t)$ requires a further differentiation step, compounding noise amplification and potentially masking small fault signatures. Moreover, the angular position enters only through the candidate library rather than the physics residual, limiting the tightness of the physical constraints.

A further source of uncertainty arises from the sequential structure of the framework itself: the identified healthy parameters $\hat{J}$, $\hat{B}$, and $\hat{T}_c$ carry residual estimation errors from the PINN stage, and these errors propagate into the fault residual $R_f(t)$, where they may be interpreted by the sparse regression as spurious fault contributions. Since the healthy identification and the fault diagnosis are decoupled into two independent stages, there is no mechanism to jointly correct for such inconsistencies, and the reliability of the identified fault terms therefore depends on the accuracy of the healthy baseline being well below the magnitude of the fault signatures of interest.

These limitations motivate the angle-based reformulation presented in Chapter 5, where the dynamics are expressed directly in terms of the encoder angle, all differentiation is handled by automatic differentiation within the network, and angle-periodic fault signatures are embedded into the physics residual itself.

5 Rotation-Angle-Based Interpretable Fault Diagnosis Method

In the rotation-speed-based method presented in Chapter 4, the neural network directly outputs the angular velocity $\omega(t)$, which is internally computed by the servo drive via numerical differentiation of the encoder readings. This signal is therefore subject to noise amplification and quantization artifacts introduced by the drive's internal processing. Moreover, the fault residual $R_f(t)$ requires a further differentiation step applied to the already noisy speed signal, compounding these effects and potentially masking small fault signatures.

In addition, the fault identification step requires constructing candidate library terms involving trigonometric functions of the shaft angle, such as $\cos\theta(t)$ and $\sin\theta(t)$. Under constant-speed operation, the angle can be approximated as $\theta(t) \approx \omega t$; however, this linear approximation is no longer valid under variable-speed profiles involving acceleration and deceleration phases, which form a substantial part of the operating conditions considered in this work. It is therefore not used here, and the true angular position is provided as an external input throughout the speed-based fault identification stage. These limitations motivated the development of an alternative formulation in which the neural network directly predicts $\cos\theta(t)$ and $\sin\theta(t)$ from the raw encoder signals, making the angle information inherently available throughout the entire workflow without relying on any speed-based approximation, and delegating all differentiation to automatic differentiation within the network.

The overall workflow is illustrated in Figure 5–1 and follows the same three-stage structure as the speed-based approach: rotor dynamics system modeling, healthy system identification, and fault diagnosis. The main contributions of this chapter are: (i) a SIREN-based PINN formulation that predicts the trigonometric projections $(\cos\theta, \sin\theta)$ and recovers all kinematic quantities via analytic differentiation, bypassing the noise-prone numerical differentiation of the discontinuous encoder angle; (ii) a three-component loss function combining data fidelity, a unit-circle geometry constraint, and a normalized second-order physics residual, together with a phase-transition normalization step that estimates the angular velocity scale from the pre-trained network to ensure balanced residual terms; and (iii) a PINNs-embedded sparse fault identification procedure that incorporates candidate fault terms with trainable amplitudes and phases directly into the physics residual, applying an alternating ℓ_0/ℓ_1 scheme within the automatic-differentiation framework to discover interpretable fault signatures without requiring a separate post-processing pipeline.

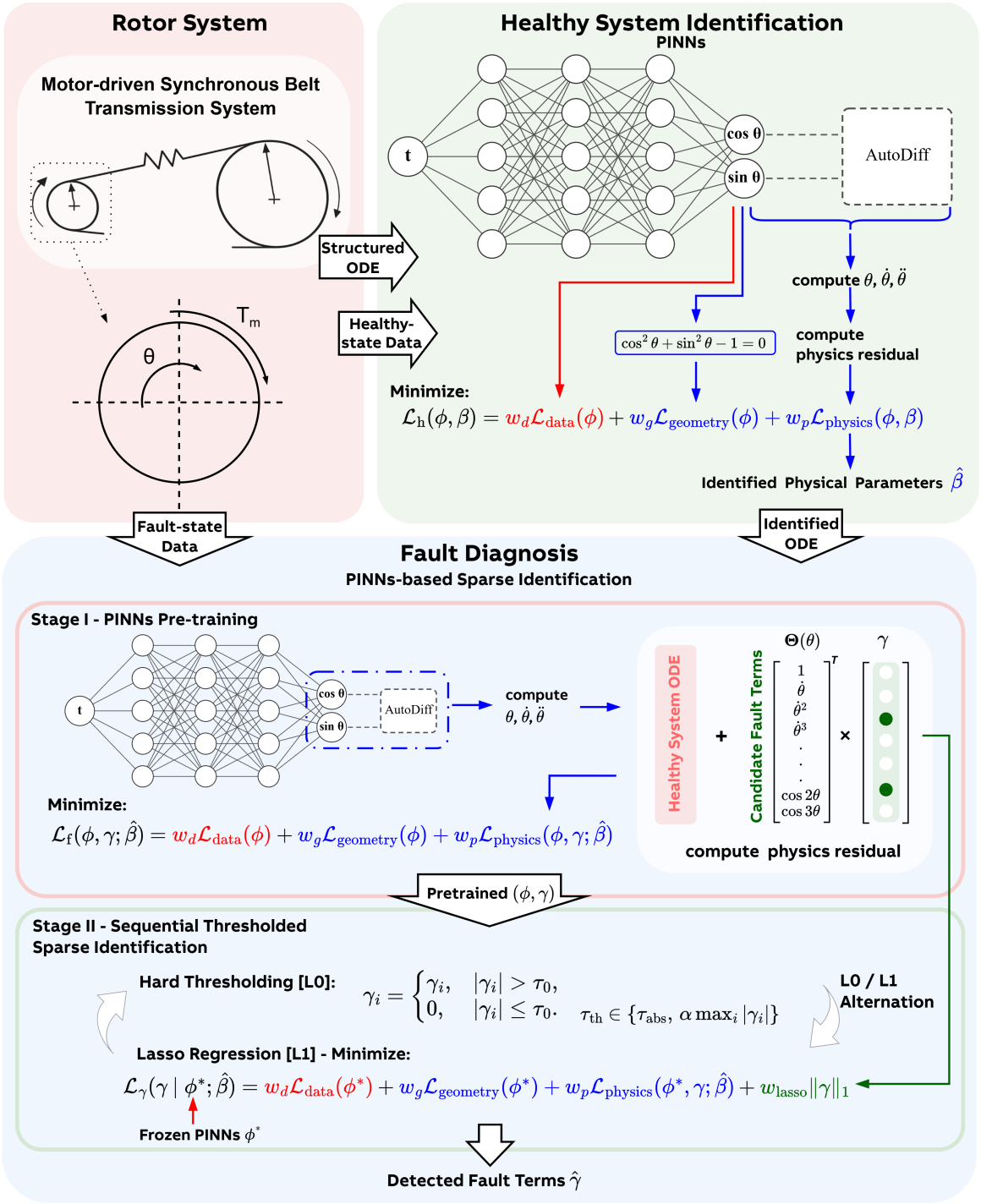


Figure 5–1: Workflow of the angle-based fault diagnosis method: (i) rotor system modeling, (ii) SIREN-based PINN healthy system identification, and (iii) PINN-embedded sparse fault identification with alternating ℓ_0/ℓ_1 sparsity promotion.

5.1 Physical Model for PINNs-based Healthy System Identification

The angle-based method operates on the second-order ODE governing the angular position $\theta(t)$, as established in Equation (3–2), in contrast to the speed-based formulation, which models the first-order dynamics of the angular velocity $\omega(t)$. The angular velocity $\dot{\theta}(t)$ and acceleration $\ddot{\theta}(t)$ required by this formulation are not obtained from the raw encoder angle, whose 2π-discontinuities would corrupt any numerical derivative estimate, but are instead recovered analytically from the continuous trigonometric projections $c(t) = \cos\theta(t)$ and $s(t) = \sin\theta(t)$ introduced in Section 3.2.2. In the PINN framework developed in this chapter, a neural network takes the normalized time t^* as input and predicts the trigonometric pair (c, s) as its two outputs. The angular velocity and acceleration are then reconstructed from the network outputs and their automatic-differentiation derivatives using the analytic expressions **(3–6)** and **(3–8)**.

The validity of these simplified derivative expressions relies on the unit-circle identity **(3–4)** holding exactly. When c and s are predicted by a neural network rather than computed from a true angle, this identity is not automatically satisfied. As shown in Section 3.2.2, violation of this constraint introduces a non-vanishing cross-coupling term in the general acceleration expression **(3–7)**, invalidating the simplified forms used throughout the physics residual. The unit-circle identity is therefore formulated as a geometry residual:

$$\mathcal{R}_{\text{geo}}(t^*) = \hat{c}(t^*)^2 + \hat{s}(t^*)^2 - 1 \,, \tag{5–1}$$

which is enforced during training by penalizing its squared magnitude at collocation points in the same manner as the physics residual, as detailed in Section 5.2. Since the network outputs c and s are bounded in $[-1, 1]$ by construction, the geometry residual requires no additional scaling. This bounded range is also well suited to neural network training: as discussed in Section 3.3.3, network outputs of order unity keep activations within the responsive region of the nonlinearity, prevent saturation in the bounded region of the activation function, and allow the data and physics losses to operate on comparable scales without further weighting [35].

For the physics residual, the governing ODE **(3–2)** must be expressed in terms of normalized quantities to ensure balanced training, following the general normalization principles established in Section 3.3.3. The time variable is mapped to $t^* \in [0, 1]$ with time span $t_s = t_{\text{end}} - t_0$, and the motor torque is scaled by its standard deviation: $T_m^*(t) = T_m(t)/\sigma_{T_m}$. The angular velocity and acceleration, reconstructed analytically from the network outputs, are expressed in physical units and can take values that are orders of magnitude larger than unity. These quantities must therefore be normalized before entering the residual.

The angular velocity standard deviation $\sigma_{\dot{\theta}}$ is selected as the normalizing scale for both the velocity and the acceleration terms. The choice of $\sigma_{\dot{\theta}}$ for the velocity follows the standard practice established in Section 3.3.3, where each signal is divided by its own standard deviation so that the scaled signal is of order unity. For the acceleration $\ddot{\theta}$, the same factor $\sigma_{\dot{\theta}}$ is reused

rather than introducing a separate $\sigma_{\ddot{\theta}}$ for two reasons. First, applying $\sigma_{\dot{\theta}}$ to both derivatives keeps the residual dimensionally consistent within a single unified scaling, which directly leads to the compact set of scaled physical parameters in **(5–3)** where each parameter is associated with a single, well-defined power of t_s. Second, with the time variable mapped to $[0, 1]$, derivatives with respect to t^* are intrinsically of order unity, so the inertial term $J\,\ddot{\theta}$ in the residual is automatically of comparable magnitude to the viscous term $B\,\dot{\theta}$ once the inertia and damping coefficients are scaled by the appropriate powers of t_s, eliminating the need for an additional independent acceleration scale. Since $\dot{\theta}$ is not directly measured but computed from the network outputs, $\sigma_{\dot{\theta}}$ is not available before training begins. It is instead estimated at the transition between Phase I and Phase II of the two-phase training strategy: at the end of Phase I, the network output closely matches the measured trigonometric signals from the data loss alone, and the angular velocity is sampled at equally spaced points by evaluating **(3–6)** using the trained network outputs and their automatic-differentiation derivatives. The standard deviation $\sigma_{\dot{\theta}}$ is computed from these samples and held fixed throughout Phase II.

With this normalization, the scaled angular velocity and acceleration entering the physics residual are $\dot{\theta}^* = \dot{\theta}/\sigma_{\dot{\theta}}$ and $\ddot{\theta}^* = \ddot{\theta}/\sigma_{\dot{\theta}}$. Substituting $\dot{\theta} = \sigma_{\dot{\theta}}\,\dot{\theta}^*$, $\ddot{\theta} = \sigma_{\dot{\theta}}\,\ddot{\theta}^*$, and $T_m = \sigma_{T_m}\,T_m^*$ into the governing ODE **(3–2)** and dividing through by σ_{T_m} yields the normalized physics residual:

$$\mathcal{R}_\theta^*(t^*) = J^*\,\ddot{\theta}^*(t^*) + B^*\,\dot{\theta}^*(t^*) + T_c^*\,\operatorname{sgn}(\dot{\theta}^*(t^*)) - T_m^*(t^*)\,, \tag{5–2}$$

where the scaled physical parameters are related to their dimensional counterparts by:

$$J^* = \frac{J\,\sigma_{\dot{\theta}}}{\sigma_{T_m}\,t_s^2}\,, \qquad B^* = \frac{B\,\sigma_{\dot{\theta}}}{\sigma_{T_m}\,t_s}\,, \qquad T_c^* = \frac{T_c}{\sigma_{T_m}}\,. \tag{5–3}$$

These scaled parameters are the external trainable variables in the healthy system identification stage, optimized jointly with the neural network weights during PINN training. Since J, B, and T_c are strictly positive physical quantities, a positivity constraint is imposed on J^*, B^*, and T_c^* via the softplus function throughout training. The sgn function in the Coulomb friction term is approximated by $\tanh(20\,\dot{\theta}^*)$ to ensure differentiability: $\tanh$ is a smooth surrogate that is bounded in $[-1, 1]$ and tends to ± 1 for non-vanishing inputs, recovering the sign behavior of sgn, while remaining infinitely differentiable so that gradient-based optimization through automatic differentiation is well defined. The factor 20 controls the steepness of the transition near $\dot{\theta}^* = 0$, with larger values yielding a tighter approximation of the discontinuous sgn at the cost of larger gradient magnitudes around zero; the value 20 is chosen as a standard empirical compromise between fidelity and numerical stability.

5.2 Healthy System Identification

The healthy system identification stage follows the same inverse-problem philosophy as the speed-based method (Section 4.2): PINNs are employed to simultaneously approximate the system trajectory and estimate the scaled physical parameters $\beta^* = \{J^*,\, B^*,\, T_c^*\}$ from measurement data recorded under healthy operating conditions. The key difference lies in the network output representation: instead of predicting the angular velocity directly, the network predicts the trigonometric pair $(\cos\theta,\, \sin\theta)$, from which the velocity and acceleration are derived analytically. This change in representation introduces additional considerations in the network architecture and loss function design, which are detailed in the following. The network learns the mapping from normalized time t^* to the trigonometric pair $(\hat{c},\, \hat{s})$, and the physical parameters are identified jointly with the network weights through the minimization of a composite loss function $\mathcal{L}_{\mathsf{h}}$, where the subscript denotes that this loss is used exclusively in the healthy system identification stage and is distinguished from the fault diagnosis loss $\mathcal{L}_{\mathsf{f}}$ introduced in Section 5.3. The healthy identification loss combines a data fidelity term, a geometry constraint term, and a physics residual term:

$$\mathcal{L}_{\mathsf{h}}(\boldsymbol{\phi},\, \boldsymbol{\beta}) = w_d\, \mathcal{L}_{\mathsf{data}}(\boldsymbol{\phi}) + w_g\, \mathcal{L}_{\mathsf{geometry}}(\boldsymbol{\phi}) + w_p\, \mathcal{L}_{\mathsf{physics}}(\boldsymbol{\phi},\, \boldsymbol{\beta})\ , \tag{5–4}$$

where $\mathcal{L}_{\mathsf{data}}$ penalizes the mismatch between the predicted and measured trigonometric signals separately for the cosine and sine components at N observation points:

$$\mathcal{L}_{\mathsf{data}} = \frac{1}{N}\sum_{i=1}^{N} \left|\hat{c}(t_i^*) - c_i\right|^2 + \frac{1}{N}\sum_{i=1}^{N} \left|\hat{s}(t_i^*) - s_i\right|^2\ , \tag{5–5}$$

$\mathcal{L}_{\mathsf{geometry}}$ penalizes the squared geometry residual **(5–1)** at M collocation points:

$$\mathcal{L}_{\mathsf{geometry}} = \frac{1}{M}\sum_{j=1}^{M} \left|\mathcal{R}_{\mathsf{geo}}(t_j^*)\right|^2\ , \tag{5–6}$$

and $\mathcal{L}_{\mathsf{physics}}$ penalizes the squared normalized physics residual **(5–2)** at the same collocation points:

$$\mathcal{L}_{\mathsf{physics}} = \frac{1}{M}\sum_{j=1}^{M} \left|\mathcal{R}_{\theta}^*(t_j^*)\right|^2\ . \tag{5–7}$$

The weights are set to $w_d = 1.0$, $w_g = 10^{-4}$, and $w_p = 10^{-3}$ in this work. The relative ordering $w_d \gg w_p \gg w_g$ reflects the role of each term: the data loss anchors the network to the measurements and is therefore kept at unit weight as the primary objective, while the physics loss

is weighted two orders of magnitude below the data loss to prevent the residual from dominating early in Phase II, in line with the gradient-balancing principle established in Section 3.3.3 and the gradient pathology discussion in Section 5.2. The geometry loss receives the smallest weight because the data loss already implicitly drives the outputs toward the unit circle: substituting the targets $c_i = \cos\theta_i$ and $s_i = \sin\theta_i$ into **(5–5)** shows that minimizing $\mathcal{L}_{\text{data}}$ pushes $\hat{c}^2 + \hat{s}^2$ to 1, so $\mathcal{L}_{\text{geometry}}$ acts only as a soft correction once the data fit is established. These values are tunable and may require adjustment depending on the data characteristics.

Neural Network Architecture

The neural network takes the normalized time $t^* \in [0, 1]$ as the single input and outputs two values: the predicted cosine and sine of the shaft angle, $\hat{c}(t^*) = \widehat{\cos\theta}(t^*)$ and $\hat{s}(t^*) = \widehat{\sin\theta}(t^*)$. As established in Section 5.1, the angular velocity and acceleration required by the physics residual are obtained from these two outputs and their automatic-differentiation derivatives via the analytic expressions **(3–6)** and **(3–8)**. The motor torque $T_m^*(t^*)$, required for evaluating the physics residual, is provided as an exogenous input through radial basis function interpolation of the measured torque signal.

The architecture adopts a SIREN, as introduced in Section 3.5, in place of the conventional $\tanh$-activated feedforward network used in the speed-based method. This choice is motivated by the periodic nature of the target signals: the cosine and sine projections of the shaft angle are inherently oscillatory, and their frequency content spans a broad range that depends on the rotational speed of the motor. Conventional feedforward networks with $\tanh$ or ReLU activations fail to represent these signals adequately at comparable network sizes, suffering from spectral bias [25], the tendency of standard networks to preferentially fit low-frequency components, as discussed in Section 3.5. The effect is most pronounced at the extrema of the trigonometric functions, where the second derivative attains its maximum magnitude $|\ddot{c}| = |\dot{\theta}|^2$ (and analogously for $\ddot{s}$); accurately reproducing the curvature in these regions requires the network to represent high-frequency content, which is precisely the regime suppressed by spectral bias.

The frequency hyperparameters and weight initialization follow the scheme described in Section 3.5, with values selected through empirical tuning.

Two-Phase Training Strategy

Training follows the same two-phase rationale as the speed-based method in Section 4.2, with an additional normalization step at the phase transition.

In Phase I, the physics residual is deactivated ($w_p = 0$, $w_g = 0$) and the physical parameters are excluded from optimization. The network is trained purely on the data loss $\mathcal{L}_{\text{data}}$ using the Adam optimizer with a learning rate of 10^{-3} for E_1 epochs. The objective of this phase

is twofold: first, to bring the network output and its automatic-differentiation derivatives into a physically meaningful range before the physics loss is introduced; second, to provide a trained network from which the angular velocity standard deviation $\sigma_{\dot{\theta}}$ can be estimated, as described in Section 5.1.

At the transition between phases, the angular velocity is sampled at equally spaced points using the analytic derivative formula **(3–6)** evaluated with the Phase I network. Sample $\hat{c}$, $\hat{s}$ and their derivatives with respect to normalized time, $\dot{\hat{c}}_s = \mathrm{d}\hat{c}/\mathrm{d}t^*$ and $\dot{\hat{s}}_s = \mathrm{d}\hat{s}/\mathrm{d}t^*$, at equally spaced points via the Phase I network. The standard deviation $\sigma_{\dot{\theta}}$ is computed from these samples and fixed for the remainder of training.

In Phase II, all three loss components are activated with their respective weights, and the scaled physical parameters $\{J^*, B^*, T_c^*\}$ are registered as external trainable variables. Both the network weights and the physical parameters are optimized jointly using the Adam optimizer at the same learning rate for E_2 epochs. The positivity constraint on the physical parameters is enforced through the softplus function throughout training.

After training converges, the dimensional physical parameters are recovered by inverting Equation (5–3):

$$J = \frac{J^*\,\sigma_{T_m}\,t_s^2}{\sigma_{\dot{\theta}}}\,, \qquad B = \frac{B^*\,\sigma_{T_m}\,t_s}{\sigma_{\dot{\theta}}}\,, \qquad T_c = T_c^*\,\sigma_{T_m}\,. \tag{5–8}$$

These identified parameters define the healthy baseline model and are carried forward to the fault diagnosis stage.

The complete procedure is summarized in Algorithm 5–1.

Algorithm 5–1 Angle-Based Healthy System Identification

Input: Measured angle $\{\theta_i\}_{i=1}^{N}$, measured torque $\{T_{m,i}\}_{i=1}^{N}$, time stamps $\{t_i\}_{i=1}^{N}$, randomly initialized SIREN $(\hat{c},\, \hat{s}) = f_{\text{SIREN}}(t^*;\, \boldsymbol{\phi})$, learning rate η.

Compute $c_i = \cos\theta_i$, $s_i = \sin\theta_i$; compute scaling factors σ_{T_m}, t_s and normalize data.

Phase I: Data-driven pre-training

for epoch $= 1$ to E_1 **do**

Calculate $\mathcal{L}_{\text{data}}$ with Eq. **(5–5)**;

$\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \eta\, \nabla_{\boldsymbol{\phi}}\, \mathcal{L}_{\text{data}}(\boldsymbol{\phi})$;

end

Phase transition: Estimate $\sigma_{\dot{\theta}}$

Sample $\hat{c}$, $\hat{s}$, $\dot{\hat{c}}_s$, $\dot{\hat{s}}_s$ at equally spaced points via the Phase I network;

$\dot{\theta}_k \leftarrow \frac{1}{t_s}\left(\hat{c}_k\, \dot{\hat{s}}_{s,k} - \hat{s}_k\, \dot{\hat{c}}_{s,k}\right)$; $\quad \sigma_{\dot{\theta}} \leftarrow \text{std}(\{\dot{\theta}_k\})$;

Phase II: Physics-informed training

for epoch $= E_1 + 1$ to $E_1 + E_2$ **do**

$\dot{\theta}^*,\, \ddot{\theta}^* \leftarrow$ analytic reconstruction from $(\hat{c},\, \hat{s})$ via Eqs. **(3–6)**, **(3–8)**, normalized by $\sigma_{\dot{\theta}}$;

$\mathcal{R}_{\theta}^{*} \leftarrow J^*\, \ddot{\theta}^* + B^*\, \dot{\theta}^* + T_c^*\, \text{sgn}(\dot{\theta}^*) - T_m^*$;

Calculate $\mathcal{L}_{\text{h}}$ with Eq. **(5–4)**;

$\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \eta\, \nabla_{\boldsymbol{\phi}}\, \mathcal{L}_{\text{h}}(\boldsymbol{\phi},\, \boldsymbol{\beta}^*)$;

$\boldsymbol{\beta}^* \leftarrow \boldsymbol{\beta}^* - \eta\, \nabla_{\boldsymbol{\beta}^*}\, \mathcal{L}_{\text{h}}(\boldsymbol{\phi},\, \boldsymbol{\beta}^*)$;

end

$\hat{J},\, \hat{B},\, \hat{T}_c \leftarrow$ recover via Eq. **(5–8)**.

return $\hat{\boldsymbol{\beta}} = \{\hat{J},\, \hat{B},\, \hat{T}_c\}$

5.3 Fault Diagnosis

The fault diagnosis stage of the angle-based method differs fundamentally from the speed-based approach. In the speed-based method (Section 4.3), the fault residual is computed offline from measurements and the identified healthy model, and a separate STLSQ is applied as a post-processing step. In the angle-based method, the sparse identification is instead embedded directly into the PINN training framework. The candidate fault terms are incorporated into the physics residual as trainable coefficients, and sparsity is enforced through an alternating ℓ_0/ℓ_1 scheme applied within the gradient-based optimization loop. This PINNs-based approach to sequential thresholded sparse identification avoids the need for numerical differentiation of measured signals and leverages the automatic-differentiation infrastructure already established in the healthy identification stage.

5.3.1 Fault Diagnosis Procedure

The fault diagnosis procedure consists of two stages, whose roles are outlined below before the detailed formulation is presented.

Stage I: PINN Pre-training with Candidate Fault Terms

In Stage I, a PINN is trained on fault-state measurement data with the augmented physics residual containing both the known healthy dynamics and a full library of candidate fault terms. The network weights ϕ, the fault amplitude coefficients γ, and the phase parameters φ of the harmonic terms are optimized jointly. No sparsity penalty is applied at this stage; all candidate coefficients are free to take non-zero values so that the optimization can explore the full candidate space without premature elimination of potentially relevant terms. The training adopts a two-phase structure similar to the healthy identification stage, but without the intermediate normalization step: Phase I trains the network purely on the data loss $\mathcal{L}_{\text{data}}$ for E_1 epochs to learn a faithful approximation of the measured trigonometric signals; Phase II directly activates all three loss components of $\mathcal{L}_{\text{f}}$ and optimizes the network weights, fault coefficients, and phase parameters jointly for E_2 epochs. No phase-transition normalization of the angular velocity is required, since the augmented residual **(5–12)** operates in physical units with the healthy dynamics block scaled by Δ_{T_m} rather than through non-dimensionalized parameters. At the end of Stage I, the network has learned a faithful representation of the faulty trajectory, the fault coefficients carry initial estimates of their amplitudes, and the phase parameters have converged to values that align each harmonic with the measured disturbance.

Stage II: PINNs-Based Sequential Thresholded Sparse Identification

In Stage II, the pre-trained network weights ϕ^* and phase parameters φ^* are frozen, and only the fault coefficients γ remain trainable. Freezing the network is essential for the integrity of the sparse identification. To see this concretely, consider the augmented residual **(5–12)**: the trainable fault terms $\boldsymbol{\Theta}(\theta,\, \dot{\theta})\, \gamma$ enter additively alongside the healthy dynamics block reconstructed from the network outputs $(\hat{c},\, \hat{s})$. If, upon pruning a candidate term $\gamma_k\, \Theta_k$, the network were free to update ϕ, it could adjust $(\hat{c},\, \hat{s})$ such that the corresponding contribution reappears within the $\hat{J}\, \ddot{\theta} + \hat{B}\, \dot{\theta}$ part of the residual via the analytic derivative reconstruction **(3–6)**, **(3–8)**. Formally, the residual gradient with respect to ϕ would absorb $-\gamma_k\, \Theta_k$ into $\partial(\hat{J}\, \ddot{\theta} + \hat{B}\, \dot{\theta})/\partial\phi$, so the optimizer would minimize the loss without truly removing the pruned dynamics. Fault terms that are physically present could thus be erroneously pruned, and the resulting coefficient vector would no longer reflect the true fault structure. By fixing ϕ^*, the analytic kinematic reconstruction is held constant during Stage II, the gradient $\partial\mathcal{R}_{\text{f}}/\partial\gamma_k = \Theta_k$ depends only on the frozen library evaluation, and the sparsity-promoting updates act directly and exclusively on the fault coefficients. The phase parameters are frozen for the same reason: each cosine term $\cos(f_k\theta + \varphi_k^*)$ becomes a fixed basis function, analogous to a fixed library column in classical STLSQ (Sec-

tion 3.4); allowing φ_k to drift during pruning would change the shape of the basis itself and confound the selection.

Sparsity is enforced through an alternating ℓ_0/ℓ_1 procedure that operates iteratively for K loops. In each loop, an ℓ_0 hard thresholding step is followed by an ℓ_1 Lasso retraining step. In the ℓ_0 step, the absolute values of all active fault coefficients are compared against a prescribed threshold τ_0; coefficients whose magnitude falls below τ_0 are set to zero and permanently removed from the set of trainable variables:

$$\gamma_k \leftarrow 0 \quad \text{if} \quad |\gamma_k| < \tau_0 \, . \tag{5–9}$$

The threshold τ_0 can be defined as a fixed absolute value, as a fraction of the largest active coefficient magnitude, or as a combination of both, providing flexibility in controlling the aggressiveness of the pruning. In this work τ_0 is selected empirically: an initial value is chosen on the order of the smallest coefficient magnitude that the practitioner considers physically meaningful, and the value is refined by running the procedure on a representative fault scenario and observing the resulting active set. A threshold that is too small leaves noise-driven coefficients in the active set, while one that is too large prunes physically relevant terms; the chosen value is the one that yields a stable active set across repeated runs and across different motion profiles for the same fault type. The specific values used in this work are reported in Section 7.2.3.

In the ℓ_1 step, the surviving coefficients are retrained by minimizing the Stage II loss **(5–15)** using the Adam optimizer for a fixed number of epochs E_3. The ℓ_1 penalty $w_{\text{lasso}}\|\boldsymbol{\gamma}\|_1$ provides continuous sparsity promotion: its subgradient with respect to γ_k is $w_{\text{lasso}}\ \mathrm{sgn}(\gamma_k)$, which adds a constant inward push of magnitude w_{lasso} to every non-zero coefficient regardless of its current value, shrinking marginal coefficients towards zero so that they fall below τ_0 in the next ℓ_0 step. The two operations are therefore complementary: the ℓ_0 step provides a discrete decision boundary that produces exact zeros, while the ℓ_1 step continuously biases coefficients toward that boundary. As a result, a coefficient is pruned only if both mechanisms agree, that is, if it is small in magnitude and its physics-residual contribution is too weak to resist the constant Lasso shrinkage. This dual condition reduces sensitivity to either hyperparameter taken alone: a slightly mis-tuned τ_0 is partially compensated by the ℓ_1 shrinkage that would have moved the coefficient below threshold in a subsequent iteration, and a slightly mis-tuned w_{lasso} is checked by the discrete cutoff at τ_0.

In each successive iteration, the pool of active coefficients shrinks as insignificant terms are pruned, and the surviving coefficients are refined to better fit the fault-induced dynamics. The procedure terminates after the prescribed K iterations or when no further coefficients are pruned, at which point the set of active coefficients has stabilized, yielding a sparse set of detected fault terms $\hat{\boldsymbol{\gamma}}$.

5.3.2 Fault Diagnosis Formulation

This subsection presents the detailed mathematical formulation underlying the two-stage fault diagnosis procedure: the normalization strategy adapted for the fault identification setting, the candidate fault library and its embedding into the physics residual, the augmented residual that combines the healthy baseline with trainable fault terms, and the loss functions used in Stage I and Stage II respectively.

Normalization

The normalization strategy for the fault diagnosis stage differs from the healthy system identification stage in two respects. First, since the healthy physical parameters $\hat{\boldsymbol{\beta}} = \{\hat{J},\, \hat{B},\, \hat{T}_c\}$ have already been identified and carry physical units, the physics residual is formulated in dimensional form rather than in the fully non-dimensionalized form used during healthy identification: the angular velocity $\dot{\theta}$ and acceleration $\ddot{\theta}$, reconstructed analytically from the network outputs, enter the residual directly in physical units alongside the dimensional healthy parameters, without the intermediate normalization by $\sigma_{\dot{\theta}}$ that was required in the healthy identification stage. This also eliminates the need for the phase-transition normalization step, since no velocity scale needs to be estimated from the pre-trained network. Second, instead of scaling by the standard deviation, the motor torque is scaled by its min-max range $\Delta_{T_m} = \max(T_m) - \min(T_m)$:

$$T_m^*(t) = \frac{T_m(t)}{\Delta_{T_m}}\,, \tag{5–10}$$

so that the scaled torque spans a range of unity. This choice parallels the treatment of the time variable, which is also mapped to $[0, 1]$ by its range $t_s = t_{\text{end}} - t_0$, and serves a specific purpose in the fault diagnosis context: since the fault coefficients must be comparable across different operating conditions (varying speed profiles, load levels), normalizing by the torque range ensures that the healthy dynamics block in the residual has a consistent magnitude regardless of the absolute torque level, making the identified fault coefficients directly comparable between segments. The network outputs (c, s) require no scaling. The angular velocity and acceleration are reconstructed accounting for the time span t_s through $\dot{\theta} = (c\,\dot{s}_s - s\,\dot{c}_s)/t_s$ and $\ddot{\theta} = (c\,\ddot{s}_s - s\,\ddot{c}_s)/t_s^2$. The entire healthy dynamics block is then divided by Δ_{T_m} to match the scale of the normalized torque, ensuring that the residual terms remain balanced.

Candidate Fault Library

Unlike the speed-based method, where a discrete library matrix $\boldsymbol{\Theta}$ is explicitly constructed and passed to an external sparse regression solver, the angle-based method does not involve a separate library building step. Instead, the candidate fault terms are formulated as differentiable expressions and embedded directly into the physics residual of the PINN. Each term is multiplied by a trainable coefficient, and the sparsity selection is performed by driving insignifi-

cant coefficients to zero through the alternating ℓ_0/ℓ_1 procedure described above. Nevertheless, it is convenient to retain the notation of a coefficient vector $\boldsymbol{\gamma}$ and a corresponding set of basis functions $\boldsymbol{\Theta}(\theta,\, \dot{\theta})$ for clarity of presentation.

The candidate basis functions comprise two groups:

$$\boldsymbol{\Theta}(\theta,\, \dot{\theta}) = \Big[\; \underbrace{\cos(\theta+\varphi_1),\; \cos(2\theta+\varphi_2),\; \cos(3\theta+\varphi_3),\; \cos(\tfrac{\theta}{3}+\varphi_4),\; \cos(\tfrac{2\theta}{3}+\varphi_5)}_{\text{Fourier harmonics with phase}},\; \underbrace{\operatorname{sgn}(\dot{\theta}),\; \dot{\theta},\; \dot{\theta}^2}_{\text{polynomial terms}} \;\Big], \tag{5–11}$$

where φ_i are phase parameters associated with each harmonic term. The Fourier harmonics capture angle-periodic disturbances at integer and fractional multiples of the shaft angle. The integer-frequency terms ($\cos\theta,\ \cos 2\theta,\ \cos 3\theta$) represent periodic faults that repeat once per motor revolution, such as parallel misalignment or rotor unbalance. The fractional-frequency terms ($\cos(\theta/3)$, $\cos(2\theta/3)$) account for the 1:3 transmission ratio between the motor and load pulleys, capturing load-side periodic faults such as tooth interference or load-side misalignment whose angular period is a multiple of the load-side revolution. Each harmonic is parameterized as a single cosine with a phase shift φ_i, which provides a compact representation equivalent to a linear combination of $\sin(f\theta)$ and $\cos(f\theta)$. The polynomial terms capture speed-dependent fault effects such as changes in friction or damping characteristics.

Each candidate term is associated with a trainable amplitude coefficient γ_k, collectively forming the fault coefficient vector $\boldsymbol{\gamma}$. A positivity constraint is imposed on the Fourier amplitude coefficients via the softplus function, since each cosine term already encodes both positive and negative contributions through its phase; the polynomial coefficients are left unconstrained. The shaft angle θ required for evaluating the Fourier terms is recovered from the network outputs via $\theta = \operatorname{atan2}(s,\, c)$.

Augmented Physics Residual

Under faulty conditions, the physics residual is augmented with the candidate fault library. The healthy dynamics, evaluated with the frozen identified parameters $\hat{\boldsymbol{\beta}}$, form the baseline, and the fault terms enter as additive corrections:

$$\mathcal{R}_{\mathrm{f}}(t^*) = \frac{\hat{J}\,\ddot{\theta} + \hat{B}\,\dot{\theta} + \hat{T}_c\,\operatorname{sgn}(\dot{\theta})}{\Delta_{T_m}} - T_m^*(t^*) + \boldsymbol{\Theta}(\theta,\, \dot{\theta})\,\boldsymbol{\gamma}\,, \tag{5–12}$$

where the first group of terms represents the known healthy dynamics divided by Δ_{T_m} to match the torque scaling, and $\boldsymbol{\Theta}(\theta,\, \dot{\theta})\,\boldsymbol{\gamma}$ collects all candidate fault contributions with their trainable coefficients. The fault coefficients $\boldsymbol{\gamma}$ are learned directly at the scale set by the normalized residual; that is, the $1/\Delta_{T_m}$ scaling of the healthy dynamics block implicitly determines the magnitude of $\boldsymbol{\gamma}$, so that no explicit scaling factor needs to be applied to the fault library terms. If the healthy model perfectly describes the measured dynamics, all fault coefficients should

converge to zero; any non-zero coefficient after sparse identification indicates a fault-induced deviation from the baseline.

Loss Functions

The loss functions differ between Stage I and Stage II, reflecting the distinct optimization objectives of each stage.

In Stage I, the loss function combines data fidelity, geometry constraint, and physics residual without any sparsity penalty:

$$\mathcal{L}_{\mathrm{f}}(\boldsymbol{\phi},\, \boldsymbol{\gamma};\, \hat{\boldsymbol{\beta}}) = w_d\, \mathcal{L}_{\mathsf{data}}(\boldsymbol{\phi}) + w_g\, \mathcal{L}_{\mathsf{geometry}}(\boldsymbol{\phi}) + w_p\, \mathcal{L}_{\mathsf{physics}}(\boldsymbol{\phi},\, \boldsymbol{\gamma};\, \hat{\boldsymbol{\beta}})\ , \tag{5–13}$$

where the data and geometry losses retain the same forms as in Equations (5–5) and (5–6), and the physics loss penalizes the augmented residual **(5–12)** at M collocation points:

$$\mathcal{L}_{\mathsf{physics}} = \frac{1}{M} \sum_{j=1}^{M} \left| \mathcal{R}_{\mathrm{f}}(t_j^*) \right|^2\ . \tag{5–14}$$

The phase parameters $\boldsymbol{\varphi}$ are trainable in this stage, allowing each harmonic term to align with the angular phase of the underlying fault signature.

In Stage II, the network weights are frozen and the loss function is augmented with an ℓ_1 penalty on the fault coefficients:

$$\mathcal{L}_{\boldsymbol{\gamma}}(\boldsymbol{\gamma} \mid \boldsymbol{\phi}^*;\, \hat{\boldsymbol{\beta}}) = w_d\, \mathcal{L}_{\mathsf{data}}(\boldsymbol{\phi}^*) + w_g\, \mathcal{L}_{\mathsf{geometry}}(\boldsymbol{\phi}^*) + w_p\, \mathcal{L}_{\mathsf{physics}}(\boldsymbol{\phi}^*,\, \boldsymbol{\gamma};\, \hat{\boldsymbol{\beta}}) + w_{\mathsf{lasso}} \|\boldsymbol{\gamma}\|_1\ , \tag{5–15}$$

where w_{lasso} controls the strength of the Lasso penalty. Although the network is frozen and its outputs do not change, the data and geometry losses remain in the objective as fixed anchor terms. The physics loss, which depends on $\boldsymbol{\gamma}$ through the augmented residual **(5–12)**, provides the gradient signal that drives the coefficient updates, while the ℓ_1 penalty shrinks small coefficients towards zero, preparing them for removal in the next hard thresholding step.

5.3.3 Practical Considerations and Discussion

This subsection addresses practical aspects of the PINNs-based fault diagnosis that are important for reliable application: the balance between competing loss components during training and its implications for convergence, a comparison of the proposed approach with the classical STLSQ algorithm used in the speed-based method, and the physical interpretation of the identified fault coefficients.

Training Stability and Loss Balancing

A practical consideration in the PINNs-based fault diagnosis is the balance between the data loss and the physics loss during Stage I training. Since the data loss drives the network towards fitting the measured trajectory while the physics loss penalizes deviations from the augmented ODE, the two objectives can produce competing gradient directions, particularly during transient phases of the measured signal where rapid acceleration or deceleration leads to large instantaneous values of $\ddot{\theta}$. In such regions, the physics residual can exhibit large magnitudes that dominate the total loss, causing the optimizer to prioritize residual reduction at the expense of data fidelity. This competition between gradient contributions from the two loss components can prevent the optimization from converging to a solution that satisfies both objectives simultaneously.

In practice, the severity of this conflict depends on the dynamic content of the specific measurement segment. The physics loss is dimensionless, since each term in the residual **(5–12)** is normalized by Δ_{T_m}, and the loss is a mean squared error of normalized residuals; values therefore measure the relative residual magnitude. For the majority of data segments in this work, the physics loss can be reduced to the 10^{-4} level when the loss weight w_p is set appropriately, corresponding to a normalized residual root-mean-square of about 1% of the torque range, which is comparable to the measurement noise floor and indicates that the augmented ODE and the measured trajectory are mutually consistent. For a small number of segments containing particularly aggressive transients, the achievable physics loss plateaus at approximately 2×10^{-3}, corresponding to a residual root-mean-square of roughly $4\%{-}5\%$ of the torque range, well above the noise floor.

The diagnosis that this elevated plateau reflects a tension between the data and physics objectives, rather than, for instance, an inadequate fault library or a poorly trained network, is supported by three observations. First, the data loss in these segments converges to the same low level ($\mathcal{O}(10^{-6})$) as in the well-converged cases, so the network does represent the measured trajectory accurately; the residual is therefore not caused by a poor fit. Second, the elevated plateau is correlated specifically with segments containing rapid acceleration or deceleration phases, where $|\ddot{\theta}|$ is large and the inertial term $\hat{J}\,\ddot{\theta}$ dominates the augmented residual; segments without such transients converge normally on the same network architecture and library. Third, reducing w_p in the affected segments lowers the physics loss back toward the typical level by allowing the data loss to dominate the gradient updates, while increasing w_p raises it further; this monotonic response to the weight is the signature of a gradient-balance issue rather than a model-misspecification issue. Taken together, these observations point to a residual tension between the data and physics objectives that cannot be fully resolved at the given model complexity. In such cases, reducing w_p alleviates the gradient conflict, ensuring that the network first maintains an accurate trajectory representation; the fault coefficients identified under a reduced physics weight remain meaningful, though their numerical precision may be slightly lower than in the well-converged cases.

Monitoring the physics loss during training therefore serves as a diagnostic for the reliability of the identified fault terms. A physics loss that converges to a low plateau indicates that the candidate library adequately captures the fault-induced dynamics, while a persistently elevated physics loss suggests that either the library is incomplete or the measurement segment contains dynamics that are difficult to reconcile with the assumed ODE structure.

Comparison with SINDy-Based STLSQ

The PINNs-based sequential thresholded sparse identification developed in this section shares the iterative threshold-and-refit philosophy with the classical STLSQ algorithm used in the speed-based method (Section 4.3), but differs in several important respects. First, the regression is performed within the automatic-differentiation framework of the PINN rather than on pre-computed numerical derivative estimates, avoiding the noise amplification inherent in finite-difference or Savitzky–Golay differentiation of measured signals. Second, the candidate terms are evaluated on the continuous network output rather than at discrete measurement points, providing a denser and smoother sampling of the dynamics along the learned trajectory. Third, the ℓ_1 Lasso retraining step replaces the ordinary least-squares step of classical STLSQ, offering continuous sparsity promotion through the penalty term in addition to the discrete pruning of the ℓ_0 step. Fourth, the fault identification is performed end-to-end within a single differentiable computational graph, eliminating the need for a separate post-processing pipeline.

Interpretation of Results

Each surviving non-zero coefficient $\hat{\gamma}_k$ in the final sparse vector directly corresponds to a specific physical fault mechanism, providing an interpretable fault signature. The magnitude of $\hat{\gamma}_k$ reflects the severity of the corresponding fault contribution, and for the Fourier terms, the associated frozen phase φ_k^* indicates the angular location of the disturbance within each revolution. Together, the identified fault terms, their amplitudes, and their phases enable fault classification, severity assessment, and spatial localization of the fault along the rotational cycle.

The complete fault diagnosis procedure is summarized in Algorithm 5–2.

Algorithm 5–2 Angle-Based Fault Diagnosis via PINNs-Based Sparse Identification

Input: Faulty measurements $\{\theta_i,\, T_{m,i}\}_{i=1}^{N}$, time stamps $\{t_i\}_{i=1}^{N}$, identified healthy parameters $\hat{\boldsymbol{\beta}} = \{\hat{J},\, \hat{B},\, \hat{T}_c\}$, randomly initialized SIREN $(\hat{c},\, \hat{s}) = f_{\text{SIREN}}(t^*;\, \boldsymbol{\phi})$.

Compute $c_i = \cos\theta_i$, $s_i = \sin\theta_i$; compute scaling factor Δ_{T_m} and normalize data.
Initialize fault coefficients $\boldsymbol{\gamma}$ and phase parameters $\boldsymbol{\varphi}$.

Stage I: PINN pre-training with candidate fault terms

Phase I: Train network $\boldsymbol{\phi}$ on $\mathcal{L}_{\text{data}}$ only for E_1 epochs.

Phase II: Activate augmented physics residual **(5–12)**.

for epoch $= E_1 + 1$ to $E_1 + E_2$ **do**

$\dot{\theta},\, \ddot{\theta} \leftarrow$ analytic reconstruction from $(\hat{c},\, \hat{s})$ via Eqs. **(3–6)**, **(3–8)**;

$\theta \leftarrow \operatorname{atan2}(\hat{s},\, \hat{c})$;

$\mathcal{R}_{\text{f}} \leftarrow \frac{\hat{J}\,\ddot{\theta} + \hat{B}\,\dot{\theta} + \hat{T}_c\,\operatorname{sgn}(\dot{\theta})}{\Delta_{T_m}} - T_m^* + \boldsymbol{\Theta}(\theta,\, \dot{\theta})\,\boldsymbol{\gamma}$;

Calculate $\mathcal{L}_{\text{f}}$ with Eq. **(5–13)**;

Update $\boldsymbol{\phi}$, $\boldsymbol{\gamma}$, $\boldsymbol{\varphi}$ by $\nabla_{\boldsymbol{\phi},\boldsymbol{\gamma},\boldsymbol{\varphi}}\,\mathcal{L}_{\text{f}}$;

end

Freeze network weights $\boldsymbol{\phi}^*$ and phase parameters $\boldsymbol{\varphi}^*$.

Stage II: PINNs-based sequential thresholded sparse identification

for loop $= 1, 2, \ldots, K$ **do**

ℓ_0 *step — Hard thresholding:*

$\gamma_k \leftarrow 0$, remove from trainable set $\quad \forall\, |\gamma_k| \le \tau_0$;

ℓ_1 *step — Lasso retraining (network frozen):*

for epoch $= 1$ to E_3 **do**

Calculate $\mathcal{L}_{\gamma}$ with Eq. **(5–15)**;

Update surviving $\boldsymbol{\gamma}$ by $\nabla_{\boldsymbol{\gamma}}\,\mathcal{L}_{\gamma}$;

end

end

return Detected fault terms $\hat{\boldsymbol{\gamma}}$ and corresponding phases $\boldsymbol{\varphi}^*$

6 Experiment Setup

This chapter describes the hardware platform and the experimental protocol used to collect data for training and evaluating the proposed method. Section 6.1 presents the mechanical and electrical components of the test bench, and Section 6.2 details the fault scenarios and motion profiles under which the experiments were conducted.

6.1 Experimental Test Bench

The experimental test bench consists of a servo motor whose speed is controlled solely by electrical current. The motor shaft is directly coupled to a small synchronous pulley (Figure 6–1C), which transmits motion to a large synchronous pulley through a toothed synchronous belt (Figure 6–1A). Both pulleys are mounted on the same plane, with their shafts secured by bearings and fixed to the large pulley via a shared clamp. A cylindrical metal block (Figure 6–1D) is mounted on the opposite side of the large pulley to serve as the inertial load. The transmission ratio between the small and large pulleys is 1:3, effectively scaling the motor torque and reducing the output speed accordingly.

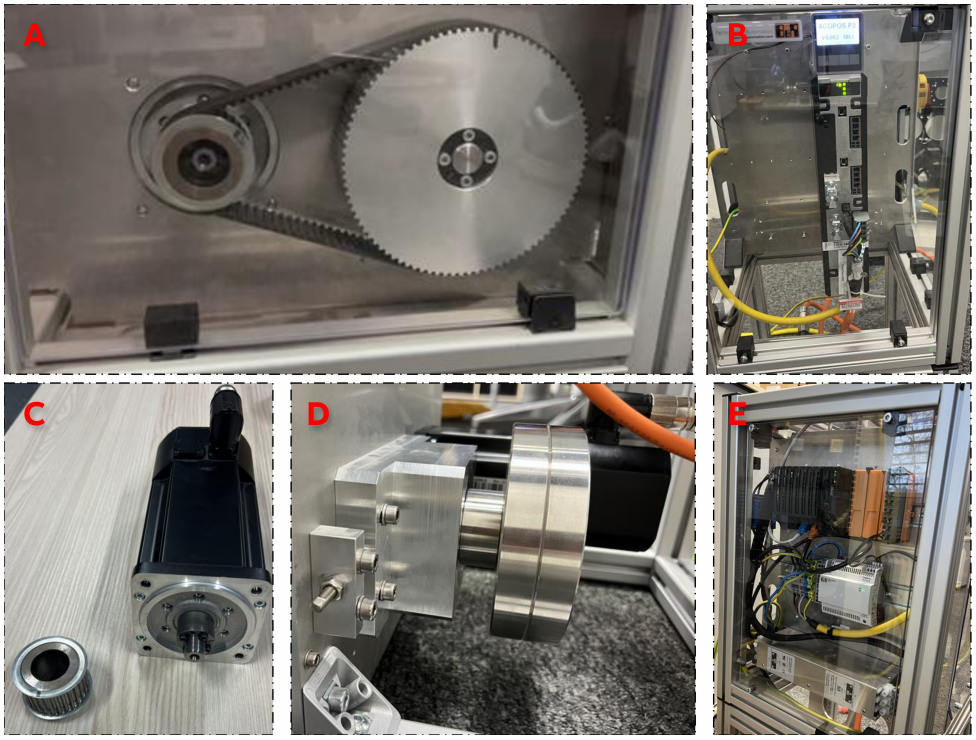


Figure 6–1: Experimental test bench. (A) Transmission mechanism. (B) Servo drive. (C) Disassembled motor with small pulley. (D) Inertial load. (E) Control cabinet.

Figure 6–1 illustrates the experimental test bench. The test bench mainly uses B&R products and is governed by a Programmable Logic Controller (PLC) [4] mounted in the control cabinet (Figure 6–1E). Motor actuation is handled by an Acopos P3 series servo drive [3] (Figure 6–1B)

connected to an 8LSA35 series servo motor [2]. Position and current feedback are provided by sensors integrated directly within the motor unit, comprising an encoder for angular position and velocity measurement and a current sensor for torque estimation. The toothed belt linking the two pulleys is an Optibelt Omega 600 5M synchronous belt [1]. The key hardware components of the test bench are summarized in Table 6–1.

Table 6–1: Test Bench Hardware Components

Component	Model / Specification
PLC	B&R X20CP3687X
Power Supply	B&R PS1042
Servo Drive	B&R Acopos P3
Servo Motor	B&R 8LSA35.DA030S200-3
Synchronous Belt	Optibelt Omega 600 5M

The **B&R X20CP3687X** is a high-performance PLC from the X20 system family, based on an Intel Atom 1.6 GHz processor. It is equipped with 2 GB LPDDR4 SDRAM and 2 GB onboard eMMC flash storage, and communicates with the servo drive over Ethernet POWERLINK V2, a deterministic real-time network protocol that enables precise synchronization of motion control tasks. The POWERLINK interface supports poll-response chaining mode, allowing cycle times in the sub-millisecond range. This tight real-time coupling between the PLC and the drive ensures that control commands and feedback signals are exchanged with consistent, bounded latency, which is essential for the accurate acquisition of the current and encoder data used as inputs to the proposed diagnostic methods.

The **B&R Acopos P3** servo drive is a compact, high-precision multi-axis drive capable of controlling up to three axes within a single housing. It covers a power range from 0.6 kW to 18 kW and achieves a control loop sampling time of 50 µs for the full current, speed, and position controller cascade. This fast sampling rate is critical for this work, as it determines the temporal resolution at which the motor current (used as a proxy for driving torque) and the encoder-derived speed signal are available for data acquisition. The drive interfaces directly with the motor via a single-cable solution that carries both power and encoder signals, reducing wiring complexity. It supports a digital multi-standard encoder interface per axis, enabling direct read-out of the high-resolution EnDat encoder integrated in the motor.

The **B&R 8LSA35.DA030S200-3** is a three-phase permanent magnet synchronous motor from the 8LS series, designed for high-performance motion control applications. Its key parameters are a nominal speed of 3,000 rpm, a stall torque of 2.3 Nm, four pole pairs, and a DC link voltage of 560 VDC. The motor is equipped with an integrated DA 2.2 inductive 32-line single-turn EnDat encoder, which provides the absolute angular position measurement used throughout this work. All relevant motor parameters are stored directly in the EnDat encoder and transmitted to the drive at startup, enabling automatic motor identification without manual parameter configuration. The motor operates with IP64 protection and self-cooling, and includes a holding

brake with 4 Nm holding torque. Crucially for the proposed diagnostic framework, no external sensors of any kind, such as accelerometers, torque transducers, or load-side encoders, are present on the test bench. The only signals available for measurement and diagnosis are the motor control current and the encoder angle, both of which are read directly from the servo drive's internal interfaces.

6.2 Experimental Design

To evaluate the proposed fault diagnosis framework, experiments were conducted under three scenarios: a healthy baseline condition, parallel misalignment, and tooth interference. The fault conditions are illustrated in Figure 6–2.

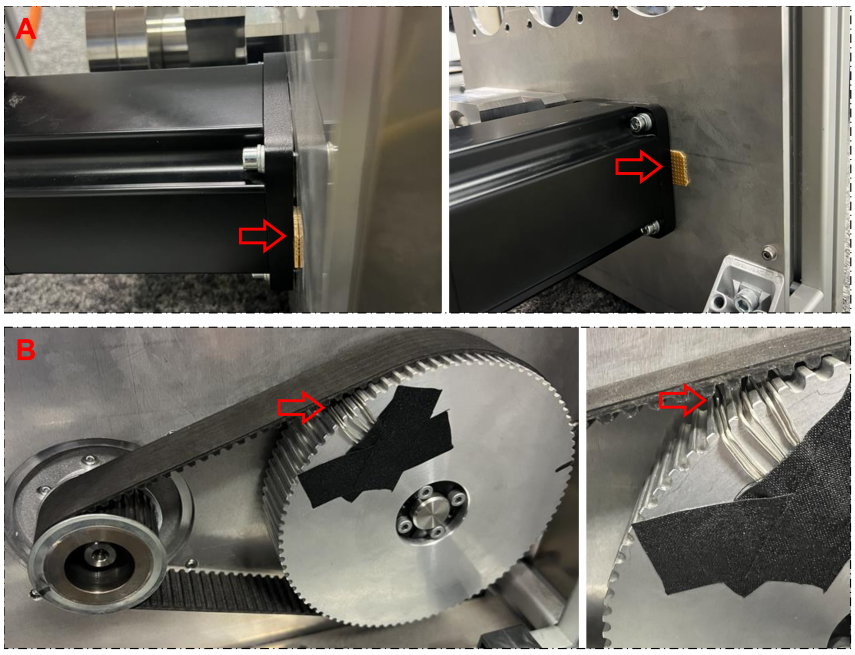


Figure 6–2: Fault Terms Configuration. (A) Parallel misalignment between the drive and load-side shafts. (B) Tooth interference simulated by wire across 3 consecutive grooves on the load-side pulley.

Parallel misalignment refers to the lateral offset error introduced by deliberate non-parallelism between the drive shaft and the load-side shaft. In a correctly aligned system, the two shafts are coplanar and parallel, ensuring symmetric belt loading across the full pulley width. To introduce a controlled misalignment, a small, thin fiberglass reinforced resin sheet was inserted beneath one of the pulley shaft bearings, tilting the load-side shaft out of plane relative to the motor shaft. This induces an asymmetric belt tension distribution and periodic lateral belt deflection forces that repeat once per motor revolution, producing a characteristic angle-periodic torque disturbance detectable in the motor current and encoder signals.

Tooth interference (debris entrapment) refers to the disruption of normal belt-tooth engagement caused by foreign material lodged in the belt-pulley mesh. In industrial deployments, such interference typically arises from debris, swarf, or accumulated contamination becoming trapped

between the belt teeth and the pulley grooves, causing a localized increase in meshing resistance each time the affected groove comes into contact with the belt. To replicate this effect in a controlled and repeatable manner, fine wire was placed across three consecutive toothed grooves on the load-side pulley and secured with tape. The wire and tape assembly introduces a localized mechanical obstruction at a fixed angular position on the load-side pulley, producing a periodic impulse-like torque disturbance that repeats once per load-side revolution, corresponding to every third motor revolution given the 1:3 transmission ratio. The mass of the wire and tape is negligible relative to the system inertia and does not measurably affect the rotational dynamics beyond the intended interference effect.

Each scenario was tested under two motion profiles. The first is a relative movement profile, in which the target rotation angle (distance) on the load side, maximum velocity, and acceleration are prescribed, causing the system to accelerate and decelerate rapidly. The second is a constant speed movement profile, in which the system accelerates to a prescribed target velocity and maintains it, with an optional jerk limit. For both profiles, different combination of rotational speed and acceleration configurations were applied. For the relative movement profile, six configurations were tested, combining load-side rotation distances of $\pm 180^\circ$ and $\pm 720^\circ$ with accelerations ranging from 10 000 °/s^2 to 15 000 °/s^2, all at a maximum velocity of 2400 °/s. For the constant speed movement profile, eight configurations were tested, covering target velocities of 360 °/s, 720 °/s, 1080 °/s, and 1440 °/s, all at an acceleration of 2400 °/s^2, with and without a jerk limit applied.

6.3 Experiment Result

Figures 6–3 to 6–8 present representative measured signals, namely rotation speed, control torque, and raw encoder angle, for the two motion profiles across all three system conditions. The relative movement profile captures the full acceleration–cruise–deceleration cycle, while the constant speed profile isolates the steady-state dynamics. Fault-induced disturbances are visible but subtle in the torque signal: misalignment produces a low-amplitude angle-periodic oscillation, while tooth interference introduces localized impulse-like spikes repeating at every third motor revolution. Both signatures are small relative to the dominant inertial and friction torques, and the signals exhibit considerable noise, necessitating appropriate filtering during preprocessing. Additionally, the sampling process is subject to aliasing if the acquisition rate is insufficient relative to the fault-characteristic frequencies, a consideration to be addressed in future data collection. The raw encoder angle is reported in native sampling units, where the full range of 0 to 524288 corresponds to one complete shaft revolution of 0 to 2π radians; this linear mapping is applied in the preprocessing stage. The resulting signal displays the expected 2π-wrapping discontinuities as the shaft completes successive revolutions, confirming the need for the trigonometric representation introduced in Section 3.2.2. These two motion profiles, namely the relative movement profile and the constant speed movement profile shown in this section, are also the representative working conditions used for evaluating both fault diagnosis methods in Chapter 7.

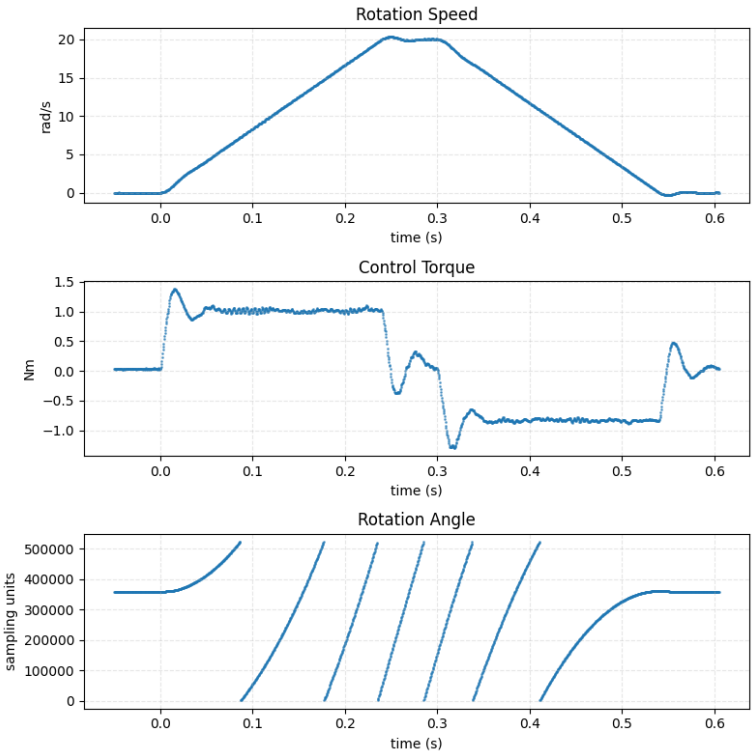


Figure 6–3: Relative Movement in healthy scenario - load-side rotation distances of $+720°$, accelerations of 10 000 °/s^2, maximum velocity of 2400 °/s.

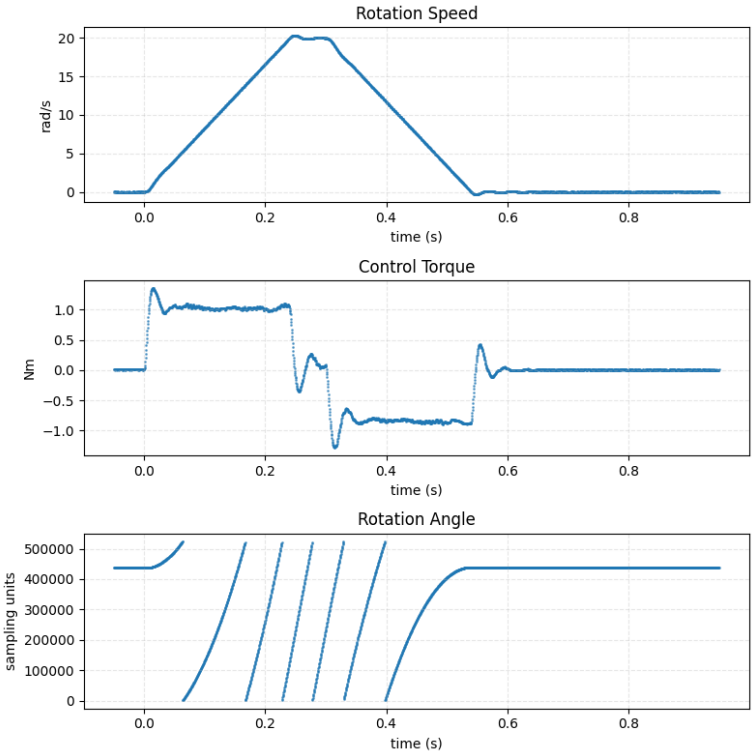


Figure 6–4: Relative Movement in misalignment scenario - load-side rotation distances of $+720°$, accelerations of 10 000 °/s^2, maximum velocity of 2400 °/s.

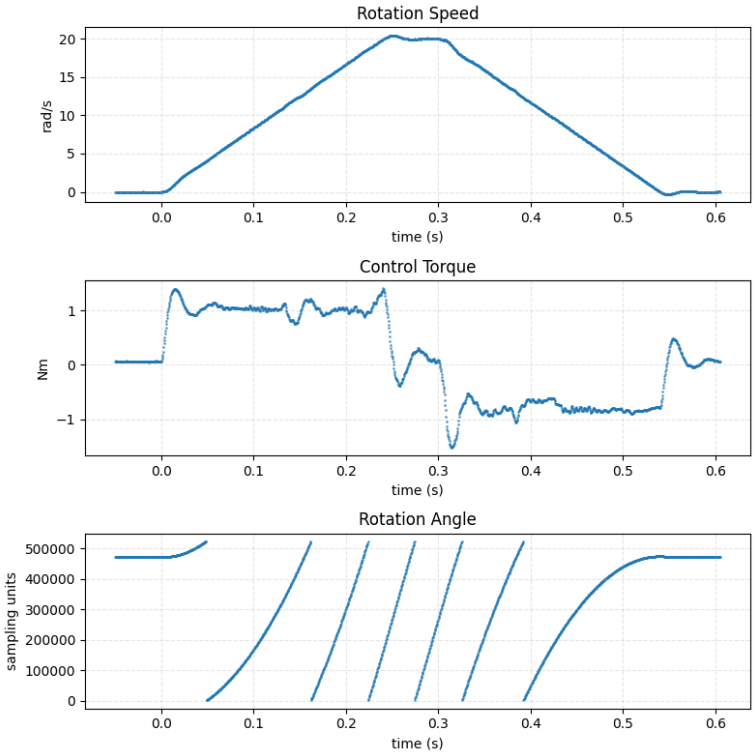


Figure 6–5: Relative Movement in tooth interference scenario - load-side rotation distances of $+720°$, accelerations of 10 000 °/s², maximum velocity of 2400 °/s.

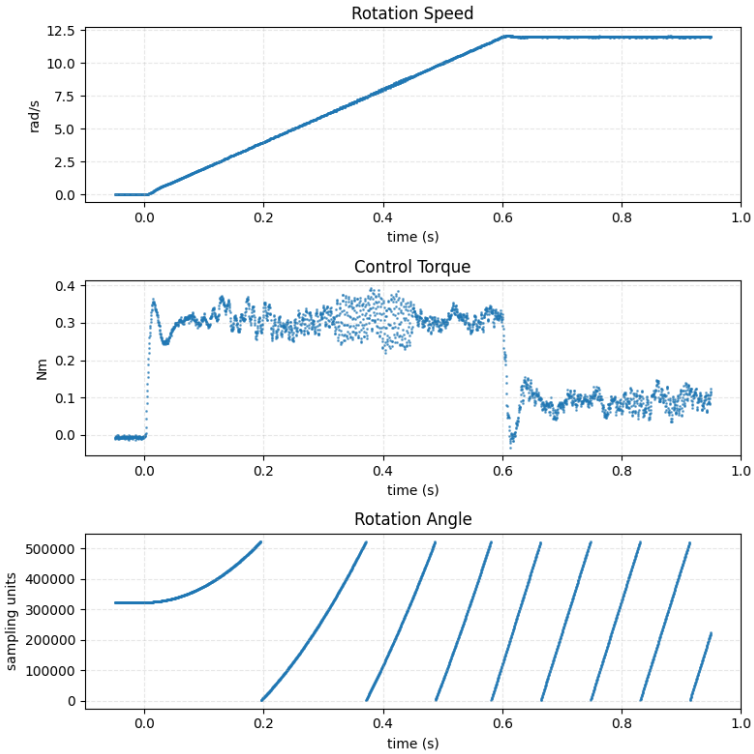


Figure 6–6: Constant Speed Movement in healthy scenario - target velocity of 1440 °/s, acceleration of 2400 °/s², and without jerk.

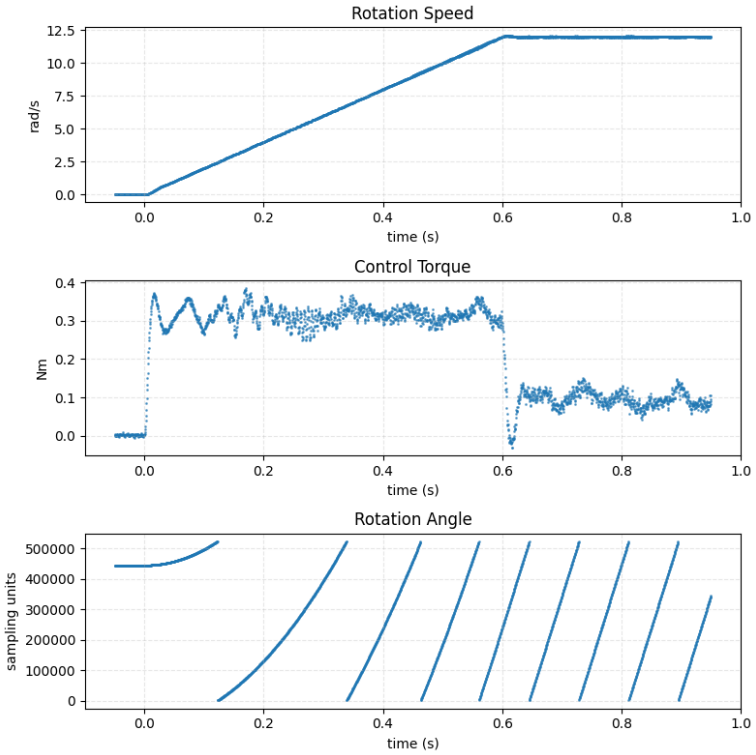


Figure 6–7: Constant Speed Movement in misalignment scenario - target velocity of 1440 °/s, acceleration of 2400 °/s^2, and without jerk.

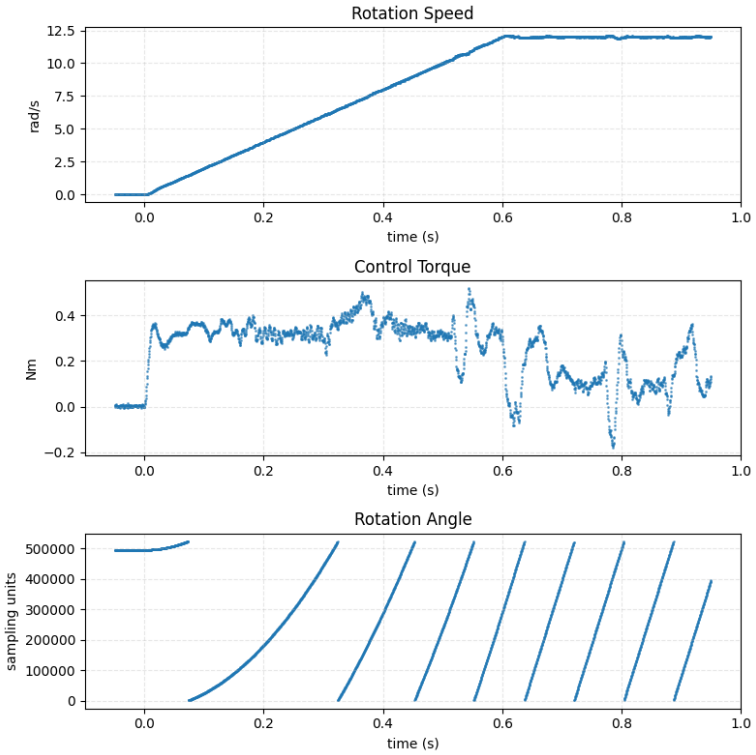


Figure 6–8: Constant Speed Movement in tooth interference scenario - target velocity of 1440 °/s, acceleration of 2400 °/s^2, and without jerk.

In summary, this chapter has described the hardware platform and experimental protocol underlying the data collection for this work. The test bench combines a B&R PMSM servo drive system with a synchronous belt transmission at a 1:3 pulley ratio. The only signals available for diagnosis are the motor control current and the integrated encoder angle, both read directly from the servo drive without any additional instrumentation. The experimental protocol, designed and carried out as part of this thesis, covers three system conditions, namely healthy, parallel misalignment, and tooth interference, each evaluated across fourteen motion profile configurations spanning variable-speed and constant-speed operation. This dataset forms the basis for the training and evaluation of both fault diagnosis methods presented in Chapter 4 and Chapter 5.

7 Result and Discussion

This chapter presents the results obtained from the experimental data described in Chapter 6 by applying the two proposed methods, namely the interpretation method based on rotation-speed-based fault diagnosis introduced in Chapter 4 and the interpretation method based on rotation-angle-based fault diagnosis introduced in Chapter 5. In addition, this chapter discusses the remaining limitations of these methods, potential improvements, and practical considerations that should be taken into account in real-world applications.

7.1 Result of Rotation-Speed-Based Interpretable Fault Diagnosis

This section will focus on presenting the implementation and results based on the method described in Chapter 4.

7.1.1 Data Preprocessing

The raw experimental data are acquired as comma-separated value (CSV) files, each corresponding to one continuous recording under a specific operating condition and system state. Every file contains five time-aligned channels sampled at 2500 Hz (sampling interval 0.4 ms): the velocity setpoint, the measured angular velocity, the q-axis current reference, the actual torque, and the raw encoder position. Of these, the speed-based method uses only the angular velocity $\omega(t)$ and the motor torque $T_m(t)$ as inputs to the PINN framework.

The preprocessing pipeline proceeds as follows. First, the motor current and torque signals are filtered using a two-stage scheme: a third-order zero-phase Butterworth low-pass filter with a cutoff frequency of 600 Hz, followed by a moving-average filter with a window size of 15 samples. The cutoff frequency is chosen to retain the highest electrical frequency component expected during operation, corresponding to the product of the maximum rotational speed and the number of pole pairs of the motor. The zero-phase implementation (forward–backward filtering) ensures that no phase distortion is introduced, which is important because the torque signal enters the physics residual **(4–1)** as a known input and any temporal shift would corrupt the residual evaluation. Second, time intervals during which the velocity setpoint is zero are removed, since the motor is stationary in these regions and the Coulomb friction term $T_c\ \mathrm{sgn}(\omega)$ in the governing ODE **(3–1)** is not designed to model the zero-speed regime. This step also eliminates the initial and final dwell periods present in some motion profiles. For files whose total duration exceeds 1.0 s after cleaning, only the first 1.0 s of data is retained, ensuring a consistent temporal window across all recordings. It's important to emphasize that current and torque are theoretically proportional; we are using torque, which is directly derived from the motor controller.

The normalization follows the per-trajectory, physics-aware scheme established in Section 3.3.3.

The time variable is mapped to $t^* \in [0, 1]$, the angular velocity and the motor torque are each divided by their respective per-trajectory standard deviations σ_ω and σ_{T_m}, and the mean is not subtracted so that the zero crossing of the speed signal is preserved. The resulting normalized signals $\omega^*(t^*)$ and $T_m^*(t^*)$ are passed to the PINN, where ω^* serves as the training target in the data loss **(4–6)** and T_m^* is supplied to the physics residual through radial basis function interpolation. The scaling factors σ_ω, σ_{T_m}, and t_s are recorded alongside each trajectory and used after training to recover the dimensional physical parameters via Equation (4–3).

7.1.2 Result of Rotation-Speed-Based Healthy System Identification

PINN Implementation

The inverse problem is solved using a physics-informed neural network implemented with the DeepXDE library [19]. The implementation details are described below in terms of network architecture, exogenous input handling, physics residual construction, training configuration, and the two-phase training strategy introduced in Section 4.2.

The network is a fully connected feedforward architecture with three hidden layers of 64 neurons each, using hyperbolic tangent activation functions and Glorot uniform initialization. The single input is the normalized time $t^* \in [0, 1]$ and the single output is the normalized angular velocity $\omega^*(t^*)$. Several alternative configurations with varying depths and widths were evaluated, including a two-hidden-layer variant with 256 neurons per layer; all yielded comparable identification accuracy, and the chosen architecture represents a balance between expressiveness and computational cost.

The normalized motor torque $T_m^*(t^*)$, which enters the physics residual **(4–1)** as a known exogenous input, is not predicted by the network but must be available at arbitrary query times, including at collocation points that do not coincide with the measurement grid. To this end, a thin-plate-spline radial basis function interpolant is fitted to the measured torque samples and used to evaluate T_m^* at any requested time point. This interpolant is passed to the DeepXDE framework as an auxiliary variable function, so that the torque value at each collocation point is automatically supplied to the residual during training.

The three scaled physical parameters J^*, B^*, and T_c^* are registered as trainable scalar variables, each initialized to 1.0. A positivity constraint is enforced throughout training by passing each parameter through the softplus function before it enters the physics residual, ensuring that the identified inertia, viscous friction coefficient, and Coulomb friction torque remain strictly positive as required by their physical definitions. The Coulomb friction term $\mathrm{sgn}(\omega^*)$ in the normalized residual **(4–1)** is approximated by $\tanh(20\,\omega^*)$ to maintain differentiability for gradient-based optimization.

The physics residual is evaluated at 500 collocation points uniformly distributed over the nor-

malized time domain $[0, 1]$. The data loss is enforced at all measurement points through a point-set boundary condition, and the initial condition is imposed separately at $t^* = 0$. The collocation grid is augmented with 500 additional anchor points equally spaced over $[0, 1]$ to ensure that the physics residual is evaluated at a sufficiently dense and regular set of time instants.

Training follows the two-phase strategy described in Section 4.2. In Phase I, only the data loss is active: the physics residual weight is set to zero and the physical parameters are excluded from optimization. The network is trained for $N_1 = 5\,000$ iterations using the Adam optimizer with a learning rate of 10^{-3} and a loss weight configuration of $(w_{\text{physics}}, w_{\text{IC}}, w_{\text{data}}) = (0, 0.001, 1)$. At the end of Phase I, the network provides a faithful approximation of the measured speed trajectory, and its automatic-differentiation derivative $\mathrm{d}\omega^*/\mathrm{d}t^*$ carries physically meaningful information.

In Phase II, the physics residual is activated and the three physical parameters are registered as external trainable variables optimized jointly with the network weights. The loss weights are set to $(w_{\text{physics}}, w_{\text{IC}}, w_{\text{data}}) = (0.001, 0.001, 1.0)$, keeping the data loss dominant while providing sufficient gradient signal from the physics residual to drive parameter convergence. Training continues for $N_2 = 20\,000$ iterations with the same Adam optimizer and learning rate. The evolution of J^*, B^*, and T_c^* is monitored every 1 000 iterations to verify convergence. After training completes, the dimensional parameters $\hat{J}$, $\hat{B}$, and $\hat{T}_c$ are recovered from their scaled counterparts via the inverse scaling relations in Equation (4–3) and saved for use in the subsequent fault diagnosis stage.

Healthy System Identification Result

The PINN was trained on a healthy constant speed movement profile at a target velocity of 1440 °/s, an acceleration of 2400 °/s², and without jerk limitation. The training converged successfully, yielding the following identified dimensional parameters:

$$\hat{J} = 1.75 \times 10^{-3}\ \mathrm{kg \cdot m^2}\ , \qquad \hat{B} = 2.22 \times 10^{-4}\ \mathrm{N \cdot m \cdot s/rad}\ , \qquad \hat{T}_c = 7.78 \times 10^{-2}\ \mathrm{N \cdot m}\ . \quad \textbf{(7–1)}$$

To assess the robustness of the identification, the PINN was additionally trained on a second representative working condition, namely a relative movement profile with a load-side rotation distance of $+720°$ and an acceleration of 10 000 °/s². The parameter estimates obtained from the two profiles are consistent within a 5 % relative deviation across all three parameters, confirming that the identified values are not artifacts of a particular motion profile but reflect the intrinsic physical properties of the system.

The identified parameters are subsequently validated by forward numerical simulation: the governing ODE **(3–1)** is solved as an initial value problem using a classical Fourth-Order Runge–Kutta (RK4) integrator, with the measured motor current as the torque input and the first measured speed sample as the initial condition. At each integration step, the torque is linearly

interpolated between adjacent measurement samples. The simulated speed trajectory is then compared against the measured trajectory using the Root Mean Square Error (RMSE) and the relative RMSE, defined as the ratio of the RMSE to the Root Mean Square (RMS) value of the measured speed. This validation is performed across all 70 healthy trajectories recorded under the full range of operating conditions described in Section 6.2, including both relative movement and constant speed profiles at various speed and acceleration settings. It is important to note that the numerical simulation introduces its own error sources, including the accumulation of integration error over long time horizons and the sensitivity of the explicit RK4 scheme to high-frequency fluctuations in the torque input. These numerical effects are inherent to the validation procedure and do not reflect inaccuracies in the identified parameters themselves; a detailed analysis of numerical integration error is outside the scope of this work.

Figure 7–1 shows a representative best-case result from the relative movement profile. The simulated and measured speed trajectories are virtually indistinguishable across the full acceleration–cruise–deceleration cycle, achieving a relative RMSE of 0.429 %. The close agreement during both the transient acceleration and deceleration phases indicates that the identified inertia $\hat{J}$ accurately captures the system's resistance to changes in rotational speed, while the agreement during the cruise phase confirms that the friction parameters $\hat{B}$ and $\hat{T}_c$ correctly describe the resistive torques acting on the shaft at constant velocity.

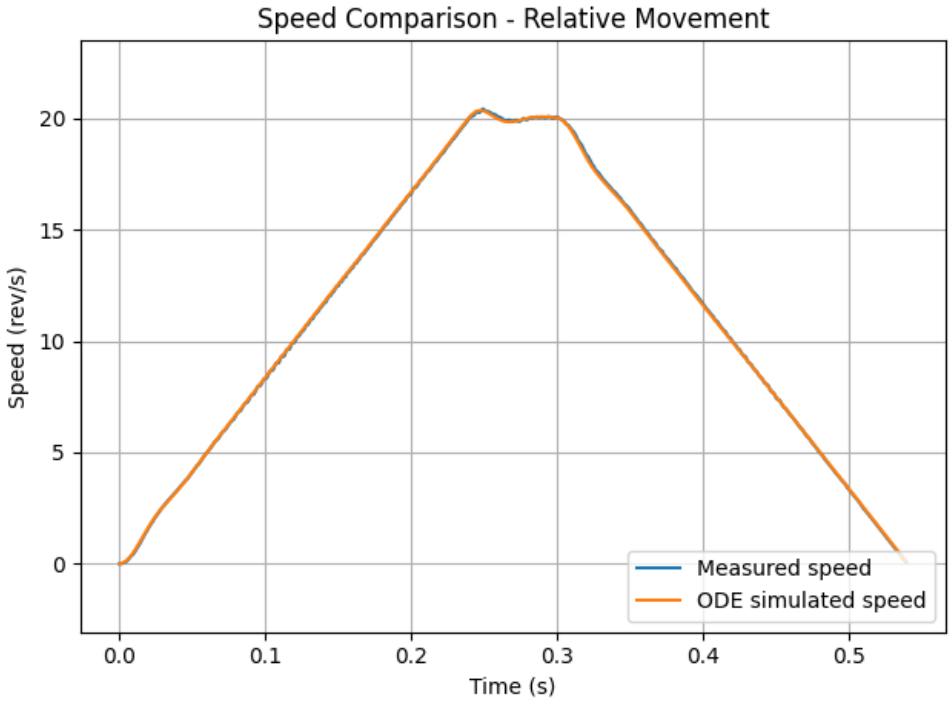


Figure 7–1: Forward simulation validation on a relative movement profile: load-side rotation distance of $+720°$, acceleration of 10 000 °/s², maximum velocity of 2400 °/s. Relative RMSE: 0.429 %.

Figure 7–2 illustrates a moderately degraded case from the constant speed movement profile. While the initial acceleration phase is well captured, the simulation error grows progressively

during the steady-state phase, with the relative error increasing from below 1 % near $t = 0.3\,\mathrm{s}$ to approximately 5 % by $t = 2.5\,\mathrm{s}$. The error exhibits a persistent positive bias overlaid with oscillatory fluctuations, indicating that the simulated speed drifts slightly above the measured value over time. This drift is characteristic of error accumulation in explicit numerical integration schemes applied over long time horizons: small per-step errors in the torque interpolation and the RK4 approximation compound monotonically, particularly during the constant speed phase where the net torque is small and the dynamics are sensitive to minor imbalances between the driving torque and the friction terms. The resulting relative RMSE of 3.552 % remains within an acceptable range for the purpose of establishing a healthy baseline.

The distribution of relative RMSE across all 70 healthy trajectories is summarized in Figure 7–3, ranked in descending order. The majority of trajectories fall below the 3 % threshold, with a substantial portion achieving relative RMSE below 1 %, demonstrating that the identified three-parameter model captures the dominant dynamics of the system across a wide range of operating conditions.

Among the small number of trajectories with relative RMSE exceeding 4 %, a common characteristic is observed: all such cases correspond to the constant speed movement profile at the lowest tested target velocity of 360 °/s on the load side, equivalent to 3 rev/s on the motor side. This operating point is well below the nominal speed of the motor. At such low rotational speeds, the measured speed signal has a small absolute magnitude, so that the signal-to-noise ratio is inherently poor. Furthermore, the uniform low-pass filtering pipeline applied across all experiments is designed for the full speed range and does not provide additional noise suppression tailored to this low-speed regime. As a result, the filtered speed signal at 360 °/s is likely still dominated by measurement noise and quantization artifacts from the drive's internal numerical differentiation, which propagate through the RK4 integrator and accumulate over the trajectory. The elevated simulation error in these cases is therefore attributed to limitations of the measurement and filtering at low speed rather than to inaccuracies in the identified physical parameters.

Overall, the forward simulation validation confirms that the identified parameters $\hat{J}$, $\hat{B}$, and $\hat{T}_c$ provide a reliable healthy baseline model for the subsequent fault diagnosis stage. The parameters are used without modification in the fault residual computation described in Section 4.3.

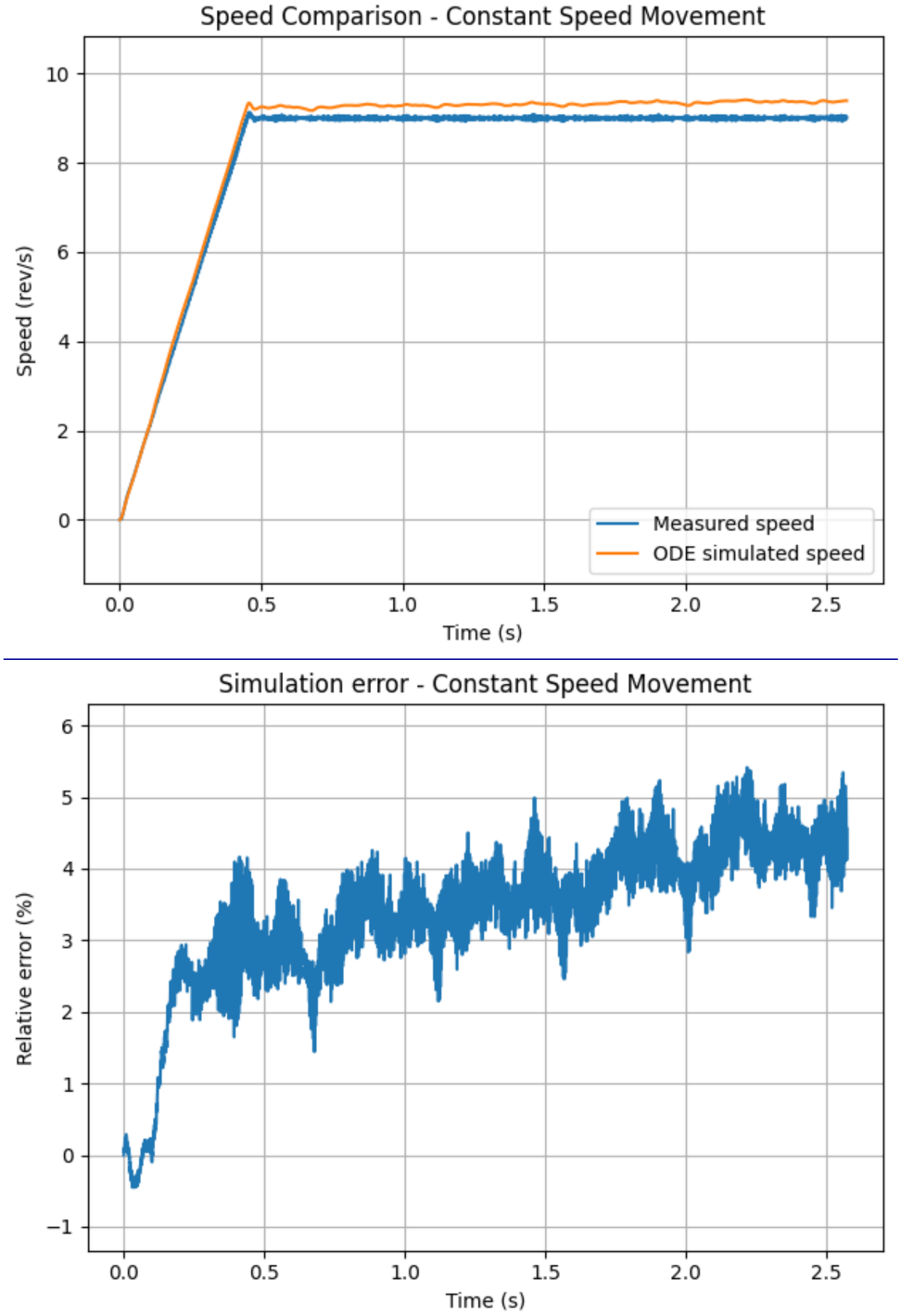


Figure 7–2: Forward simulation validation on a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s^2, without jerk limitation. Relative RMSE: 3.552 %. Top: speed comparison; bottom: relative simulation error.

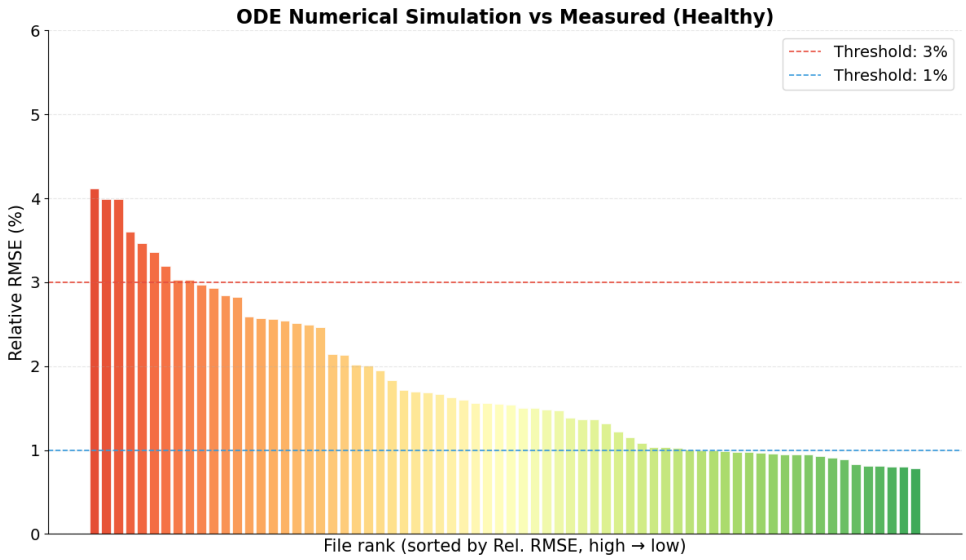


Figure 7–3: Relative RMSE of forward ODE simulation against measured speed for all 70 healthy trajectories, ranked in descending order. The red and blue dashed lines indicate thresholds of 3 % and 1 %, respectively.

7.1.3 Result of Rotation-Speed-Based Fault Diagnosis

The fault diagnosis is implemented using the PySINDy library [29], following the procedure described in Section 4.3.

The fault residual $R_f(t)$ defined in Equation (4–10) requires the measured angular acceleration $\dot{\omega}(t)$, which is not directly available and must be obtained by differentiating the measured rotation speed signal. A Savitzky–Golay filter [27] is applied for this purpose, performing simultaneous smoothing and first-order differentiation in a single pass with a window length of 51 samples and a third-order polynomial. This configuration provides sufficient smoothing to suppress the high-frequency noise amplified by differentiation while preserving the signal dynamics within the bandwidth of interest. The healthy model prediction $\dot{\omega}_{\mathrm{healthy}}(t)$ is then computed from Equation (4–9) using the identified parameters $\hat{J}$, $\hat{B}$, and $\hat{T}_c$, with the Coulomb friction signum approximated by $\tanh(20\,\omega)$ consistently with the PINN training. The fault residual is obtained as the pointwise difference between the measured and healthy-predicted accelerations.

The candidate fault library follows the structure defined in Equation (4–12). The Coulomb and polynomial terms operate on the measured rotation speed signal, while the Fourier and fractional-harmonic terms operate on the measured encoder angle. Both signals are supplied to the respective library groups through radial basis function interpolation of the measurement data, ensuring continuous evaluation at all sample points.

The STLSQ optimizer is configured with a sparsity threshold of $\lambda = 1.5$ and a ridge regularization parameter of $\alpha = 1.0$. Column normalization is enabled to prevent scale-dependent bias

in the thresholding step, ensuring that the threshold acts uniformly across library terms with different magnitudes. The SINDy model is fitted by supplying the two-column feature matrix $[\omega(t),\, \theta(t)]$ as the state variable and the fault residual $R_f(t)$ as the target derivative, bypassing PySINDy's internal numerical differentiation. As a preliminary validation step, the algorithm is first applied to data from the healthy scenario; the optimizer returns no active fault terms, confirming that the identified healthy baseline adequately explains the measured dynamics and that the fault residual does not contain spurious structure.

Fault Diagnosis Result

The STLSQ sparse regression is applied to the fault residual for each combination of fault scenario and motion profile. For each case, the identified sparse coefficient vector $\boldsymbol{\xi}$ is reported alongside a bar chart showing the absolute magnitudes of all library terms, allowing the active (non-zero) and inactive (pruned) terms to be distinguished at a glance. As a baseline check, the algorithm is first applied to data from the healthy scenario under both motion profiles; in both cases the optimizer returns an identically zero coefficient vector, confirming that the healthy baseline model adequately explains the measured dynamics and that the fault residual contains no spurious structure.

For the parallel misalignment scenario, the results are shown in Figure 7–4 (relative movement) and Figure 7–5 (constant speed movement). Under both motion profiles, the STLSQ algorithm identifies a single active term: the first-order cosine harmonic $\cos\theta$, with all remaining library entries pruned to zero. The identified coefficient magnitude is $|\xi| = 0.413$ under the relative movement profile and $|\xi| = 0.163$ under the constant speed profile. The consistency of the selected term across two fundamentally different motion profiles — one dominated by transient acceleration and deceleration, the other by steady-state dynamics — provides confidence that the identification reflects a genuine physical effect rather than an artifact of a particular operating condition.

For the tooth interference scenario, the results are shown in Figure 7–6 (relative movement) and Figure 7–7 (constant speed movement). Under the relative movement profile (Figure 7–6), the dominant active terms are the fractional-harmonic components at $\theta/3$ and $2\theta/3$, with coefficients reaching absolute magnitudes of 10.494 and 11.227 respectively — substantially larger than the integer-harmonic terms. The Coulomb term $\mathrm{sgn}(\omega)$ is also activated with a coefficient of 2.653, together with integer-harmonic terms at $\sin\theta$, $\cos\theta$, $\sin 2\theta$, $\cos 2\theta$, and $\sin 3\theta$ at moderate magnitudes. No polynomial terms in ω are selected. Under the constant speed profile (Figure 7–7), the fractional-harmonic and integer-harmonic terms remain active, but the identified model additionally includes the Coulomb term, all three polynomial terms (ω, ω^2, ω^3), and a broader set of integer-harmonic components. The total number of active terms increases from 9 under relative movement to 13 under constant speed operation.

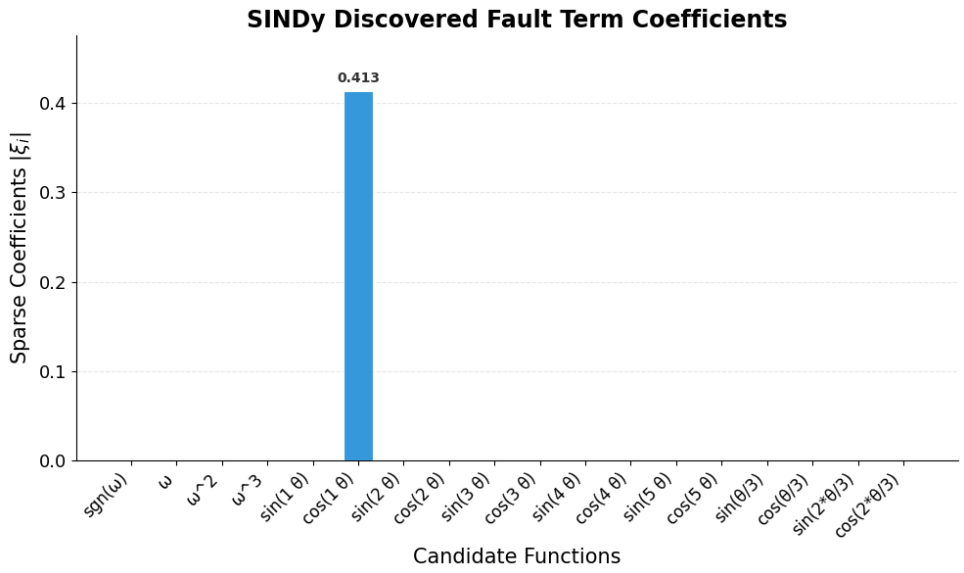


Figure 7–4: Identified fault coefficients for parallel misalignment under a relative movement profile: load-side rotation distance of $+720°$, acceleration of 10 000 °/s², maximum velocity of 2400 °/s.

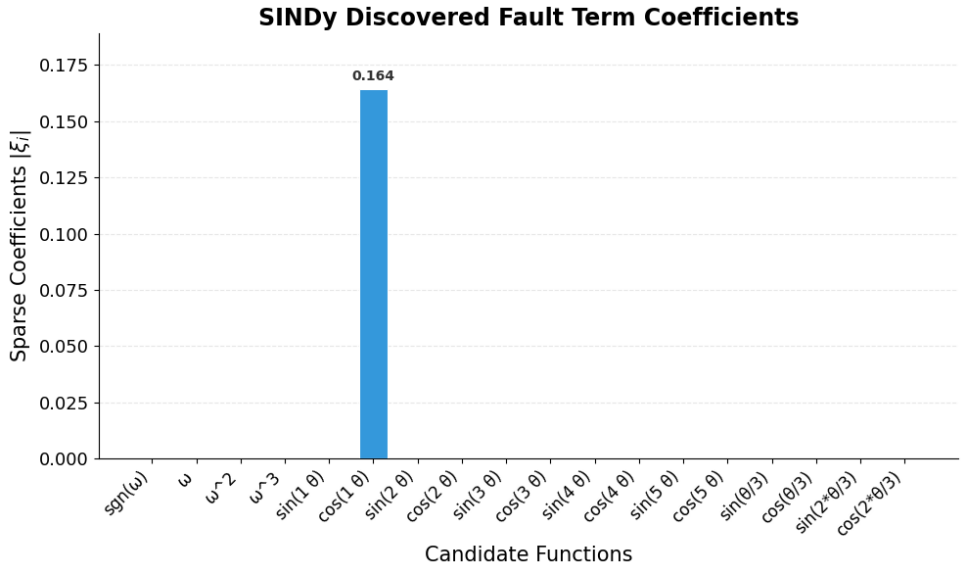


Figure 7–5: Identified fault coefficients for parallel misalignment under a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s², without jerk limitation.

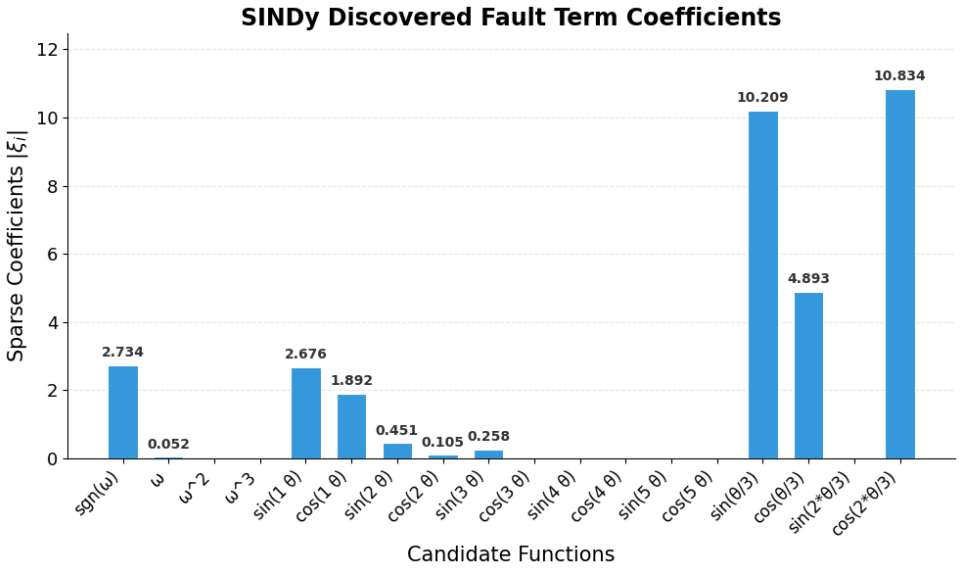


Figure 7–6: Identified fault coefficients for tooth interference under a relative movement profile: load-side rotation distance of $+720°$, acceleration of 10 000 °/s^2, maximum velocity of 2400 °/s.

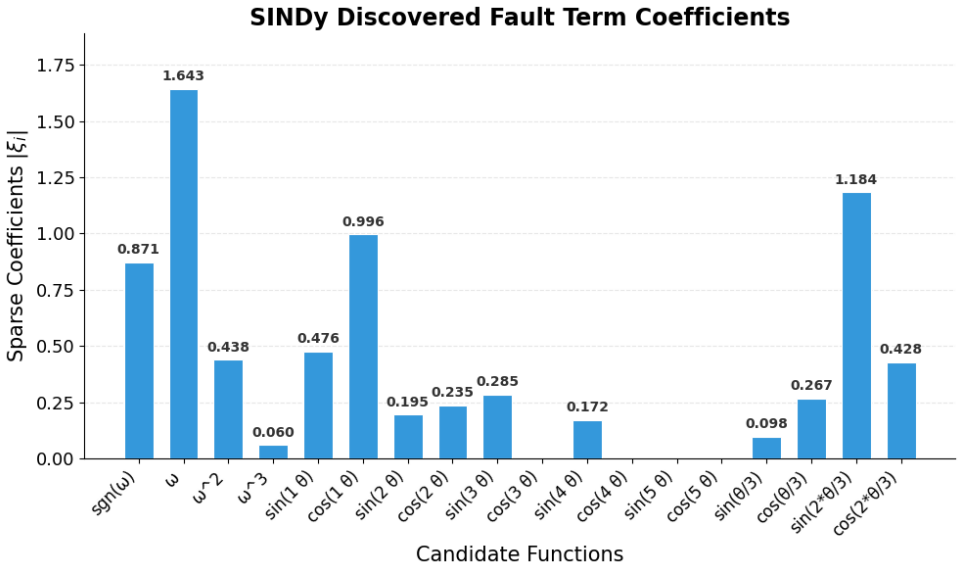


Figure 7–7: Identified fault coefficients for tooth interference under a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s^2, without jerk limitation.

Diagnosis Result Analysis

The physical interpretation of the identified fault terms is discussed below for each fault scenario.

For parallel misalignment, the exclusive identification of the first-order harmonic $\cos\theta$ is consistent with the established vibration signature of shaft misalignment in belt transmission systems. Prior spectral analysis studies [20, 28] have reported that parallel misalignment excites vibrations at the first, second, and third orders of the shaft frequency, though higher-order components are typically weaker and not always observable. In the present experiment, the applied misalignment was deliberately mild, producing only a small lateral offset between the drive and load shafts. Under this weak excitation, only the dominant first-order component exceeds the STLSQ sparsity threshold, while higher-order contributions remain below the detection level. The result demonstrates that the proposed framework can isolate the characteristic misalignment signature even at low fault severity, using only the signals available in the servo drive.

For tooth interference, the identified fault structure is richer and differs between the two motion profiles, reflecting the interplay between the mechanical fault excitation and the servo drive's closed-loop control behavior.

Under the relative movement profile (Figure 7–6), where transient acceleration and deceleration dominate, the most prominent terms are the fractional harmonics at $\theta/3$ and $2\theta/3$. These sub-harmonics correspond to the fundamental and second harmonic of the load-side shaft rotation, given the 1:3 transmission ratio between the motor and load pulleys: one revolution of the large pulley corresponds to three revolutions of the motor shaft, so that a localized disturbance repeating once per load-side revolution appears at a period of $3 \times 2\pi$ in the motor angle, corresponding to angular frequencies of $\theta/3$ and $2\theta/3$. This signature is consistent with the physical setup, in which the wire obstruction was placed at a fixed angular position on the load-side pulley. The activation of these fractional harmonics can be understood through the nonlinear belt stiffness model. Under tooth interference, the fault introduces a periodic perturbation Δx onto the nominal belt elastic displacement x_0:

$$x(\theta) = x_0 + \Delta x(\theta) \ , \quad \Delta x(\theta) = a\cos(n\theta) \ , \tag{7–2}$$

where a is the perturbation amplitude and n denotes the harmonic order referred to the motor shaft. The belt force $F(x)$ is nonlinear in displacement:

$$F(x) = k_1 x + k_2 x^2 + k_3 x^3 \ , \tag{7–3}$$

where k_1, k_2, and k_3 are the linear, quadratic, and cubic stiffness coefficients. Expanding around the nominal operating point, the fault-induced nonlinear perturbation force is:

$$F_{\text{nonlinear}} = F(x_0 + \Delta x) - F(x_0) \ . \tag{7–4}$$

Applying trigonometric identities to the powers of the cosine perturbation recovers harmonic components at $\theta/3$, $2\theta/3$, and θ [23, 36], which matches the structure identified by the sparse regression.

Under the constant speed profile (Figure 7–7), where steady-state operation dominates, the fractional and integer harmonics remain active, but the identified model additionally includes polynomial terms in ω up to third order. This difference is attributed to the closed-loop rotation speed regulation of the servo drive: during constant speed operation, the controller continuously modulates the motor current to reject disturbances and maintain the target velocity. The periodic torque impulses caused by the tooth interference induce rotation-speed-dependent corrections through the controller, which manifest as polynomial terms in ω within the fault residual. Under the relative movement profile, the large commanded accelerations dominate the torque signal and mask this closed-loop effect, resulting in a cleaner identification that isolates only the angle-periodic fault terms.

Overall, the identified fault terms remain physically consistent across scenarios and motion profiles, supporting the interpretability of the proposed speed-based framework. However, the reconstruction quality also exposes an inherent limitation of the two-step approach. As a representative example, Figure 7–8 presents the tooth interference scenario under the relative movement profile. The integrated trajectory of the complete identified dynamics matches the measured rotation speed with a relative RMSE of 1.83 %, which is acceptable but not negligible. More importantly, the comparison of measured and predicted angular acceleration reveals substantial discrepancies, particularly during the transient phases around $t = 0.25\,\mathrm{s}$ and $t = 0.30\,\mathrm{s}$, where the prediction clearly over- or undershoots the measured values.

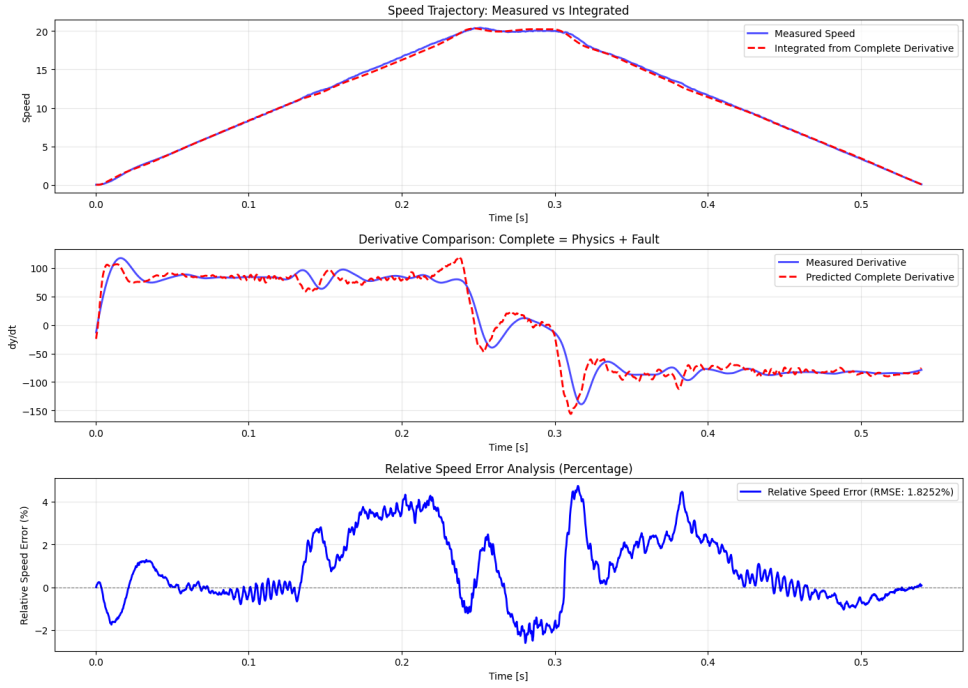


Figure 7–8: Validation of the identified fault model for tooth interference under the relative movement profile. Top: measured rotation speed versus the trajectory reconstructed by integrating the complete identified dynamics. Middle: measured angular acceleration versus the predicted complete derivative (healthy model plus sparse fault terms). Bottom: relative rotation speed error over time. Relative RMSE: 1.83 %.

This reconstruction error originates from the noise amplification chain inherent in the speed-based method. The angular acceleration $\dot{\omega}(t)$ used both for computing the fault residual **(4–10)** and as the regression target is obtained by numerically differentiating the measured rotation speed signal via Savitzky–Golay filtering. Although the filter suppresses high-frequency noise, the differentiation step inevitably amplifies measurement noise and quantization artifacts present in the rotation speed signal, which itself is already a numerical derivative of the encoder angle computed internally by the servo drive. The fault residual therefore inherits these amplified errors, and the STLSQ regression fits a sparse model to a target that is corrupted by differentiation noise. The result is that the identified coefficients are biased by the noise content of the derivative estimate, and the reconstructed dynamics deviate from the true measured behavior, particularly during transient phases where the acceleration changes rapidly and the Savitzky–Golay approximation is least accurate.

This fundamental dependence on numerical differentiation of the rotation speed signal, together with the resulting noise amplification, is one factor motivating the development of the rotation-angle-based method presented in Chapter 5, which eliminates all numerical differentiation by operating directly on the raw encoder angle and recovering the angular velocity and acceleration analytically through automatic differentiation within the neural network.

7.2 Result of Rotation-Angle-Based Interpretable Fault Diagnosis

This section will focus on presenting the implementation and results based on the method described in Chapter 5.

7.2.1 Data Preprocessing

The raw data acquisition, signal filtering, and removal of zero-velocity-setpoint intervals follow the same pipeline described in Section 7.1.1. The key difference lies in which signals are extracted and how they are normalized for the PINN.

The angle-based method uses the motor torque $T_m(t)$ and the encoder angle $\theta(t)$ as inputs to the PINN framework, rather than the angular velocity used in the speed-based method. The raw encoder position, recorded as an integer in the range 0 to 2^{19}, is converted to radians by a linear mapping. From this angle, the trigonometric projections $c(t) = \cos\theta(t)$ and $s(t) = \sin\theta(t)$ are computed, which serve as the two network outputs and training targets as established in Section 3.2.2. These projections are continuous and bounded in $[-1, 1]$ by construction and therefore require no additional scaling.

The normalization is applied per trajectory, following the principles of Section 3.3.3. The time variable is mapped to $t^* \in [0, 1]$ as in the speed-based method. The motor torque is divided by its per-trajectory standard deviation σ_{T_m}, again without subtracting the mean. The angu-

lar velocity standard deviation $\sigma_{\dot{\theta}}$, which is needed to normalize the velocity and acceleration terms in the physics residual **(5–2)**, is not available before training and is instead estimated at the transition between Phase I and Phase II of the two-phase training strategy, as described in Section 5.2. The scaling factors σ_{T_m}, $\sigma_{\dot{\theta}}$, and t_s are recorded for each trajectory and used after training to recover the dimensional physical parameters via Equation (5–8).

7.2.2 Result of Rotation-Angle-Based Healthy System Identification

PINN Implementation

The inverse problem is solved using a SIREN-based PINN implemented with the DeepXDE library [19]. The network takes normalized time $t^* \in [0, 1]$ as input and outputs the two trigonometric projections $(\hat{c},\, \hat{s}) = (\widehat{\cos\theta},\, \widehat{\sin\theta})$. The architecture follows the SIREN design described in Section 3.5, with three hidden layers of 64 neurons each. The frequency hyperparameter is set to $\omega_0 = 15.0$ for the first layer and $\omega_0 = 3.0$ for subsequent layers, with weight initialization following the scheme of Equation (3–24). The final output layer is linear with near-zero initialization. Several configurations with varying depths, widths, and frequency settings were evaluated; the chosen architecture balances representational capacity with training stability.

The motor torque is supplied as an exogenous input through a thin-plate-spline radial basis function interpolant, identically to the speed-based method. The three scaled physical parameters J^*, B^*, and T_c^* are registered as trainable scalars initialized to 1.0, with positivity enforced via the softplus function. The Coulomb friction signum is approximated by $\tanh(20\,\dot{\theta}^*)$. The angular velocity and acceleration entering the physics residual **(5–2)** are reconstructed analytically from the network outputs using Equations (3–6) and (3–8), with the required first- and second-order time derivatives of $\hat{c}$ and $\hat{s}$ computed via automatic differentiation through the network graph. The physics residual is evaluated at 500 collocation points, augmented with 500 equally spaced anchor points. The unit-circle constraint **(5–1)** is enforced as a separate residual term at the same collocation points.

Training follows the two-phase strategy of Section 5.2. In Phase I, the network is trained for $E_1 = 3\,000$ iterations on the data loss alone with loss weights $(w_{\cos},\, w_{\sin}) = (1.0,\, 1.0)$, using the Adam optimizer at a learning rate of 10^{-3}. At the phase transition, the angular velocity is sampled at 1 000 equally spaced points using the trained network, and $\sigma_{\dot{\theta}}$ is computed from these samples and held fixed thereafter. In Phase II, the physics and geometry residuals are activated and the physical parameters are optimized jointly with the network weights for $E_2 = 20\,000$ iterations, with loss weights $(w_{\text{physics}},\, w_{\text{geometry}},\, w_{\cos},\, w_{\sin}) = (0.001,\, 10^{-4},\, 1.0,\, 1.0)$. The parameter evolution is monitored every 1 000 iterations. After convergence, the dimensional parameters are recovered via Equation (5–8).

Healthy System Identification Result

The PINN was trained on the same healthy constant speed movement profile used in the speed-based identification (Section 7.1.2): a target velocity of 1440 °/s, an acceleration of 2400 °/s^2, and without jerk limitation. The training converged successfully, yielding the following identified dimensional parameters:

$$\hat{J} = 1.76 \times 10^{-3}\ \mathrm{kg \cdot m^2}\ , \qquad \hat{B} = 2.16 \times 10^{-4}\ \mathrm{N \cdot m \cdot s/rad}\ , \qquad \hat{T}_c = 7.90 \times 10^{-2}\ \mathrm{N \cdot m}\ . \quad \textbf{(7–5)}$$

These values are in close agreement with those obtained from the speed-based method ($\hat{J} = 1.75 \times 10^{-3}$, $\hat{B} = 2.22 \times 10^{-4}$, $\hat{T}_c = 7.78 \times 10^{-2}$), with relative deviations of less than 3 % for all three parameters. This consistency confirms that the two fundamentally different formulations — one operating on the rotation speed signal with a $\tanh$-activated feedforward network, the other on the trigonometric projections of the encoder angle with a SIREN architecture — converge to the same underlying physical properties of the system. The agreement is expected, since both methods solve the same inverse problem for the same governing ODE under healthy conditions; the two formulations differ only in the choice of state variable and differentiation strategy, which become consequential in the fault diagnosis stage rather than in the healthy identification.

The identified parameters are validated by the same forward simulation procedure described in Section 7.1.2: the governing ODE **(3–1)** is solved as an initial value problem using an RK4 integrator with the measured motor torque as input, and the simulated rotation speed is compared against the measured trajectory across all 70 healthy recordings. The distribution of relative RMSE is shown in Figure 7–9. The overall pattern is consistent with the speed-based result (Figure 7–3): the majority of trajectories fall below the 3 % threshold, with a substantial portion achieving below 1 %. The same low-speed outliers at 360 °/s are observed for the same reasons discussed previously. The comparable validation performance across both methods further confirms the reliability of the identified healthy baseline.

Since both methods yield essentially identical healthy parameters and comparable validation accuracy, the practical distinction between them lies not in the system identification stage but in the subsequent fault diagnosis stage, where the angle-based formulation offers the advantages of end-to-end sparse identification within the PINN framework and the elimination of numerical differentiation, as presented in the following subsection.

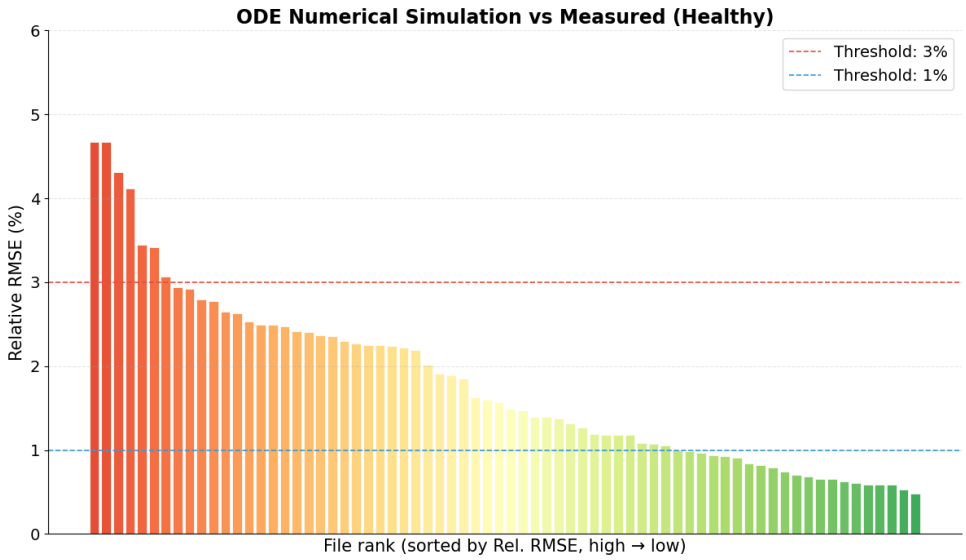


Figure 7–9: Relative RMSE of forward ODE simulation against measured rotation speed for all 70 healthy trajectories using the angle-based identified parameters, ranked in descending order. The red and blue dashed lines indicate thresholds of 3 % and 1 %, respectively.

7.2.3 Result of Rotation-Angle-Based Fault Diagnosis

PINNs-based Sparse Identification Implementation

The fault diagnosis is implemented within the same DeepXDE framework used for the healthy identification, following the two-stage procedure described in Section 5.3. The SIREN architecture, torque interpolation, and unit-circle constraint are identical to the healthy identification setup (Section 7.2.2), with the network re-initialized for each faulty trajectory. The key additions are the embedding of candidate fault terms into the physics residual and the alternating ℓ_0/ℓ_1 sparsity enforcement. The normalization in this stage differs from the healthy identification: the motor torque is scaled by its min-max range Δ_{T_m} rather than by its standard deviation, and the angular velocity and acceleration enter the residual in physical units, as described in Section 5.3.2. The identified healthy parameters $\hat{J}$, $\hat{B}$, and $\hat{T}_c$ from the preceding stage are loaded and held fixed throughout.

The candidate fault library follows the structure defined in Equation (5–11). Five cosine harmonics at frequencies 1, 2, 3, $1/3$, and $2/3$ of the shaft angle are included, each parameterized with a trainable amplitude γ_k and a trainable phase φ_k. Three speed-dependent terms are included: $\mathrm{sgn}(\dot{\theta})$, $\dot{\theta}$, and $\dot{\theta}^2$, each with a trainable coefficient. All amplitude coefficients are constrained to be non-negative. The amplitudes and phases are initialized to small values (10^{-3} and 0.1, respectively) to avoid biasing the optimization toward any particular fault structure at the start of training.

Stage I: PINN pre-training with candidate fault terms.

Training proceeds in two phases as described in Section 5.3.1. In Phase I, the network is trained on the data loss alone for up to $E_1 = 10\,000$ iterations with early stopping at a loss threshold of 10^{-6}, using the Adam optimizer at a learning rate of 10^{-3}. At the phase transition, the angular velocity is sampled at 1 000 equally spaced points to estimate the velocity range $\Delta_{\dot{\theta}}$. In Phase II, the augmented physics residual **(5–12)** is activated together with the geometry constraint. All network weights, fault amplitude coefficients, and phase parameters are optimized jointly for $E_2 = 22\,000$ iterations (2 000 minimum followed by up to 20 000 with early stopping at 4×10^{-6}), with loss weights $(w_{\text{physics}}, w_{\text{geometry}}, w_{\cos}, w_{\sin}) = (0.001, 10^{-4}, 1.0, 1.0)$. No sparsity penalty is applied in this stage.

Stage II: PINNs-based sequential thresholded sparse identification.

At the start of Stage II, the network weights and phase parameters are frozen, and only the fault amplitude coefficients remain trainable. The alternating ℓ_0/ℓ_1 procedure described in Section 5.3.1 is executed for $K = 3$ loops. In each loop, the ℓ_0 hard thresholding step applies two criteria: coefficients below a hard threshold of 10^{-4} are pruned unconditionally, and among the remaining coefficients, the three smallest that fall below a relative threshold of $|\bar{\gamma}|/20$ (where $|\bar{\gamma}|$ is the mean absolute magnitude of the active coefficients) are also pruned. Pruned coefficients are set to zero and permanently removed from the trainable set. The subsequent ℓ_1 retraining step optimizes the surviving coefficients for $E_3 = 3\,000$ iterations using the Adam optimizer at a learning rate of 10^{-3}, with an ℓ_1 penalty weight of $w_{\text{lasso}} = 0.005$ and loss weights $(w_{\text{physics}}, w_{\text{geometry}}, w_{\cos}, w_{\sin}, w_{\ell_1}) = (1.0, 10^{-4}, 1.0, 1.0, 0.005)$. After the final loop, the surviving non-zero coefficients and their associated frozen phases constitute the identified fault signature.

Fault Diagnosis Result

The PINNs-based sparse identification is applied to each combination of fault scenario and motion profile described in Section 6.2. As in the speed-based evaluation, the algorithm is first applied to healthy data; all fault coefficients are pruned to zero after the alternating ℓ_0/ℓ_1 procedure, confirming that the healthy baseline leaves no residual structure for the sparse identification to capture.

For the parallel misalignment scenario, the results are shown in Figure 7–10 (relative movement) and Figure 7–11 (constant speed movement). Under the relative movement profile (Figure 7–10), three angle-periodic terms survive the sparsity pruning: $\cos(\theta + \varphi_0)$, $\cos(2\theta + \varphi_1)$, and $\cos(3\theta + \varphi_2)$, with comparable magnitudes on the order of 10^{-3}. Under the constant speed profile (Figure 7–11), only the first-order harmonic $\cos(\theta + \varphi_0)$ is retained, with a coefficient magnitude of approximately 0.042. In both cases, all speed-dependent terms — $\text{sgn}(\dot{\theta})$, $\dot{\theta}$, and

$\dot{\theta}^2$ — as well as the fractional-harmonic terms at $\theta/3$ and $2\theta/3$ are pruned to zero. The identified fault signature is therefore exclusively composed of integer-order angle-periodic terms, consistent with a disturbance that repeats once per motor revolution as expected from the misalignment configuration described in Section 6.2, where a shim inserted beneath one bearing tilts the load-side shaft and produces a periodic lateral belt deflection force at the motor shaft frequency.

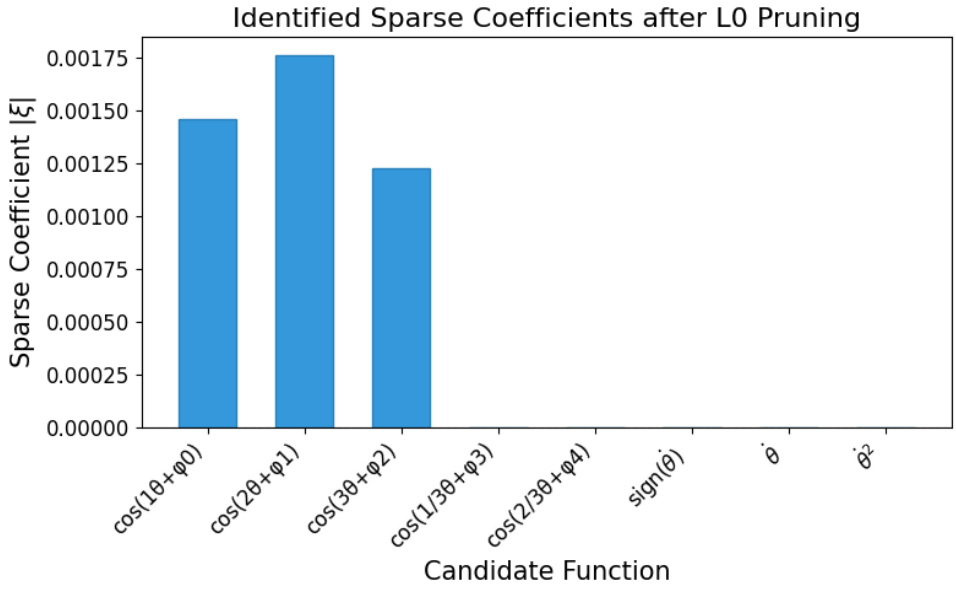


Figure 7–10: Identified fault coefficients for parallel misalignment under a relative movement profile: load-side rotation distance of $+720°$, acceleration of 10 000 °/s², maximum velocity of 2400 °/s.

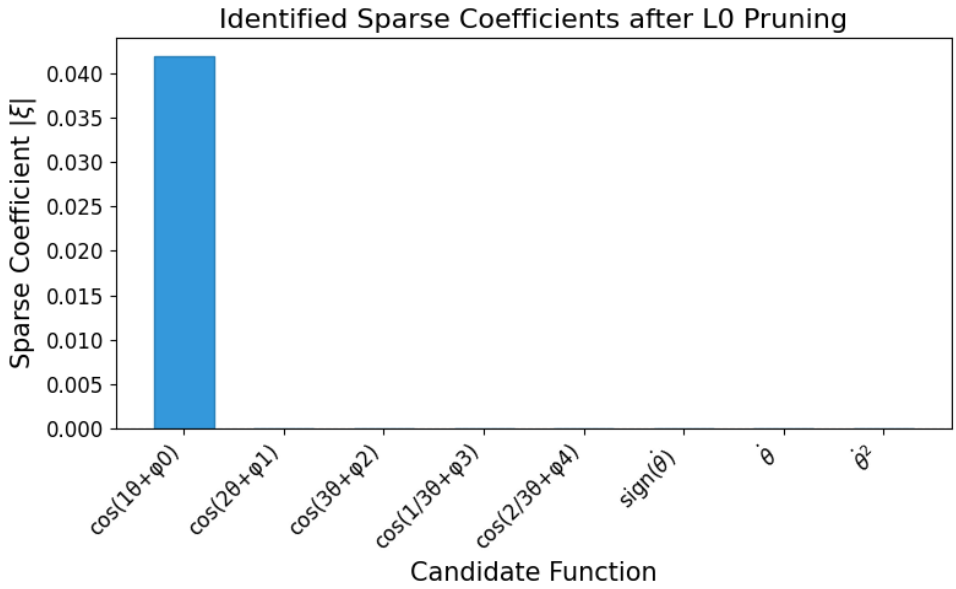


Figure 7–11: Identified fault coefficients for parallel misalignment under a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s², without jerk limitation.

For the tooth interference scenario, the results are shown in Figure 7–12 (relative movement)

and Figure 7–13 (constant speed movement). Under the relative movement profile (Figure 7–12), two terms are retained: the first-order harmonic $\cos(\theta + \varphi_0)$ and the fractional harmonic $\cos(\theta/3 + \varphi_3)$. Under the constant speed profile (Figure 7–13), the surviving terms are $\cos(2\theta + \varphi_1)$ and $\cos(2\theta/3 + \varphi_4)$, with the fractional harmonic exhibiting the largest coefficient magnitude of approximately 0.10. In both cases, all speed-dependent polynomial terms are pruned to zero, and the identified fault structure consists exclusively of angle-periodic terms. The presence of fractional harmonics at $\theta/3$ or $2\theta/3$ is the distinguishing feature compared to the misalignment scenario: these terms correspond to disturbances repeating once per load-side revolution, which is consistent with the wire obstruction placed at a fixed angular position on the large pulley as described in Section 6.2. Given the 1:3 transmission ratio, the obstruction engages with the belt every third motor revolution, producing the characteristic sub-harmonic signature at multiples of $\theta/3$.

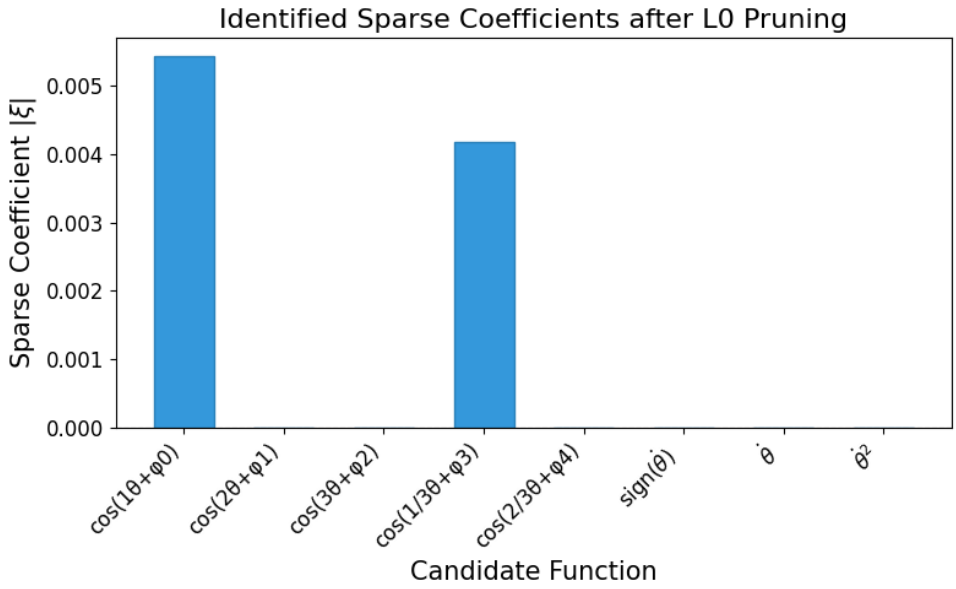


Figure 7–12: Identified fault coefficients for tooth interference under a relative movement profile: load-side rotation distance of $+720°$, acceleration of 10 000 °/s², maximum velocity of 2400 °/s.

A notable observation across all four cases is the sparsity of the identified models: the angle-based method retains between one and three active terms per scenario, compared to the speed-based method which retained up to 13 terms for the tooth interference case under constant speed operation (Figure 7–7). Furthermore, no speed-dependent polynomial terms appear in any of the angle-based results, whereas the speed-based method identified Coulomb and polynomial terms in several cases. This substantially sparser identification is a direct consequence of the end-to-end formulation: by embedding the fault terms into the physics residual and optimizing them jointly with the network within the automatic-differentiation framework, the angle-based method avoids the noise amplification inherent in the numerical differentiation of the rotation speed signal, enabling the sparsity-promoting procedure to prune terms that were retained in the speed-based method due to fitting differentiation noise rather than genuine fault dynamics.

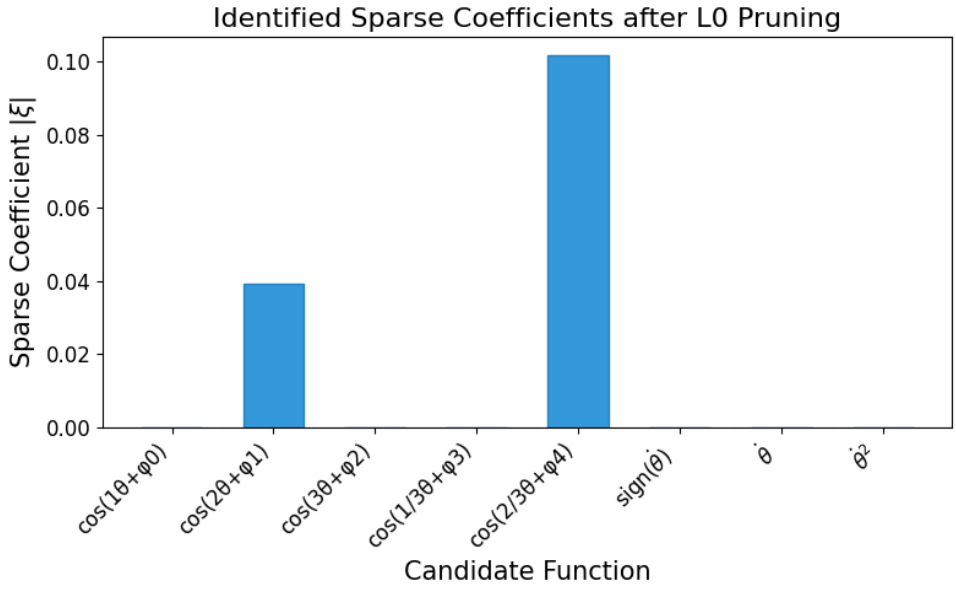


Figure 7–13: Identified fault coefficients for tooth interference under a constant speed movement profile: target velocity of 1440 °/s, acceleration of 2400 °/s^2, without jerk limitation.

Diagnosis Result Analysis

The physical interpretation of the identified fault terms follows the same reasoning established in the speed-based analysis (Section 7.1.3); this subsection focuses on the differences in the identified fault structure and the practical considerations specific to the angle-based method.

For parallel misalignment, the constant speed result identifies the same single first-order harmonic $\cos(\theta + \varphi_0)$ as the speed-based method, consistent with the expected vibration signature of mild shaft misalignment discussed previously. Under the relative movement profile, however, the angle-based method additionally resolves the second- and third-order harmonics $\cos(2\theta+\varphi_1)$ and $\cos(3\theta+\varphi_2)$, which were not detected by the speed-based STLSQ. This is consistent with the spectral analysis literature [20, 28], which predicts that parallel misalignment excites vibrations at the first, second, and third orders of the shaft frequency. The ability to resolve these higher-order components, despite the deliberately mild misalignment applied in the experiment, suggests that the angle-based method achieves higher sensitivity to weak fault signatures by avoiding the noise amplification that obscures small terms in the speed-based approach.

For tooth interference, the fractional harmonics at $\theta/3$ or $2\theta/3$ are consistently activated across both motion profiles, confirming the load-side origin of the fault through the 1:3 transmission ratio, as discussed in the speed-based analysis. The physical mechanism — nonlinear belt stiffness generating sub-harmonic components from a localized periodic perturbation — remains the same as described in Section 7.1.3. A notable difference is that the angle-based method identifies no polynomial terms in $\dot{\theta}$ under either motion profile, whereas the speed-

based method activated polynomial terms up to third order under constant speed operation. The absence of these terms indicates that the end-to-end formulation, by eliminating the numerical differentiation chain, avoids conflating the closed-loop controller response with the fault-induced dynamics, yielding a cleaner separation between the mechanical fault signature and the drive's regulation behavior.

A practical consideration for the angle-based method is the sensitivity of the identified fault terms to training convergence. Reliable fault identification requires that the data loss converges to the order of 10^{-6} and the physics loss reaches the 10^{-4} level. When both conditions are met, the identified fault structure is stable and reproducible across repeated runs with different random initializations. If the physics loss plateaus above this level — for instance around 10^{-3} — the fault coefficients become less predictable and may vary between runs due to the inherent stochasticity of neural network training (random weight initialization and mini-batch sampling). In practice, the majority of trajectories in this work achieved the required convergence levels. For the small number of segments containing particularly aggressive transients where the physics loss could not be reduced below approximately 5×10^{-4}, the identified fault terms remained qualitatively consistent (the same candidate terms were selected) but exhibited slightly larger variability in the coefficient magnitudes. Monitoring the data and physics loss values therefore serves as a reliability indicator for the identified fault signature, as noted in Section 5.3.3.

7.3 Discussion of Diagnostic Results

This section synthesizes the findings from both methods, discusses practical insights gained during implementation and experimentation, and addresses the current limitations of the work.

7.3.1 Comparative Assessment of the Two Methods

Both the rotation-speed-based and the rotation-angle-based methods successfully identify the healthy system parameters $\hat{J}$, $\hat{B}$, and $\hat{T}_c$ with mutual agreement within 3 %, and both yield forward simulation errors below 3 % for the majority of the 70 healthy trajectories. The healthy system identification stage therefore performs equivalently in both formulations, confirming that the underlying inverse problem is well-posed and that the PINN framework reliably recovers the physical parameters regardless of whether the state variable is the angular velocity or the angular position.

The decisive differences emerge in the fault diagnosis stage. The speed-based method, which applies STLSQ as a standalone post-processing step on a numerically differentiated fault residual, suffers from two compounding sources of noise: the angular velocity signal is itself a numerical derivative of the encoder angle computed internally by the servo drive, and the fault residual requires a further differentiation via Savitzky–Golay filtering. This double differentia-

tion amplifies measurement noise and quantization artifacts, causing the sparse regression to retain terms that fit noise rather than genuine fault dynamics. The consequence is visible in the tooth interference results, where the speed-based method identifies up to 13 active terms under constant speed operation (Figure 7–7), including polynomial terms in ω that reflect the servo controller's closed-loop response rather than the mechanical fault itself. The reconstruction error shown in Figure 7–8 further illustrates this limitation: the predicted angular acceleration deviates substantially from the measured values during transient phases, indicating that the identified fault model is corrupted by differentiation noise.

The angle-based method resolves these issues through three design choices. First, by predicting the trigonometric projections $(\cos\theta, \sin\theta)$ and recovering all kinematic quantities via automatic differentiation, the entire numerical differentiation chain is eliminated. Second, by embedding the candidate fault terms directly into the physics residual and optimizing them jointly with the network, the fault identification operates on the continuous, smooth network output rather than on discrete noisy measurements. Third, the alternating ℓ_0/ℓ_1 sparsity scheme provides more aggressive and robust pruning than the single-pass STLSQ used in the speed-based method. The result is a substantially sparser identification: the angle-based method retains between one and three active terms per scenario, all of which are angle-periodic and physically interpretable. No speed-dependent polynomial terms appear in any case, confirming that the end-to-end formulation cleanly separates the mechanical fault signature from the drive's regulation behavior.

For parallel misalignment, the angle-based method additionally resolves the second- and third-order harmonics under the relative movement profile, which the speed-based method fails to detect. For tooth interference, the fractional harmonics at $\theta/3$ and $2\theta/3$ are consistently identified by both methods, but the angle-based results are free of the spurious polynomial and Coulomb terms that contaminate the speed-based identification. Overall, the angle-based method yields more accurate, more robust, and more physically consistent fault signatures across all tested scenarios and motion profiles.

7.3.2 Practical Insights from Implementation

The development and experimental evaluation of both methods revealed several practical considerations that are important for reliable application and that may guide future implementations.

Network Architecture and Activation Functions

The choice of activation function has direct consequences for both the network's representational capacity and its compatibility with the physics residual. In the speed-based method, the $\tanh$ activation is well suited because its output range naturally matches the normalized

time input $t^* \in [0, 1]$ and the normalized speed output, both of which are of order unity. In the angle-based method, the trigonometric projections $(\cos\theta, \sin\theta)$ are inherently periodic and oscillatory, making SIREN the natural architectural choice: the sinusoidal activations are structurally aligned with the target signals, and the derivatives of the network output remain within the same functional class, ensuring that the automatic-differentiation derivatives used in the physics residual are smooth and well-behaved.

However, the SIREN configuration requires careful tuning. An overly expressive SIREN, with excessively high frequency hyperparameters ω_0 or too many hidden layers, can fit the measurement data with very high fidelity but produce derivatives that are dominated by high-frequency oscillations unrelated to the true physics. In such cases, the physics residual becomes noisy and the fault identification loses reliability. The network should be expressive enough to faithfully represent the trigonometric signals and their dominant frequency content, but not so expressive that it overfits to noise in the data, since any spurious structure in the network output propagates directly into the physics residual through automatic differentiation.

Loss Balancing and Gradient Dynamics

A recurring and initially counterintuitive finding is that assigning a large weight to a particular loss component does not necessarily cause that component to decrease faster. In PINNs, the data loss and the physics loss can produce competing gradient directions, particularly during early training. The root cause is as follows: the physics residual $\mathcal{R}^*$ is a function of both the network output and its automatic-differentiation derivatives, which in turn depend on the network weights ϕ. When the data loss has not yet converged, that is, when the network output deviates significantly from the measured trajectory, the derivatives $\mathrm{d}\omega^*/\mathrm{d}t^*$ or $\mathrm{d}^2\hat{c}/\mathrm{d}t^{*2}$ reflect the current (incorrect) weight configuration rather than the true system kinematics. The physics residual then evaluates the governing ODE on an incorrect trajectory, and its gradients $\nabla_\phi \mathcal{L}_{\text{physics}}$ drive the network weights toward reducing this incorrectly evaluated residual. Since these gradients are generally not aligned with $\nabla_\phi \mathcal{L}_{\text{data}}$, the two objectives interfere: the physics loss pulls the weights away from the data-fitting direction, which in turn increases the data loss, further degrading the trajectory that the physics residual is evaluated on. This feedback loop can prevent convergence entirely when the physics loss weight is large relative to the data loss weight in the early training phase.

The two-phase training strategy adopted in both methods directly addresses this issue. Phase I trains the network exclusively on the data loss until the network output closely approximates the measured trajectory and the data loss has reached a sufficiently low level ($< 10^{-4}$). At this point, the automatic-differentiation derivatives correspond to the kinematics of the measured trajectory rather than to an arbitrary function, and the physics residual gradients carry information that is consistent with the governing ODE. Phase II then activates the physics loss at a moderate weight, ensuring that the two gradient contributions reinforce rather than oppose each other.

Beyond the balance between data and physics losses, the magnitude of individual terms within the physics residual itself is equally important. In standard neural network practice, attention is typically focused on the scale of inputs and outputs, but in PINNs the internal structure of the residual matters just as much. If the terms in the governing ODE span different orders of magnitude, for instance, if the inertial torque $J\ddot{\theta}$ is orders of magnitude larger than the Coulomb friction torque $T_c \operatorname{sgn}(\dot{\theta})$, the optimizer prioritizes the dominant terms and the smaller parameters remain poorly identified. The normalization strategy adopted in this work addresses this by scaling all signals so that the residual terms are of comparable magnitude, rather than converting to non-dimensional form and losing the direct physical correspondence. This physics-aware normalization is essential for reliable multi-parameter identification and extends naturally to the fault diagnosis stage, where the balance between the healthy dynamics block and the candidate fault terms in the augmented residual determines whether small fault contributions can be detected.

Fault Library Design

The parameterization of the candidate fault library differs between the two methods in a way that has practical consequences for the reliability of the sparse identification.

In the speed-based method, the PySINDy library represents each angular frequency by a pair of sine and cosine terms ($\sin f\theta$ and $\cos f\theta$). This is a structural constraint of the PySINDy framework: the library entries are fixed basis functions evaluated on the data, and there is no mechanism to introduce trainable phase parameters that could be updated during the regression. A sinusoidal disturbance at frequency f with an arbitrary phase φ is therefore decomposed into $a \sin f\theta + b \cos f\theta$, and the STLSQ algorithm selects or prunes each coefficient independently. The problem arises when the physical disturbance has a phase such that one of the two components carries a small amplitude. For example, if the true fault term is $A\cos(f\theta + \varphi)$ with φ close to zero, the sine component $a = -A\sin\varphi$ is small and may fall below the sparsity threshold at some iteration, while the cosine component $b = A\cos\varphi$ is retained. Once the sine term is pruned, the remaining cosine term alone cannot represent the correct phase, and the identified amplitude is biased. This is not a tuning issue but a structural consequence of using fixed sine-cosine pairs without phase degrees of freedom.

In the angle-based method, each frequency is represented as a single cosine with a trainable phase, $\gamma_k \cos(f_k\theta + \varphi_k)$, which is mathematically equivalent to the sine-cosine pair but consolidates the phase information into a single parameter. However, this parameterization introduces its own constraint on the sparse identification procedure. In the alternating ℓ_0/ℓ_1 Stage II, if the sine-cosine representation were used instead, the same problem as in the speed-based method would reappear in a more severe form: the ℓ_0 hard thresholding would prune the smaller of the two components, and in the subsequent ℓ_1 retraining loop, the surviving terms would attempt to compensate for the lost phase information by redistributing amplitude among other retained library entries, corrupting the identification of neighboring frequencies. With the

cosine-plus-phase parameterization, this failure mode is avoided by design. The phase φ_k converges during Stage I to align with the angular location of the measured disturbance, and is then frozen at the transition to Stage II. Each candidate term thereby becomes a fixed basis function, analogous to the fixed library columns in classical STLSQ, whose shape does not change during the sparsity selection. The ℓ_0/ℓ_1 procedure then operates exclusively on the amplitude γ_k, making the pruning decision unambiguous: a term is either physically present (non-zero amplitude) or absent, without the risk of partial cancellation between coupled sine and cosine coefficients.

The scale of the fault terms relative to the healthy dynamics block in the augmented residual is also critical. If the candidate terms are poorly scaled, small but physically genuine fault contributions may be masked by the residual of the healthy dynamics, and the sparsity-promoting procedure prunes them as insignificant. The min-max torque normalization adopted in the fault diagnosis stage ensures that the healthy dynamics block has a consistent magnitude across operating conditions, making the fault coefficients directly comparable between segments and enabling the threshold parameters to be set once rather than adjusted per trajectory.

Sparse Identification: STLSQ, STRidge, and Embedded Sparsity

In the PySINDy implementation used for the speed-based method, the distinction between STLSQ and Sequentially Thresholded Ridge Regression (STRidge) is controlled by the ridge regularization parameter α: setting $\alpha = 0$ yields the original STLSQ algorithm of Brunton et al. [6], which alternates between ordinary least squares and hard thresholding, while $\alpha > 0$ introduces ridge regularization into each least-squares step, producing the STRidge variant. The ridge penalty improves numerical conditioning when the library columns are nearly collinear, but it also biases all coefficients towards zero, which can interact unpredictably with the sparsity threshold λ.

The angle-based method sidesteps this issue by embedding the sparse identification into the PINN training loop. The ℓ_1 Lasso penalty used in Stage II serves a role analogous to ridge regularization in providing continuous shrinkage, but it promotes sparsity rather than merely regularizing the solution. A practical subtlety is that the ℓ_1 penalty competes with the physics residual loss: the Lasso pushes coefficients towards zero while the physics loss pushes them towards values that minimize the ODE residual. This competition is the reason why the network weights must be frozen in Stage II. If the network were free to adapt, its high expressive capacity would allow it to absorb the contributions of pruned fault terms into its own parameters, undermining the sparse identification. Freezing the network decouples the trajectory representation from the coefficient optimization, ensuring that the sparsity-promoting updates act directly on the fault amplitudes without interference from the network's representational flexibility.

Data Segment Selection for Healthy Identification

The quality of the identified healthy parameters depends critically on the choice of data segment used for training. In this work, a constant speed profile at a sufficiently high velocity (1440 °/s on the load side) was used, providing a trajectory with high signal-to-noise ratio that exercises both the inertial and friction terms of the governing ODE during the acceleration phase and the steady-state phase, respectively. In an industrial deployment, this corresponds to the need for a dedicated system identification experiment, that is, a controlled motion profile executed under known healthy conditions before the system enters regular operation. This requirement is well established in the drive technology literature: parameter identification and self-commissioning procedures for AC motor drives [34] and identification of system parameters in magnetically supported PMSM systems [38] both rely on carefully designed excitation sequences to ensure that the relevant dynamic modes are sufficiently excited for reliable parameter estimation. Integrating the proposed PINN-based identification into such a self-commissioning phase would provide the healthy baseline parameters without requiring additional effort from the operator, provided that the motion profile is designed to excite the relevant dynamic modes of the system.

7.3.3 Current Limitations

Despite the demonstrated effectiveness of both methods, several limitations constrain the current state of the work.

First, all experiments were conducted offline: the measured data were recorded, preprocessed, and analyzed after the fact, with no real-time diagnostic capability. The PINN training, particularly for the angle-based method with its two-stage fault identification procedure, requires computation times on the order of minutes per trajectory on a standard workstation, which is not compatible with the millisecond-scale control loop of the servo drive. Deploying the framework in an online setting would require either significant acceleration of the training process, for instance through transfer learning from previously trained networks or through reduced-order surrogate models, or a hybrid architecture in which the healthy parameters are identified offline during commissioning and only the fault detection is performed online using a lightweight residual monitor.

Second, the experimental validation was limited to a single test bench with a single belt transmission configuration. To avoid risking damage to the available hardware, only two relatively mild fault conditions were implemented: a small parallel misalignment introduced by a thin shim, and a localized tooth interference simulated by wire in three grooves. More severe fault conditions, such as large misalignment offsets, extensive tooth damage, belt wear, or bearing degradation, as well as combined fault scenarios where multiple faults occur simultaneously, were not tested. The generalizability of the identified fault signatures to different transmission ratios, belt types, load configurations, and fault severities remains to be established through

broader experimental campaigns.

Third, the hyperparameters of both methods, including network architecture, loss weights, SIREN frequency parameters, sparsity thresholds, and the number of alternating ℓ_0/ℓ_1 loops, were tuned empirically for the specific test bench and operating conditions used in this work. While the general framework is transferable, the specific parameter settings may require re-tuning for different systems, and no systematic hyperparameter optimization was performed.

8 Conclusion

This thesis presented two physics-informed and interpretable fault diagnosis frameworks for PMSM-driven transmission systems using only electrical current and encoder measurements. The proposed methods identify first the healthy system via PINN-based parameter estimation and afterwards recognizes additional fault relevant terms by sparse regression. By separating healthy system identification from fault term discovery, it enables physically consistent and explainable diagnosis.

8.1 Summary

The main work of this thesis is briefly reviewed by chapter as follows.

Chapter 2 reviewed existing fault diagnosis approaches for rotating machinery and transmission systems, covering signal-based, model-based, data-driven, and physics-informed machine learning methods. The review identified three open challenges that motivate this work: the reliance of most diagnostic methods on dedicated sensors beyond the standard servo drive instrumentation, the lack of physical interpretability in purely data-driven approaches, and the limited applicability of frequency-domain methods under variable-speed operation.

Chapter 3 established the theoretical foundations underlying both proposed methods. The rotational dynamics of the PMSM-driven belt transmission were formulated as a first-order ODE in angular velocity and as a second-order ODE in angular position. A trigonometric representation of the encoder angle was introduced to eliminate the 2π-wrapping discontinuities inherent in the raw encoder signal, together with analytic expressions for recovering the angular velocity and acceleration from the continuous projections $(\cos\theta, \sin\theta)$. The chapter further presented the PINNs framework for inverse parameter identification, the physics-aware normalization strategy for balanced multi-parameter estimation, the SINDy/STLSQ sparse regression methodology, and the SIREN architecture for representing oscillatory signals.

Chapter 4 developed the first fault diagnosis method, formulated in terms of the angular velocity. A $\tanh$-activated feedforward network predicts the speed signal, and a two-phase training strategy first fits the measured trajectory and then jointly identifies the healthy physical parameters J, B, and T_c through the normalized physics residual. The fault diagnosis stage computes a fault residual by subtracting the healthy model prediction from the numerically differentiated measured acceleration and applies STLSQ sparse regression over a physics-motivated candidate library comprising Coulomb, polynomial, and integer- and fractional-order Fourier terms.

Chapter 5 developed the second fault diagnosis method, formulated directly in terms of the angular position. A SIREN-based network predicts the trigonometric projections $(\cos\theta, \sin\theta)$, from which the angular velocity and acceleration are recovered via automatic differentiation,

bypassing all numerical differentiation. The healthy system identification follows the same two-phase inverse problem structure with an additional phase-transition normalization step. The fault diagnosis stage embeds candidate fault terms with trainable amplitudes and phases directly into the augmented physics residual, and enforces sparsity through an alternating ℓ_0/ℓ_1 scheme within the PINN training loop, yielding an end-to-end fault identification without separate post-processing.

Chapter 6 described the B&R PMSM servo test bench with its 1:3 synchronous belt transmission and detailed the experimental protocol designed and carried out as part of this work. Three system conditions (healthy, parallel misalignment, and tooth interference) were tested across fourteen motion profile configurations spanning relative movement and constant speed operation at varying speed and acceleration settings, using exclusively the motor current and encoder angle available from the servo drive.

Chapter 7 presented the implementation, results, and comparative analysis of both methods. Both methods reliably identified the healthy system parameters with close mutual agreement and produced accurate forward simulation results across the tested operating conditions. In the fault diagnosis stage, the speed-based method correctly captured the dominant fault signatures but retained a large number of active terms in certain scenarios, including spurious contributions attributed to numerical differentiation noise and closed-loop controller effects. The angle-based method yielded substantially sparser and physically cleaner fault identifications, demonstrating that the end-to-end formulation achieves higher sensitivity and more robust fault separation. Practical insights on loss balancing, SIREN tuning, fault library parameterization, and data segment selection were discussed alongside the current limitations of the work.

8.2 Future Work

Several directions remain open for extending the work presented in this thesis.

The most immediate step is the transition from offline to online fault diagnosis. The current PINN training requires computation times on the order of minutes per trajectory, which is incompatible with the real-time control loop of the servo drive. A practical deployment would separate the computational effort across two phases: an offline commissioning phase in which the healthy physical parameters are identified once through a dedicated system identification experiment, analogous to the self-commissioning routines already established in modern AC drive technology, and an online monitoring phase in which only the sparse fault coefficients are updated. In the online phase, the pre-trained network weights serve as a fixed representation of the healthy baseline, and a lightweight residual monitor evaluates the augmented physics residual over short measurement windows as they are acquired. Transfer learning across operating conditions, where a network trained on one motion profile is fine-tuned for another, offers an additional means of reducing the online computational burden.

Broader experimental validation is necessary to establish the generalizability of the proposed framework. The current evaluation was confined to a single test bench with two relatively mild fault conditions, chosen to avoid risking damage to the available hardware. Future experiments should include more severe fault levels such as large misalignment offsets, extensive tooth damage, and progressive belt wear, as well as additional fault types relevant to industrial belt transmissions, including bearing defects, pulley eccentricity, and belt tension loss. Progressive fault scenarios, in which a fault develops gradually over time, would further test the method's ability to detect incipient faults and track fault evolution, which is of particular relevance for predictive maintenance applications. Combined fault scenarios, where multiple faults occur simultaneously, would assess whether the sparse identification framework can disentangle superimposed fault signatures through its selection of physically distinct candidate terms. Validation across different transmission ratios, belt types, and load configurations would confirm that the identified fault signatures reflect physical mechanisms rather than properties specific to the current setup.

A further direction concerns the systematic tuning of the framework's hyperparameters. The network architecture, loss weights, SIREN frequency parameters, sparsity thresholds, and the number of alternating ℓ_0/ℓ_1 loops were selected empirically in the present work. Automated hyperparameter optimization, for example through Bayesian search or meta-learning approaches, would reduce the manual effort required to apply the framework to new systems and could identify parameter settings that improve robustness across operating conditions. In particular, adaptive strategies for the loss weights, such as those based on gradient magnitude balancing, could further mitigate the gradient conflict between data and physics losses during training.

Finally, the candidate fault library used in the sparse identification stage could be extended and refined based on deeper physical modeling of the belt transmission. The current library captures speed-dependent friction effects and angle-periodic disturbances at integer and fractional multiples of the shaft angle, which cover the dominant fault mechanisms observed in this work. More sophisticated libraries could incorporate terms derived from detailed belt-pulley contact models, backlash dynamics, or bearing fault frequencies, enabling the identification of fault types that fall outside the current scope. Such extensions would move the framework closer to a comprehensive diagnostic tool capable of characterizing the full spectrum of faults encountered in industrial PMSM-driven transmission systems.

Acknowledgements

There are many people I wish to thank, not only for this thesis, but for the entire journey of my master's studies.

First, I would like to thank my thesis supervisor, Dr. Agnes Steinert, and my advisor, Mr. Zoe Mbikayi. They took a genuine interest in my research from the very beginning, and they were always patient in helping me work through the algorithms whenever I got stuck. They also pointed out the places where my academic writing fell short, which I truly appreciated. Our meetings were held regularly, and after each one, I walked away with my thoughts much more organized than before. Their guidance shaped the thesis in a way I am deeply grateful for.

Furthermore, I would like to thank Prof. Dr. Florian Holzapfel. The rigorous academic culture he has cultivated at the Institute of Flight System Dynamics left me both nervous and excited. The dense, carefully reasoned theories and algorithms developed there are the foundation that allows stable commercial airliners and high-performance supersonic fighter jets to do their jobs well. I understood this in principle before, but studying here and writing this thesis has made me appreciate it on a much deeper level.

My special thanks go to the ABB German Research Center. Every colleague there was kind, and the working environment was truly first-rate; the months I spent there were both productive and genuinely enjoyable. A big thank you to my supervisor there, Dr. Chen Song, a senior scientist at ABB. We spoke often throughout the project, and our conversations went well beyond the immediate problem at hand. He helped me see how theory, engineering practice, and real industrial needs fit together, and through his patient explanations he deepened my understanding of AI, particularly industrial AI, and even of the underlying mathematics. I learned a great deal from him. I would also like to thank Dr. Joachim Weissbacher from ABB Austria (B&R), who introduced us to the test bench and its control methods, and who offered us so much guidance and hospitality during our experiments in Austria.

I am grateful to my teachers from middle school through my undergraduate years, who gave me far more than what was written in the textbooks. I also want to thank my friends, who have been an important source of strength throughout this journey.

Finally, I want to thank my family. My grandfather was a distinguished electrical engineer. His final two months fell during my summer break after the national college entrance exam, and I spent that time with him in the hospital. Three weeks before he passed, I was admitted to Wuhan University of Technology, which made him very happy. I am not a clever person, but he and my grandmother always believed in me, and even stood up to my primary school teacher when she dismissed my ability to learn. It has been nearly ten years, but I still think of their encouragement whenever things get hard. I believe he would be proud to know I made it to the Technical University of Munich, and I am thankful my grandmother is still with us.

My mother is a single mother. As a child, I worked hard at my studies but my grades were poor, and I was often sick. She cried many times, but she never gave up on my education, and has always believed in me, encouraged me, and supported me. My father has not been absent from my life either; he supports my pursuits in his own way, and though he is not one to express himself easily, I have felt it all the same. My aunt and uncle helped my mother guide my studies when my father was not around, and they have supported me ever since. My cousin has been like an older sister to me; we grew up together, and she could be quite strict with me at times. During COVID-19, my mother was diagnosed with breast cancer but kept it from me out of concern, because I was in Germany preparing for the TUM entrance exam. It was my cousin who stayed with her at the hospital.

The road here has not been easy. To everyone who walked part of it with me, thank you.

References

[1] Arntz Optibelt Group. *Optibelt Omega Timing Belts — Technical Data Sheet*. https://www.optibelt.com/. Accessed: 2024-XX-XX. 2024.

[2] B&R Industrial Automation. *8LS Series Synchronous Servo Motor 8LSA35.DA030S200-3 — Technical Data Sheet*. https://www.br-automation.com/. Accessed: 2024-XX-XX. 2024.

[3] B&R Industrial Automation. *ACOPOS P3 Servo Drive — Technical Documentation*. https://www.br-automation.com/. Accessed: 2024-XX-XX. 2024.

[4] B&R Industrial Automation. *X20 System Compact CPU X20CP3687X — Technical Data Sheet*. https://www.br-automation.com/. Accessed: 2024-XX-XX. 2024.

[5] Nicolo Bachschmid, Paolo Pennacchi, and A Vania. "Identification of multiple faults in rotor systems". In: *Journal of sound and vibration* 254.2 (2002), pp. 327–366.

[6] Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. "Discovering governing equations from data by sparse identification of nonlinear dynamical systems". In: *Proceedings of the national academy of sciences* 113.15 (2016), pp. 3932–3937.

[7] Pao-Hsiung Chiu, Jian Cheng Wong, Chinchun Ooi, My Ha Dao, and Yew-Soon Ong. "CAN-PINN: A fast physics-informed neural network based on coupled-automatic-numerical differentiation method". In: *Computer Methods in Applied Mechanics and Engineering* 395 (2022), p. 114909.

[8] Lucio Ciabattoni, Francesco Ferracuti, Alessandro Freddi, and Andrea Monteriu. "Statistical spectral analysis for fault diagnosis of rotating machines". In: *IEEE Transactions on Industrial Electronics* 65.5 (2017), pp. 4301–4310.

[9] R. F. Fung et al. "Dynamic analysis and system identification of an LCD glass-handling robot with nonlinear joints". In: *Applied Mathematical Modelling* (2010).

[10] Zhiwei Gao, Carlo Cecati, and Steven X Ding. "A survey of fault diagnosis and fault-tolerant techniques—Part I: Fault diagnosis with model-based and signal-based approaches". In: *IEEE transactions on industrial electronics* 62.6 (2015), pp. 3757–3767.

[11] William T Hale and George M Bollas. "Symbolic regression of uncertainty-resilient inferential sensors for fault diagnostics". In: *IFAC-PapersOnLine* 53.2 (2020), pp. 11446–11451.

[12] Fabian Heldmann, Sarah Berkhahn, Matthias Ehrhardt, and Kathrin Klamroth. "PINN Training using Biobjective Optimization: The Trade-off between Data Loss and Residual Loss". In: *Journal of Computational Physics* 488 (2023), p. 112211.

[13] Diederik P. Kingma and Jimmy Ba. "Adam: A Method for Stochastic Optimization". In: *International Conference on Learning Representations*. 2015.

[14] Yaguo Lei, Naipeng Li, Liang Guo, Ningbo Li, Tao Yan, and Jing Lin. “Machinery health prognostics: A systematic review from data acquisition to RUL prediction”. In: *Mechanical systems and signal processing* 104 (2018), pp. 799–834.

[15] Yaguo Lei, Bin Yang, Xinwei Jiang, Feng Jia, Naipeng Li, and Asoke K Nandi. “Applications of machine learning to machine fault diagnosis: A review and roadmap”. In: *Mechanical systems and signal processing* 138 (2020), p. 106587.

[16] Wangwang Liao, Xiangyun Long, and Chao Jiang. “A physics-informed neural network method for identifying parameters and predicting remaining life of fatigue crack growth”. In: *International Journal of Fatigue* 191 (2025), p. 108678.

[17] Xue Liu, Wei Cheng, Ji Xing, Xuefeng Chen, Zhibin Zhao, Lin Gao, Rongyong Zhang, Qian Huang, Hongpeng Zhou, Wei Xing Zheng, et al. “Hierarchical physics-informed neural network for rotor system health assessment”. In: *IEEE Transactions on Automation Science and Engineering* 22 (2025), pp. 10392–10405.

[18] Xue Liu, Wei Cheng, Ji Xing, Xuefeng Chen, Zhibin Zhao, Rongyong Zhang, Qian Huang, Jinqi Lu, Hongpeng Zhou, Wei Xing Zheng, et al. “Physics-informed Neural Network for system identification of rotors”. In: *IFAC-PapersOnLine* 58.15 (2024), pp. 307–312.

[19] Lu Lu, Xuhui Meng, Zhiping Mao, and George Em Karniadakis. “DeepXDE: A deep learning library for solving differential equations”. In: *SIAM Review* 63.1 (2021), pp. 208–228.

[20] Jason Mais et al. “Spectrum analysis: the key features of analyzing spectra”. In: *SKF USA, Inc* (2002).

[21] Ehsan Mazaheri-Tehrani, Jawad Faiz, Shiva Garaei, Ahmad Kalhor, and Chunyan Lai. “Physics-Informed Neural Network Approach to Diagnosing Uniform Demagnetization Faults in Permanent Magnet Synchronous Machines”. In: *IEEE Access* 13 (2025), pp. 190933–190946.

[22] Marc Parziale, Luca Lomazzi, Marco Giglio, and Francesco Cadini. “Physics-informed neural networks for the condition monitoring of rotating shafts”. In: *Sensors* 24.1 (2023), p. 207.

[23] ZK Peng, ZQ Lang, SA Billings, and GR Tomlinson. “Comparisons between harmonic balance and nonlinear output frequency response function in nonlinear system analysis”. In: *Journal of Sound and Vibration* 311.1-2 (2008), pp. 56–73.

[24] Cristian Puerto-Santana, Carlos Ocampo-Martinez, and Javier Diaz-Rozo. “Mechanical rotor unbalance monitoring based on system identification and signal processing approaches”. In: *Journal of Sound and Vibration* 541 (2022), p. 117313.

[25] Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred A. Hamprecht, Yoshua Bengio, and Aaron Courville. “On the Spectral Bias of Neural Networks”. In: *Proceedings of the 36th International Conference on Machine Learning (ICML)*. 2019.

[26] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations”. In: *Journal of Computational physics* 378 (2019), pp. 686–707.

[27] A. Savitzky and M. J. E. Golay. “Smoothing and Differentiation of Data by Simplified Least Squares Procedures”. In: *Analytical Chemistry* 36.8 (1964), pp. 1627–1639. DOI: 10.1021/ac60214a047.

[28] Wen-Bin Shangguan, Xiao Feng, Haoting Lin, and Jiawei Yang. “A calculation method for natural frequencies and transverse vibration of a belt span in accessory drive systems”. In: *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science* 227.10 (2013), pp. 2268–2279.

[29] Brian de Silva, Kathleen Champion, Markus Quade, Jean-Christophe Loiseau, J. Kutz, and Steven Brunton. “PySINDy: A Python package for the sparse identification of nonlinear dynamical systems from data”. In: *Journal of Open Source Software* 5.49 (2020), p. 2104. DOI: 10.21105/joss.02104. URL: https://doi.org/10.21105/joss.02104.

[30] Vincent Sitzmann, Julien N.P. Martel, Alexander W. Bergman, David B. Lindell, and Gordon Wetzstein. “Implicit Neural Representations with Periodic Activation Functions”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2020.

[31] Fangzheng Sun, Yang Liu, Jian-Xun Wang, and Hao Sun. “Symbolic physics learner: Discovering governing equations via Monte Carlo tree search”. In: *arXiv preprint arXiv:2205.13134* (2022).

[32] Shilin Sun, Tianyang Wang, Hongxing Yang, and Fulei Chu. “Adversarial representation learning for intelligent condition monitoring of complex machinery”. In: *IEEE Transactions on Industrial Electronics* 70.5 (2022), pp. 5255–5265.

[33] Silviu-Marian Udrescu and Max Tegmark. “AI Feynman: A physics-inspired method for symbolic regression”. In: *Science advances* 6.16 (2020), eaay2631.

[34] Peter Vas. *Parameter Estimation, Condition Monitoring, and Diagnosis of Electrical Machines*. Monographs in Electrical and Electronic Engineering. Oxford, UK: Oxford University Press, 1993. ISBN: 978-0-19-859375-1.

[35] Shengfeng Wang, Yannik Bhatt, and Siddhartha Bhatt. “On the preprocessing of physics-informed neural networks: How to better utilize data in fluid mechanics”. In: *Physics of Fluids* 36.1 (2024).

[36] Shibo Wang and Bin Tang. “A comparative study of parameter identification methods for asymmetric nonlinear systems with quadratic and cubic stiffness”. In: *Sensors* 22.15 (2022), p. 5854.

[37] Sifan Wang, Xinling Yu, and Paris Perdikaris. “When and why PINNs fail to train: A neural tangent kernel perspective”. In: *Journal of Computational Physics* 449 (2022), p. 110768.

[38] Yichen Yao, Honglei Sha, Yixin Su, Gexue Ren, and Suyuan Yu. “Identification of system parameters and external forces in AMB-supported PMSM system”. In: *Mechanical Systems and Signal Processing* 166 (2022), p. 108456. DOI: 10.1016/j.ymssp.2021.108456.

[39] Yaoxiang Yu, Liang Guo, Yongwen Tan, Hongli Gao, and Jiangquan Zhang. “Multisource partial transfer network for machinery fault diagnostics”. In: *IEEE Transactions on Industrial Electronics* 69.10 (2021), pp. 10585–10594.

[40] Lei Yuan, Yi-Qing Ni, Xiang-Yun Deng, and Shuo Hao. “A-PINN: Auxiliary physics informed neural networks for forward and inverse problems of nonlinear integro-differential equations”. In: *Journal of Computational Physics* 462 (2022), p. 111260.

[41] Wei Zhou and Y. Mei. “Data-Guided Physics-Informed Neural Networks for Solving Inverse Problems in Partial Differential Equations”. In: *arXiv preprint arXiv:2407.10836* (2024).